

ALGEBRA 2

Unit 2

The collage displays various algebra worksheets for Unit 2. Visible topics include:

- RELATIONS**: Definitions of relations and functions.
- LINEAR Relations**: Notes on graphing linear relations and finding the slope of a line.
- SCATTER PLOTS**: Worksheets on positive, negative, and no correlation, and lines of best fit.
- SYSTEMS of Linear Inequalities**: Worksheets on graphing and solving systems of linear inequalities.
- LINEAR PROGRAMMING**: Worksheets on finding the feasible region and optimizing a function.
- SYSTEMS OF EQUATIONS**: Worksheets on solving systems of equations by graphing.

LINEAR FUNCTIONS & SYSTEMS

NOTES • HOMEWORK • QUIZZES • TEST

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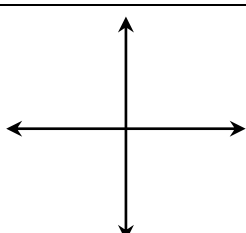
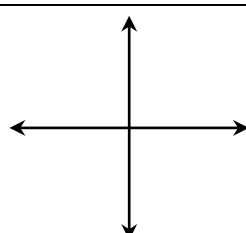
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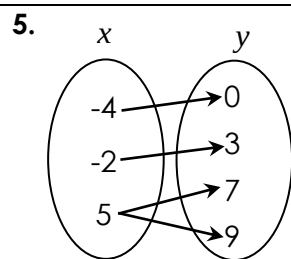


Unit 2 – Linear Functions & Systems: Sample Unit Outline

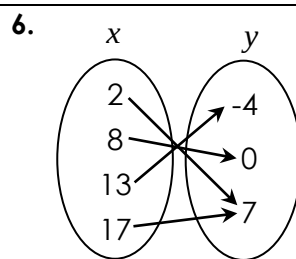
DAY	TOPIC	HOMEWORK
DAY 1	Relations & Functions Review: (Relation vs. Function, Domain, Range, Evaluating Functions)	HW #1
DAY 2	Graphing Linear Functions Review Day 1: (Slope & Slope-Intercept Form, Vertical & Horizontal Lines)	HW #2
DAY 3	Graphing Linear Functions Review Day 2: (Standard Form and Intercepts)	↓
DAY 4	Quiz 2-1	None
DAY 5	Writing Linear Functions Review: (Given a Point and Slope & Two Points, Parallel vs. Perpendicular)	HW #3
DAY 6	Applications of Linear Functions & Linear Regression	HW #4
DAY 7	Quiz 2-2	None
DAY 8	Solving Systems by Graphing & Substitution	HW #5
DAY 9	Solving Systems by Elimination & Applications	HW #6
DAY 10	Solving Systems with Three Variables	HW #7
DAY 11	Quiz 2-3	None
DAY 12	Linear Inequalities & Systems of Linear Inequalities	HW #8
DAY 13	Unit 2 Review	Study for Test
DAY 14	UNIT 2 TEST	None
DAY 15	Linear Programming	Choice Board Assignment

Name: _____	Date: _____
Topic: _____	Class: _____

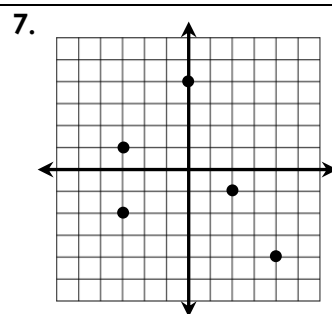
Main Ideas/Questions	Notes/Examples																									
RELATIONS	- A relation is a set of ordered pairs. Relations can be shown in a _____, _____, _____, or expressed as an _____. - The set of x-values is called the _____. - The set of y-values is called the _____. - A relation with distinct points is called _____. - A relation with connected lines or curves is called _____.																									
What is a FUNCTION?	A function is a relation in which each element of the domain is paired with exactly one element of the range. <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px; vertical-align: top;"> FUNCTION: </td> <td style="width: 50%; padding: 5px; vertical-align: top;"> NOT A FUNCTION: </td> </tr> <tr> <td colspan="2" style="text-align: center; padding: 5px;"> This means x-values do not repeat! </td> </tr> </table>		FUNCTION:	NOT A FUNCTION:	This means x-values do not repeat!																					
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This means x-values do not repeat!																										
VERTICAL LINE TEST In order to be a function, a vertical line can not pass through more than one point in the relation.	FUNCTION: 	NOT A FUNCTION: 																								
EXAMPLES	Give the domain, range, and determine whether the relation is a function. <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px; vertical-align: top;"> 1. $\{(-1, -9), (0, -5), (4, 3), (6, -2)\}$ D: _____ R: _____ Function? _____ </td> <td style="width: 50%; padding: 5px; vertical-align: top;"> 2. $\{(5, 2), (-3, 4), (5, -6), (8, 0)\}$ D: _____ R: _____ Function? _____ </td> </tr> <tr> <td style="width: 50%; padding: 5px; vertical-align: top;"> 3. <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td style="width: 10%;">x</td> <td style="width: 10%;">7</td> <td style="width: 10%;">2</td> <td style="width: 10%;">0</td> <td style="width: 10%;">-4</td> </tr> <tr> <td>y</td> <td>-8</td> <td>3</td> <td>-1</td> <td>3</td> </tr> </table> D: _____ R: _____ Function? _____ </td> <td style="width: 50%; padding: 5px; vertical-align: top;"> 4. <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td style="width: 10%;">x</td> <td style="width: 10%;">10</td> <td style="width: 10%;">15</td> <td style="width: 10%;">18</td> <td style="width: 10%;">24</td> </tr> <tr> <td>y</td> <td>0</td> <td>3</td> <td>7</td> <td>11</td> </tr> </table> D: _____ R: _____ Function? _____ </td> </tr> </table>		1. $\{(-1, -9), (0, -5), (4, 3), (6, -2)\}$ D: _____ R: _____ Function? _____	2. $\{(5, 2), (-3, 4), (5, -6), (8, 0)\}$ D: _____ R: _____ Function? _____	3. <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td style="width: 10%;">x</td> <td style="width: 10%;">7</td> <td style="width: 10%;">2</td> <td style="width: 10%;">0</td> <td style="width: 10%;">-4</td> </tr> <tr> <td>y</td> <td>-8</td> <td>3</td> <td>-1</td> <td>3</td> </tr> </table> D: _____ R: _____ Function? _____	x	7	2	0	-4	y	-8	3	-1	3	4. <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td style="width: 10%;">x</td> <td style="width: 10%;">10</td> <td style="width: 10%;">15</td> <td style="width: 10%;">18</td> <td style="width: 10%;">24</td> </tr> <tr> <td>y</td> <td>0</td> <td>3</td> <td>7</td> <td>11</td> </tr> </table> D: _____ R: _____ Function? _____	x	10	15	18	24	y	0	3	7	11
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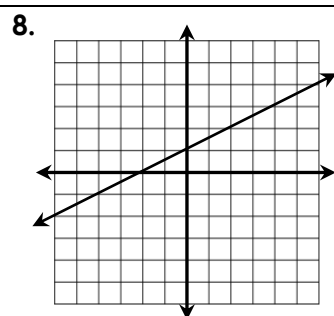
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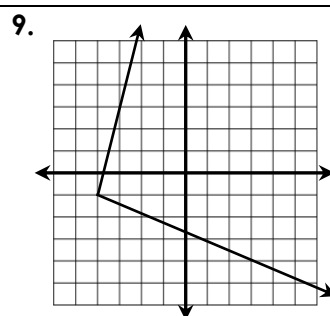
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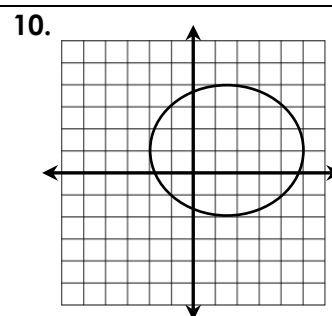
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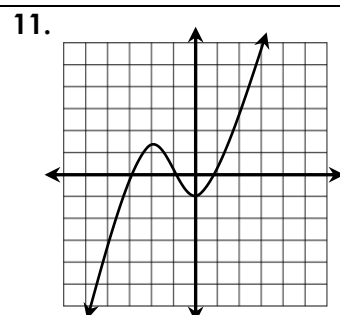
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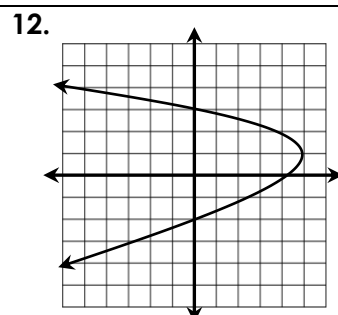
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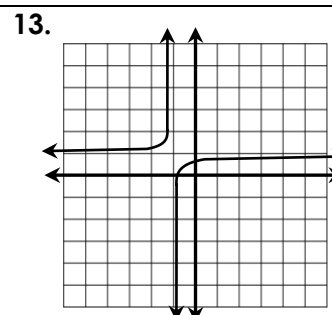
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 Function? _____



D : _____
 R : _____
 Function? _____



D : _____
 R : _____
 Function? _____



D : _____
 R : _____
 Function? _____

NOTE:

Use a graphing device for #14-17.

14. $y = -x^2 + 3$

D : _____
 R : _____
 Function? _____

15. $y = -5x - 1$

D : _____
 R : _____
 Function? _____

16. $y = -7$

D : _____
 R : _____
 Function? _____

17. $x = -1$

D : _____
 R : _____
 Function? _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
FUNCTION <i>Notation</i>	Equations that represent functions are often written in function notation . Example: $y = 2x + 7 \rightarrow$ _____ is the input and _____ is the output .	
EVALUATING <i>Functions</i> Given: $f(x) = -5x - 14$ $g(x) = -x^2 + 9x - 1$	Function notation is specifically useful for finding a value in the range that corresponds to a certain domain value.	Example: If $f(x) = 2x + 7$, find $f(-3)$
	For questions 1-6, evaluate using $f(x)$ and $g(x)$ given to the left.	
	1. $f(-9)$	2. $f\left(\frac{3}{10}\right)$
	3. $g(0.4)$	4. $g(-7)$
	5. $f(1) - f(-4)$	6. $2[g(12)]$
Given: $p(x) = \frac{1}{2} - \frac{4}{3}x$ $q(x) = 3x - x^3$	For questions 7-12, evaluate using $p(x)$ and $q(x)$ given to the left.	
	7. $p\left(-\frac{9}{16}\right)$	8. $q(5)$
	9. $q(-2)$	10. $p(18)$

	11. $7 - q(4.5)$	12. $3[p(-1)] + 5$
Given: $f(x) = 3x + 19$	For questions 13 and 14, find the value of x given $f(x)$ to the left.	
	13. If $f(x) = 34$, find x .	14. If $f(x) = -62$, find x .
Given: $f(x) = -4x + 9$ $g(x) = x^2 - 5$	For questions 15-18, evaluate using $f(x)$ and $g(x)$ given to the left.	
	15. $f(c + 2)$	16. $f(7w - 5)$
	17. $g(a - 3)$	
	18. $g(2m + 1) - f(3m - 7)$	
APPLICATION: Owen paid a lawn company \$108 for an initial lawn treatment, then \$74 per month thereafter. The function $p(t) = 108 + 74t$ represents the total amount of money he will have paid after t months.	Use the information to the left for questions 19 and 20.	
	19. Find and interpret $p(24)$.	20. How long has Owen been receiving lawn service if he has paid a total of \$2,994 to date?

Name: _____

Unit 2: Linear Functions & Systems

Date: _____ Per: _____

Homework 1: Relations & Functions

** This is a 2-page document! **

Find the domain and range of each relation. Then determine if the relation is a function.

1. $\{(-1, 4), (-1, 9), (-1, 15), (-1, 0)\}$

D: _____

R: _____

Function? _____

2.

x	5	6	7	8
y	0	-2	10	-3

D: _____

R: _____

Function? _____

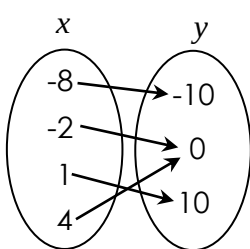
3. $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$

D: _____

R: _____

Function? _____

4.

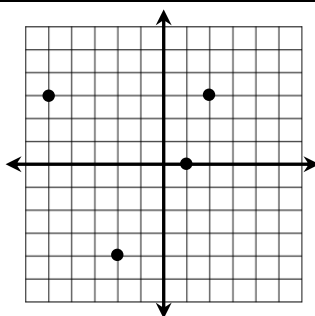


D: _____

R: _____

Function? _____

5.

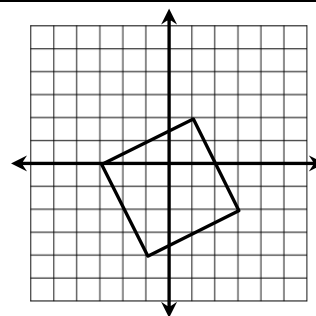


D: _____

R: _____

Function? _____

6.

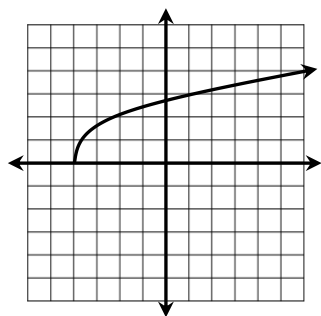


D: _____

R: _____

Function? _____

7.

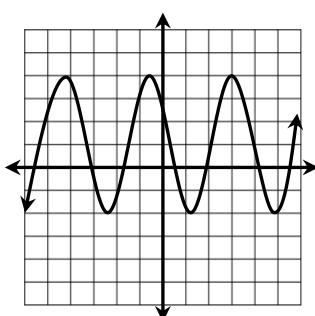


D: _____

R: _____

Function? _____

8.

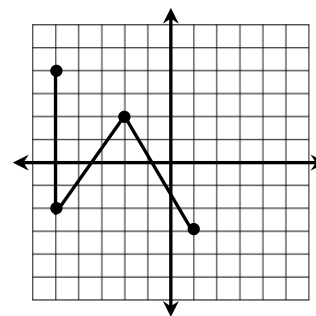


D: _____

R: _____

Function? _____

9.

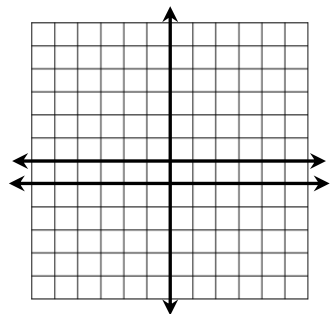


D: _____

R: _____

Function? _____

10.



D: _____

R: _____

Function? _____

11. $x = 4$

D: _____

R: _____

Function? _____

12. $y = x^2 - 4x - 5$

D: _____

R: _____

Function? _____

Use $f(x) = -\frac{3}{2}x + 11$, $g(x) = x^2 - 7x$, and $h(x) = |15 - 9x|$ to evaluate each function.

13. $f(-8)$	14. $g(-2)$	15. $h\left(\frac{2}{3}\right)$
16. $h(5) + g(9)$		17. $1-2\left[f\left(\frac{8}{9}\right)\right]$
18. If $f(x) = -7$, find x .		19. If $h(x) = 48$, find x .
20. Find $f(-8c + 2)$		21. Find $g(y - 8)$
22. The commission earned on the sale of a car can be represented by the equation $c(p) = 250 + 0.02p$ where c represents the commission and p represents the purchase price of the car.		
a) Jack sold two cars last weekend, one for \$15,075 and the other for \$21,640. Find his total commission.		b) If $c(p) = 745.80$, find the purchase price of the car.

Name:

Date:

Topic:

Class:

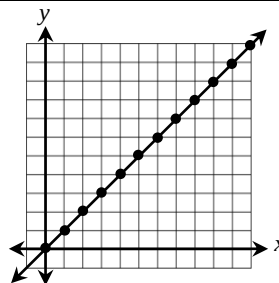
Main Ideas/Questions

Notes/Examples

LINEAR

Relations

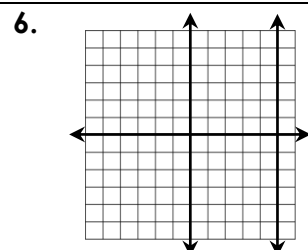
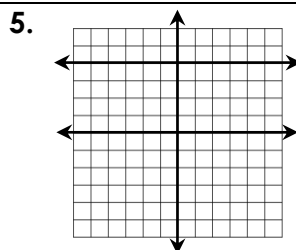
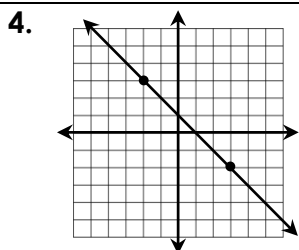
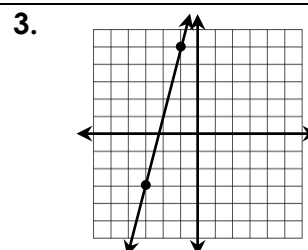
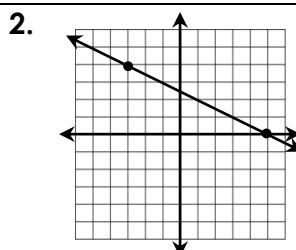
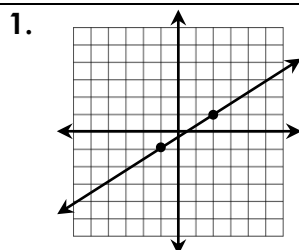
- When points on a graph lie along a **straight line**, it is called a **linear relation**.
- Linear relations can be represented by an **equation**, for example, the equation of the graph to the right can be written as $y = x$.



SLOPE

Slope is the constant **rate of change** between the points on a line.
(Remember rise/run!)

Find the slope of the line graphed.



SLOPE FORMULA:

Find the slope of the line passing through the given points.

7. $(-8, -6)$ and $(2, 14)$

8. $(8, -7)$ and $(16, -13)$

9. $(5, -6)$ and $(5, 2)$

10. $(-7, -1)$ and $(4, -1)$

SLOPE-INTERCEPT

Form

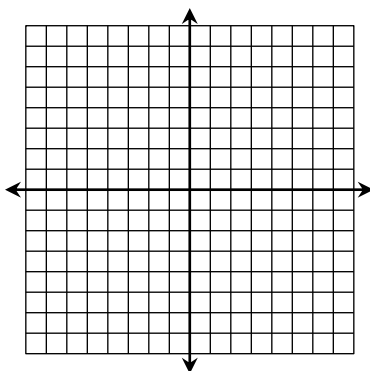
Linear equations are often written in **slope-intercept form**:

_____ (or) _____

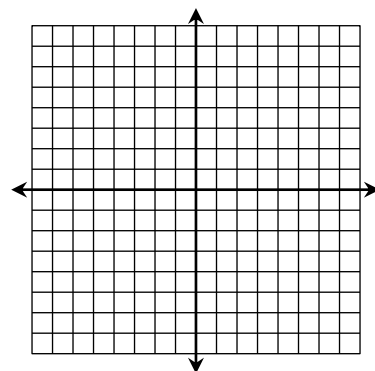
(where **m** is the _____ and **b** is the _____)

Identify the slope and y-intercept, then graph the line.

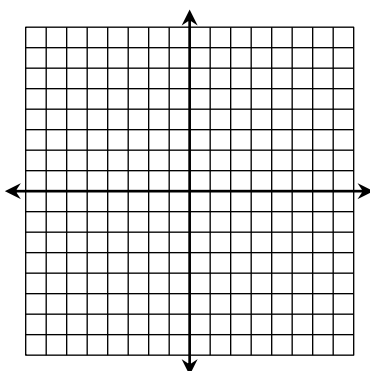
11. $y = \frac{2}{5}x + 7$



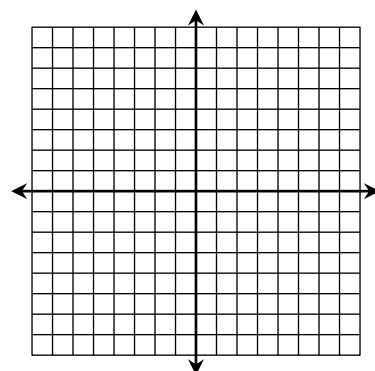
12. $y = -x + 2$



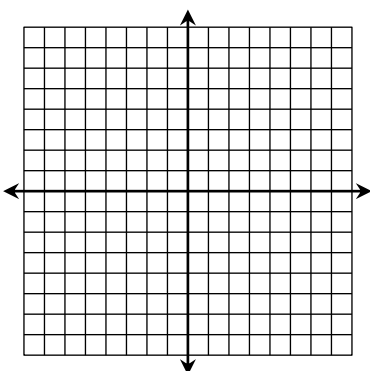
13. $y = -\frac{1}{3}x - 5$



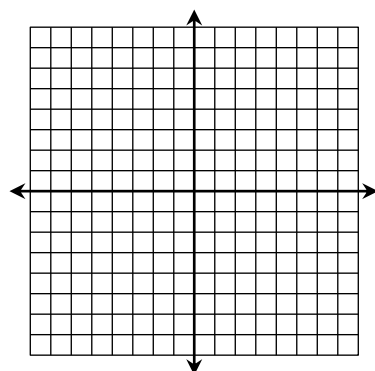
14. $f(x) = -4x + 1$



15. $f(x) = \frac{1}{2}x$



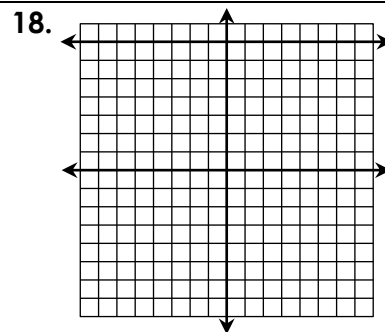
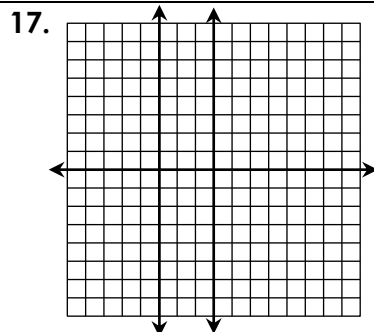
16. $f(x) = \frac{5}{6}x + 2$



**VERTICAL &
HORIZONTAL**

Lines

Write each equation to represent the line shown on the graph.



Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

STANDARD*Form*

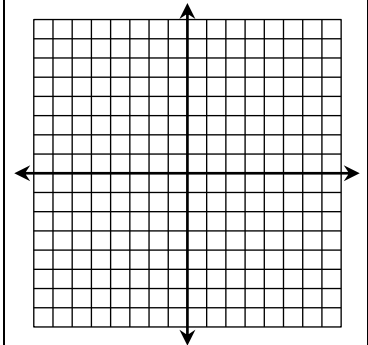
Directions: Write each equation in both standard form and slope-intercept form, then graph the line.

Linear equations are also often written in **standard form**:

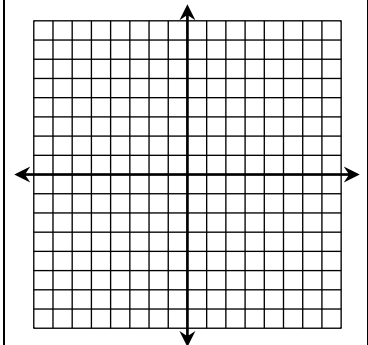
(where A , B , and C are integers, $A \geq 1$, and the **GCF** of A , B , and C is 1)

Standard Form**Slope-Intercept Form****Graph**

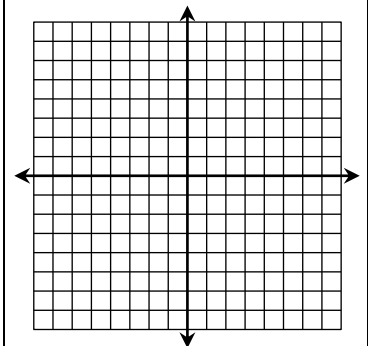
1. $-5x + 6y = -12$



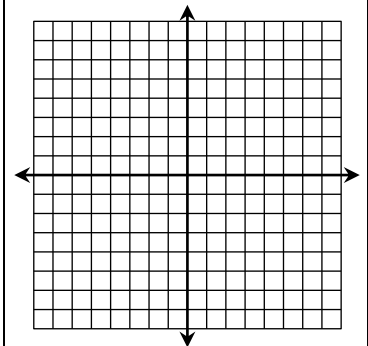
2. $9x + 6y - 24 = 0$



3. $7y = 7 - 7x$

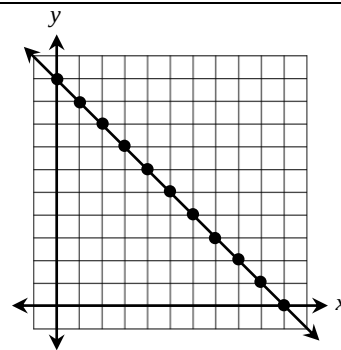


4. $\frac{3}{4}x - 3y = 1$



X- and Y-INTERCEPTS

- The point at which the graph intersects the x-axis is called the **x-intercept**. Likewise, the point at which the graph intersects the y-axis is called the **y-intercept**.
- Identify the x- and y-intercept of the graph shown on the right.



The **x-intercept** is also referred to as a _____, _____, or _____ to the equation.

FINDING INTERCEPTS

Algebraically

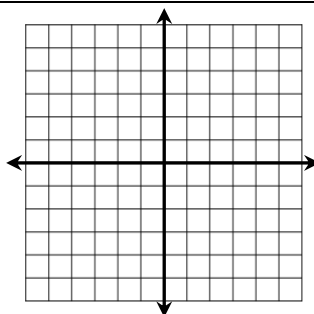
To find the x-intercept of an equation, set y equal to 0 and solve for x . To find the y-intercept of an equation, set x equal to 0 and solve for y .

Example: Find the x- and y-intercepts of the equation $y = -2x + 5$.

EXAMPLES

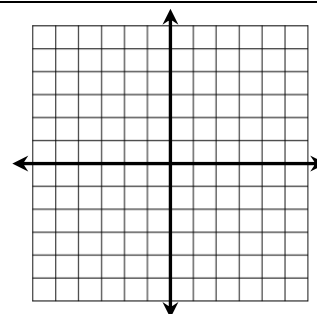
Directions: Find the x- and y-intercept of each equation. Then, graph the equation using the intercepts.

5. $y = 5x - 4$



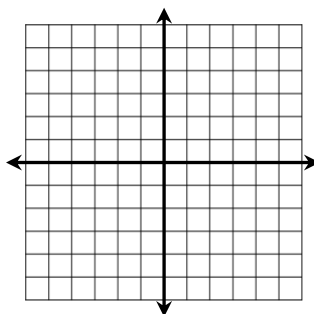
x-int: _____ y-int: _____

6. $y = -\frac{6}{5}x + 2$



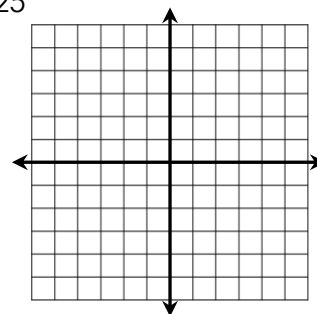
x-int: _____ y-int: _____

7. $4x - y = 6$



x-int: _____ y-int: _____

8. $-5x - 8y = -25$



x-int: _____ y-int: _____

Name: _____

Unit 2: Linear Functions & Systems

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Homework 2: Slope-Intercept & Standard Form

** This is a 2-page document! **

Find the slope of the line passing through the given points.

1. $(5, -2)$ and $(-7, -8)$

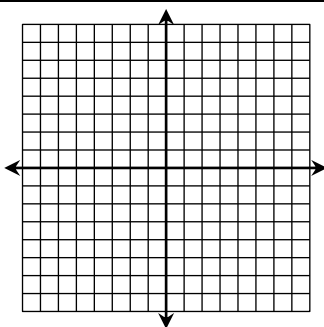
2. $(-3, 4)$ and $(-3, -5)$

3. $(-2, 1)$ and $(-10, 3)$

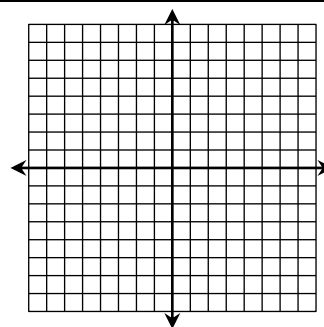
4. $(0, 8)$ and $(-6, 8)$

Graph each line. Write the equation in slope-intercept form when necessary.

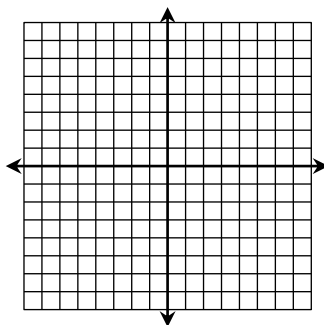
5. $y = -\frac{5}{6}x + 4$



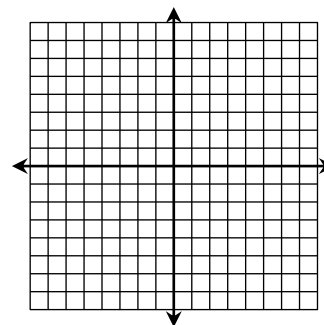
6. $y = x - 1$



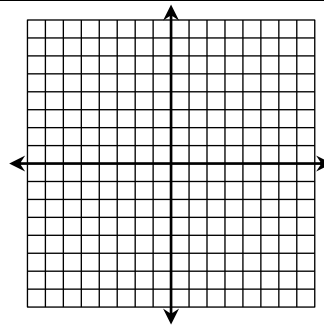
7. $f(x) = 3x - 7$



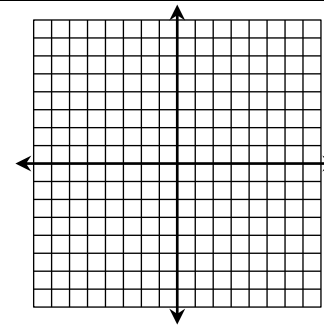
8. $f(x) = -\frac{7}{2}x + 5$



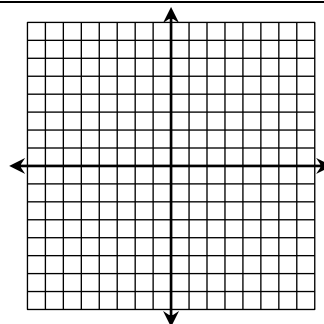
9. $x - y = 8$



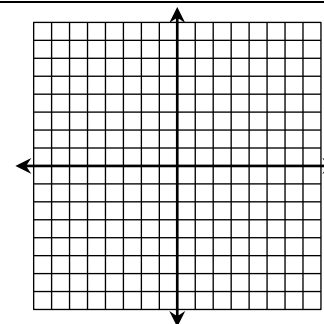
10. $12x + 8y = 28$

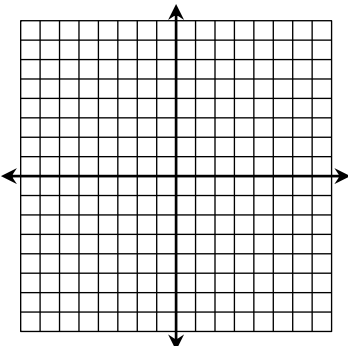
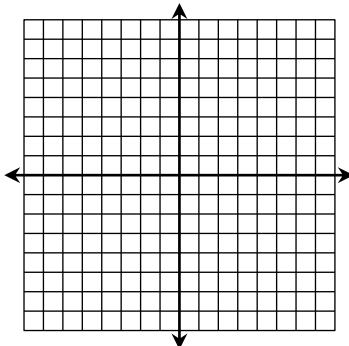


11. $x - 2y - 2 = 0$

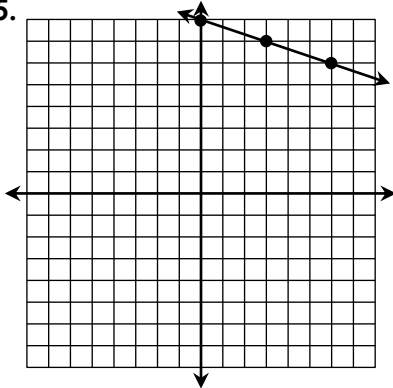
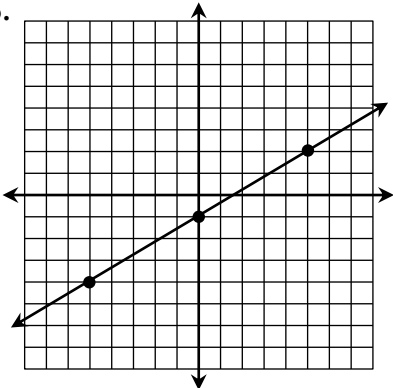


12. $-5x = 2y$



13. $y = -6$ 	14. $x = 2$ 
---	--

Write the equation of the line graphed in both slope-intercept form and standard form.

15. 	16. 
---	--

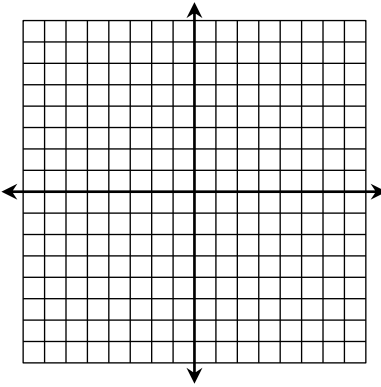
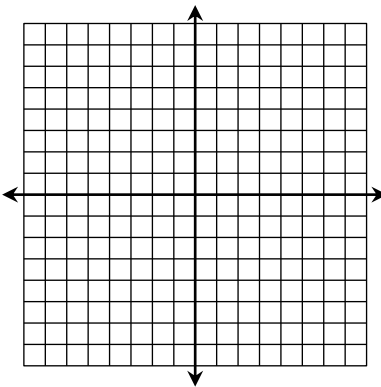
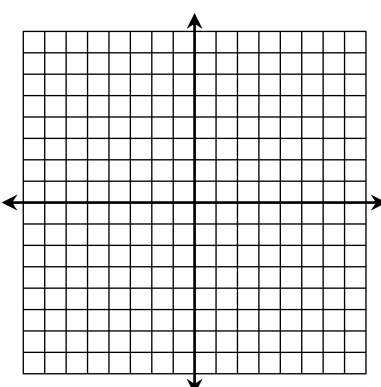
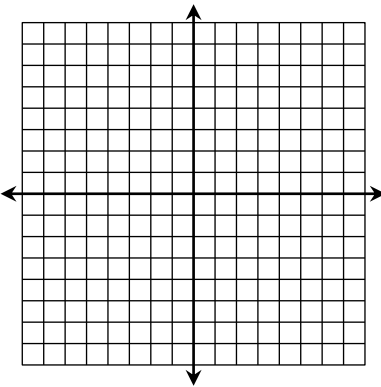
Slope-Intercept Form:

Slope-Intercept Form:

Standard Form:

Standard Form:

Find the x - and y -intercepts of each equation. Use the intercepts to graph the line.

17. $y = 2x - 2$ 	18. $6x + 5y = -15$ 
19. $20 + 12x = 4y$ 	20. $7x - 10y = 35$ 

Name: _____

Algebra 2

Date: _____ Per: _____

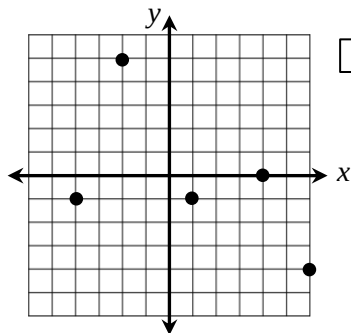
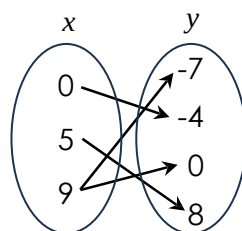
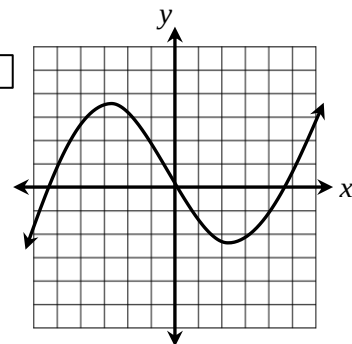
Unit 2: Linear Functions & Systems

Quiz 2-1: Relations, Functions, & Graphing Linear Functions

1. Which relations represent a function? Check all that apply.

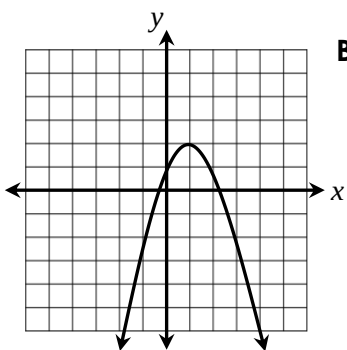
☐

x	y
-8	1
-7	1
-6	1
-5	1

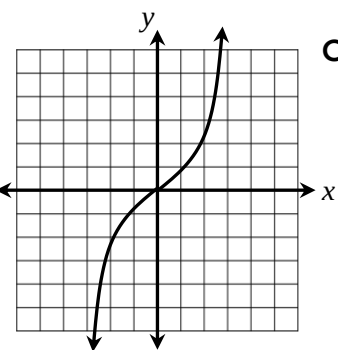
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2. Which graph does not represent a function?

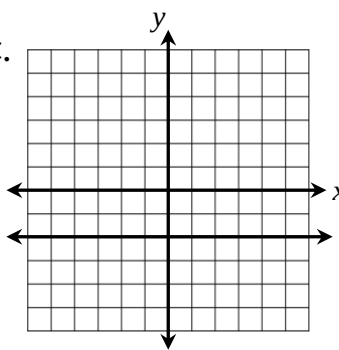
A.



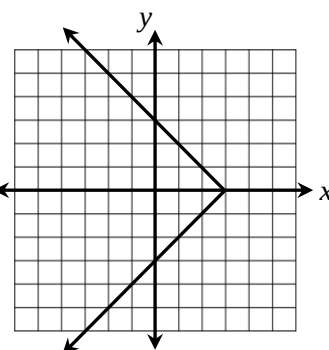
B.



C.

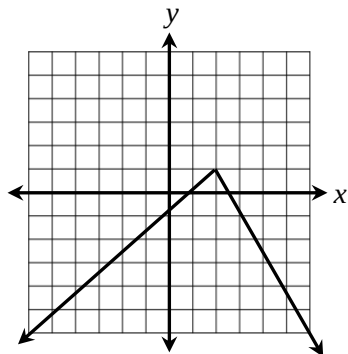


D.

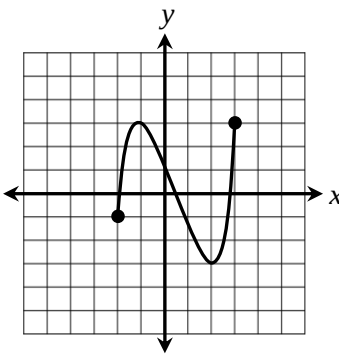


Give the domain and range of each relation below.

3.



4.



3. $D =$ _____

$R =$ _____

4. $D =$ _____

$R =$ _____

Use the functions $f(x) = 10 - \frac{9}{2}x$, $g(x) = |-3x + 16|$, and $h(x) = -x^2 + 8x$ for the questions 5 and 6.

5. Find $f\left(-\frac{1}{3}\right)$

6. Find $g(20) + h(7)$

5. _____

6. _____

Use the functions $f(x) = 10 - \frac{9}{2}x$, $g(x) = |-3x + 16|$, and $h(x) = -x^2 + 8x$ for the questions 7 and 8.

7. If $f(x) = -53$, find x .

8. If $g(x) = 17$, find x .

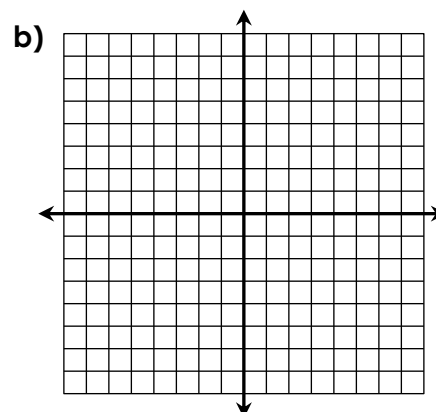
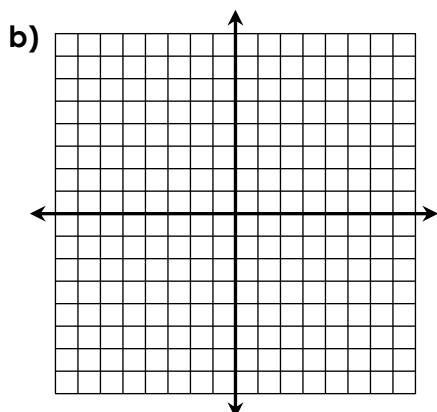
7. _____

8. _____

For questions 9 and 10, (a) write the equation in slope-intercept form, then (b) graph the line.

9. $x - 4y = 28$ a) _____

10. $9y - 36 = -15x$ a) _____



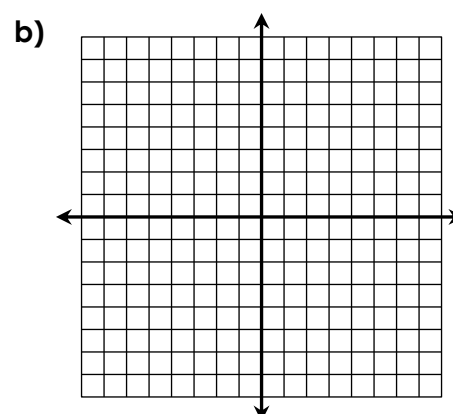
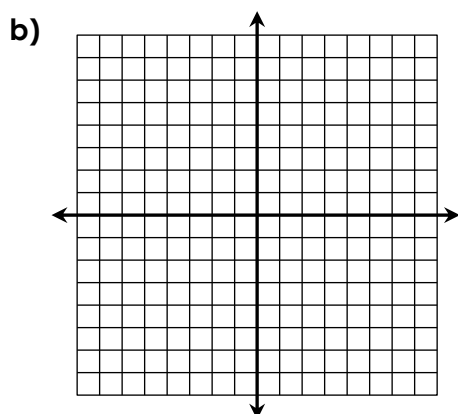
For questions 11 and 12, (a) give the x- and y-intercepts of the function, then (b) graph the line.

11. $y = -6x - 5$ a) x-intercept: _____

12. $\frac{9}{4}x - \frac{1}{2}y = -3$ a) x-intercept: _____

y-intercept: _____

y-intercept: _____



Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
WRITING LINEAR EQUATIONS Given a Point & Slope	To write a linear equation given the slope of the line (m), and a point that it passes through (x_1, y_1) , use the point-slope formula : _____	
	Write the equation of the line in slope-intercept form given a point on the line and the slope of the line.	
	1. $(-1, 7); m = -4$	2. $(-6, -5); m = \frac{2}{3}$
	3. $(4, -3); m = 1$	4. $(-8, 12); m = -\frac{5}{4}$
WRITING LINEAR EQUATIONS Given Two Points	To write a linear equation given two points (x_1, y_1) and (x_2, y_2) use the slope formula followed by the point-slope formula : _____  _____	
	5. $(-6, 1)$ and $(-3, 7)$	6. $(-2, -2)$ and $(-6, 8)$
	7. $(10, -5)$ and $(1, -8)$	8. $(-8, -3)$ and $(-2, -3)$

PARALLEL vs. PERPENDICULAR LINES

- The **slopes** of **parallel lines** are _____.
- The **slopes** of **perpendicular lines** are _____.

Determine whether the pair of lines are parallel, perpendicular, or neither.

9. $10x + 5y = 5$ and $-x = 2y + 16$

10. $8x - 6y = 42$ and $-3y = 9 - 4x$

11. $-8x + 8y = 0$ and $x - y = 4$

12. $x = 2$ and $y = 2$

Write an equation in slope-intercept form with the given information.

13. Passing through the point $(-6, 2)$ and parallel to the line $x - 3y = 21$.

14. Passing through the point $(-1, -3)$ and parallel to the line passing through the points $(-4, 11)$ and $(2, -1)$.

15. Passing through the point $(-9, 1)$ and perpendicular to the line $6x + 4y = -4$.

16. Passing through the point $(7, -16)$ and perpendicular to the line passing through the points $(4, 2)$ and $(-8, -10)$.

Name: _____

Unit 2: Linear Functions & Systems

Date: _____ Per: _____

Homework 3: Writing Linear Equations



Write a linear equation in slope-intercept form with the given information.	
1. slope = -6; passes through (-4, 1)	2. slope = $\frac{4}{3}$; passes through (-5, -6)
3. passes through (-4, 11) and (2, 8)	4. passes through (6, 3) and (14, -5)
5. passes through (-8, 3) and parallel to the line $3x - 4y = 4$	6. passes through (-2, 9) and parallel to the line passing through (1, -1) and (-2, 14)
7. passes through (4, -1) and perpendicular to the line $x - 2y = 6$	8. passes through (8, 9) and perpendicular to the line passing through (-6, 2) and (-1, -2)
Determine if the given lines are parallel, perpendicular, or neither.	
9. $x + y = 8$ and $y = -x - 1$	10. $9x - 6y = -20$ and $-3y = 2x - 11$
11. $2x = 14 + y$ and $4x + 2y = 10$	12. $y = -4$ and $y = 7$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Linear Equation APPLICATIONS TYPE I: Slope-Intercept Form	1. A cab ride costs \$4 plus \$1.75 per mile driven.	
	a) Write a linear equation to represent the cab fare. Define your variables.	b) If Clay's destination is 17 miles away and he has \$30, will he make it?
	2. The Jones' are purchasing a new home. The closing costs are \$249 plus 3% of the purchase price of the home.	
	a) Write a linear equation to represent the total closing costs. Define your variables.	b) The purchase price of their home is \$195,000. The sellers offered \$5,000 in closing cost assistance. Will this cover all their closing costs? If not, how much will the Jones' need to pay out of pocket?
	3. A 22.5-ounce candle burns at the rate of one ounce every five hours.	
	a) Write an equation to represent the amount of candle left. Define your variables.	b) How many hours will the candle last?
	4. The water level of a lake is at 80 feet. The lake will be drained at a rate of 3 inches per day each day in September, beginning on September 1.	
	a) Write an equation to represent the level of the lake each day since it was drained.	b) Find the level of the lake at the end of the day on September 30.

TYPE 2:

Standard Form

5. Josh bought 3 bags of potting soil and 10 bags of mulch from Home Depot.

a) Write an equation to represent the total cost. Define your variables.

b) If Josh spent \$40.91 and mulch cost \$2.75 per bag, how much was a bag of potting soil?

6. Kayleigh has a collection of quarters and nickels.

a) Write an equation to represent the total value of the coins. Define your variables.

b) If the total value of the coins is \$5.50 and Kayleigh has 17 quarters, how many nickels does she have?

7. The baseball concession stand sells soft pretzels for \$4.50 and bags of popcorn for \$3.25.

a) Write an equation to represent the total sales from pretzels and popcorn on a given day. Define your variables.

b) If the concession stand sold 137 soft pretzels, how many bags of popcorn would they need to sell to make at least \$1,000?

8. A math test has two sections: Section A consists of true/false questions worth 2 points each, and Section B consists of multiple choice worth 5 points each.

a) Write an equation to represent the total number of points possible a student could earn on the test. Define your variables.

b) If the highest possible score is 125 points and there are 5 questions in Section A, how many total questions are on the test?

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

SCATTER PLOTS

A **scatter plot** shows the **relationship between two variables**.
Sketch the three possible relationships below.

Positive Correlation

Negative Correlation

No Correlation

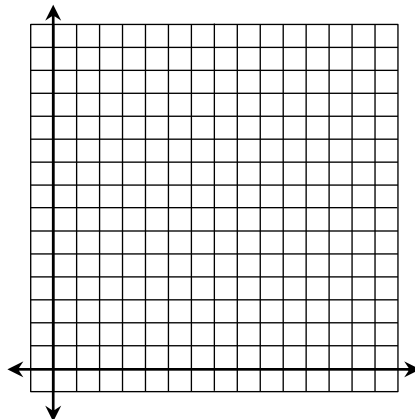
Line of BEST FIT

Given a positive or negative correlation, it's possible to write a prediction equation, called a **line of best fit**. It can be used to predict the value of a variable given the value of the other variable.

EXAMPLE

d	t
0	85
4	82
7	81
10	79
12	77
15	76

a) Make a scatter plot.



b) Use two ordered pairs to write a line of best fit.

c) Predict the temperature of the lake at 25 feet.

LINEAR Regression

Calculator Steps

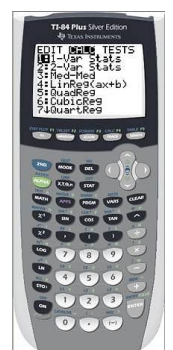
Another method to find the line of best fit is called **linear regression**. This can be done through your graphing calculator and will create the **most precise equation** possible.

Enter your data.

- Hit **STAT**, **ENTER**
- Enter **x-values** in L_1
- Enter **y-values** in L_2

Find the line of best fit.

- Hit **STAT**
- Arrow over to **CALC**
- Choose **#4-LinReg(ax+b)**
- Hit **ENTER** (possibly twice)



LINEAR REGRESSION EXAMPLES

1. The table below shows the number of people, p , and suitcases, s , on 8 airplanes.

p	63	115	153	192	224	297	331	375
s	112	209	278	302	379	500	586	712

a) Find the regression equation.

b) Predict the number of suitcases on an airplane with 500 people.

2. The table below shows the population, in thousands, during certain years since 2000.

year	0	3	6	9	12	15
population	10.7	11.1	11.9	12.5	13.2	13.6

a) Find the regression equation.

b) Predict the population in 2025.

3. The table below shows the salary, in millions, made by the highest paid major league baseball player during certain years.

year	1985	1991	1993	1996	1998	2001
salary	2.1	3.8	6.2	9.3	14.9	22

a) Find the regression equation.

b) Predict the salary of the highest paid player in 2020.

4. The table below shows the mileage, m , on six cars that were recently traded in, along with their value, v .

m	5,824	25,824	58,126	73,005	105,092	159,062
v	\$16,920	\$10,300	\$7,225	\$5,300	\$3,125	\$1,800

a) Find the regression equation.

b) Predict the value of a car with 13,258 miles.

Name: _____

Unit 2: Linear Functions & Systems



Date: _____ Per: _____

Homework 4: Applications & Linear Regression

**** This is a 2-page document! ****

Linear Equation Applications	
1. Troy's truck has a 30-gallon gas tank and gets an average of 21 miles per gallon.	
a) Write an equation to represent the amount of gas in Troy's truck after driving a certain number of miles (assuming he starts with a full tank). Define your variables.	b) If the tank is half full, approximately how many miles has Troy driven since last filling up his tank?
2. Mikayla is training for a half-marathon. At the beginning of her training, she can run 2 miles without stopping. Her goal is to increase her distance by three quarters of a mile each week.	
a) Write an equation to represent the number of miles Mikayla can run after each week. Define your variables.	b) If a half-marathon (13.1 miles) is 12 weeks after the start of Mikayla's training program, will she be ready?
3. The music store sold 12 trumpets and 19 clarinets last month.	
a) Write an equation to show the amount of money earned on trumpet and clarinet sales last month. Define your variables.	b) If clarinets sell for \$139.95 and the store made \$4,578.45 from sales on the two instruments, find the price of a trumpet.
4. Alysha has a collection of dime and nickel rolls that she'd like to take to the bank. Each dime roll has 50 dimes and each nickel roll has 40 nickels.	
a) Write an equation to represent the total value of Alysha's dime and nickel collection. Define your variables.	b) If the total value of the coins is \$71 and she has 9 rolls of dimes, how many rolls of nickels does she have?

Linear Regression

5. The table below shows the average price of a movie ticket during certain years.

<i>year</i>	2002	2003	2004	2005	2006	2007	2008	2009
<i>price</i>	5.80	6.03	6.21	6.41	6.55	6.88	7.18	7.35

a) Find the regression equation.

b) During what approximate year will the price of a movie ticket reach \$15?

6. The table below shows the recorded temperature on a certain day starting at 6:00 a.m.

<i>time</i>	6:00	6:45	7:30	8:30	9:00	9:45	10:30	11:00
<i>temp</i>	65°	67°	68°	72°	74°	75°	77°	80°

a) Find the regression equation.

b) Predict the temperature at 2:00 p.m.

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 2: Linear Functions & Systems

Quiz 2-2: Parallel & Perpendicular Lines, Writing Equations, & Applications

1. Which equations are **parallel** to the line $-3x - 6y = 30$?
Check all that apply.

☐ $2x - y = 15$

☐ $y = -3x + 1$

☐ $4y = 20 - 2x$

☐ $5x + 10y = 10$

☐ $x = 18 - 2y$

☐ $6 - 2y = 4x$

2. Give an example of a linear equation that is **perpendicular** to $y = 5$. 2. _____

Write the equation of the line in slope-intercept form with the given information.

3. Passes through $(-8, 7)$ with a slope of $-\frac{1}{4}$

3. _____

4. _____

5. _____

6. _____

7. _____

4. Passes through $(-2, -8)$ and parallel to $7x - 2y = -8$.

5. Passes through $(2, -2)$ and $(-3, -17)$

6. Passes through $(5, 2)$ and perpendicular to $5x - 2y = -6$.

7. Passes through $(-8, -7)$ and perpendicular to the line passing through the points $(-6, 3)$ and $(-1, -7)$.

8. To manage a rental property, a property manager charges \$59 per month plus 8% of the rent.

a) Write an equation to represent the total monthly fees charged by the property manager.

8a) _____

b) _____

c) _____

b) If Jack pays the property manager \$165 each month, what is the monthly rent on his property?

c) If Jack raises the rent on the house to \$1,495, how much more will he pay the property manager each month?

9. Katya tutors and works in the college bookstore to pay for her college tuition. She makes \$20 per hour tutoring and \$8.50 per hour working in the bookstore.

a) Write an equation to represent the total amount of money Katya earns tutoring and working in the bookstore.

9a) _____

b) _____

b) The tuition for one semester (16 weeks) is \$5,736. If Katya works 21 hours each week in the bookstore, how many hours per week must she tutor to pay her tuition?

10. The table below shows the average price of a personal computer during certain years.

<i>year</i>	1997	1999	2003	2007	2010	2012
<i>price</i>	\$1,950	\$1,719	\$1,420	\$1,175	\$949	\$725

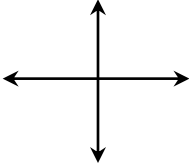
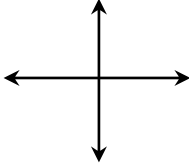
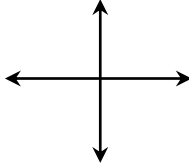
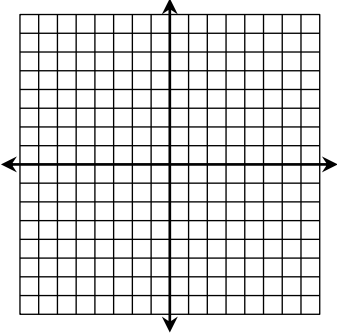
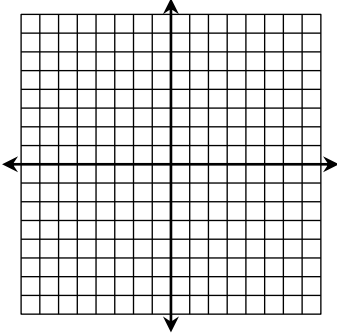
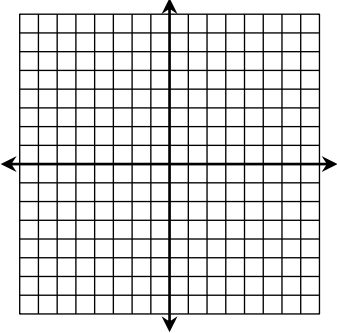
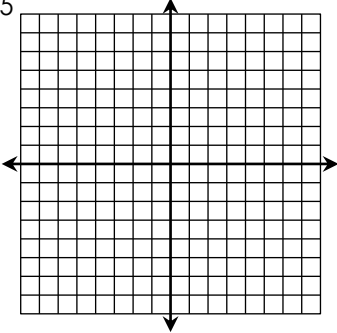
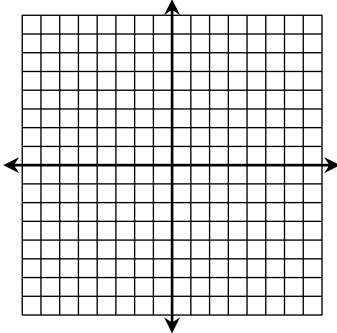
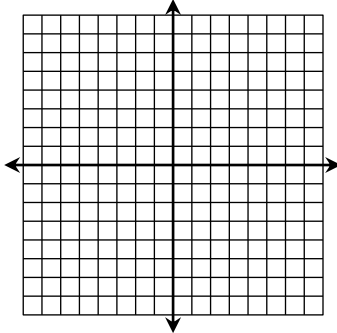
a) Write an equation to model the data using linear regression.

10a) _____

b) _____

b) Use the equation to predict the average cost of a personal computer in 2020.

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
SYSTEMS OF EQUATIONS			
<i>Types of</i> SOLUTIONS			
	One Solution	No Solution	Infinite Solution
<i>Solving by</i> GRAPHING	Solve each system of equations below by graphing. Identify the solution.		
	1. $\begin{cases} y = -\frac{5}{3}x - 6 \\ y = \frac{1}{6}x + 5 \end{cases}$ 	2. $\begin{cases} y = 5 \\ y = 2x + 7 \end{cases}$ 	
	3. $\begin{cases} 6x - 5y = -45 \\ 2x + 2y = -4 \end{cases}$ 	4. $\begin{cases} 3x + 6y = -12 \\ y = -\frac{1}{2}x - 5 \end{cases}$ 	
	5. $\begin{cases} 2y = 8x + 18 \\ 24 + 4y = x \end{cases}$ 	6. $\begin{cases} -y = -x - 6 \\ 3x + 18 = 3y \end{cases}$ 	

<i>Solving by</i> SUBSTITUTION	❶	SOLVE one equation for x or y. (Isolate a variable)
	❷	SUBSTITUTE the resulting equation from step 1 into the other equation for that variable.
	❸	SOLVE for the remaining variable.
	❹	SUBSTITUTE your answer from step 3 into either original equation to find the other variable.
Solve each system of equations below by substitution. Identify the solution.		
7. $\begin{cases} y = -7x - 1 \\ y = x - 9 \end{cases}$	8. $\begin{cases} y = -5x + 30 \\ 7x + 3y = 42 \end{cases}$	
9. $\begin{cases} 6x - 5y = -28 \\ 7x + y = 22 \end{cases}$	10. $\begin{cases} x - 7y = 53 \\ -4x - 5y = 19 \end{cases}$	
11. $\begin{cases} 2y = 6x + 10 \\ 3x - y = 5 \end{cases}$	12. $\begin{cases} 5x + 7y = -17 \\ 4x - 3y = -5 \end{cases}$	

Name: _____

Unit 2: Linear Functions & Systems

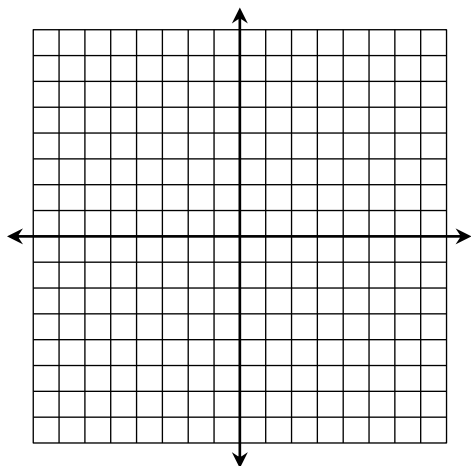
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Homework 5: Solving Systems of Equations
by Graphing & Substitution

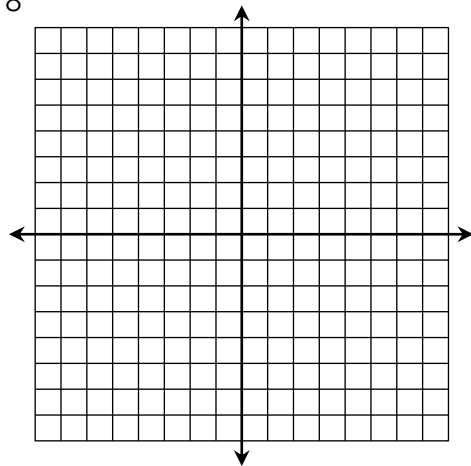
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Solve each system of equations by graphing. Clearly identify your solution.

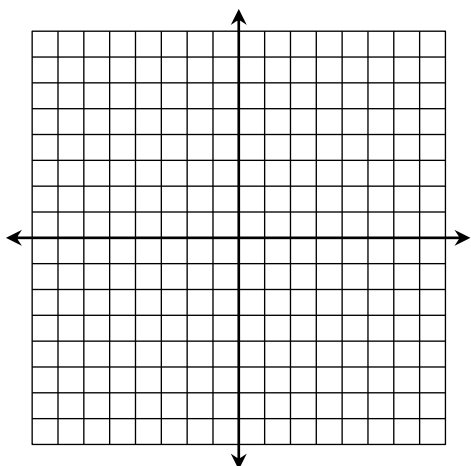
1.
$$\begin{cases} y = \frac{1}{4}x + 1 \\ y = -x - 9 \end{cases}$$



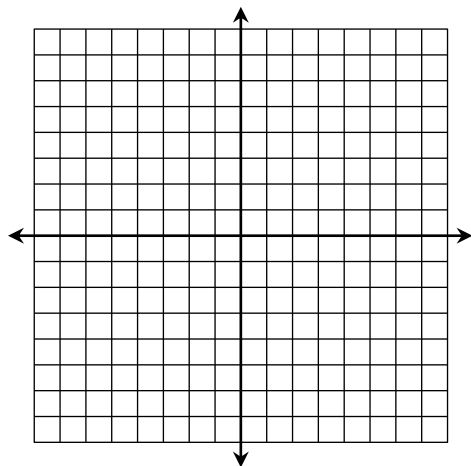
2.
$$\begin{cases} y = 2x - 8 \\ 13x + 2y = 18 \end{cases}$$



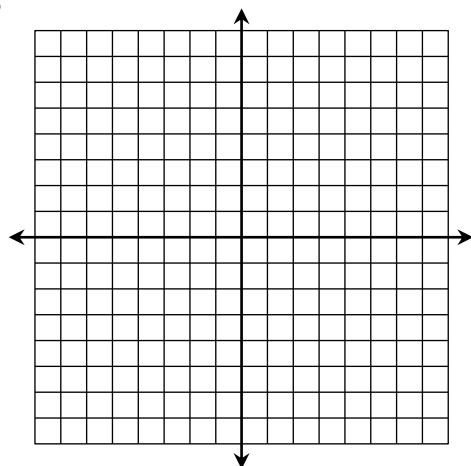
3.
$$\begin{cases} 3x + 7y = -63 \\ x - y = -1 \end{cases}$$



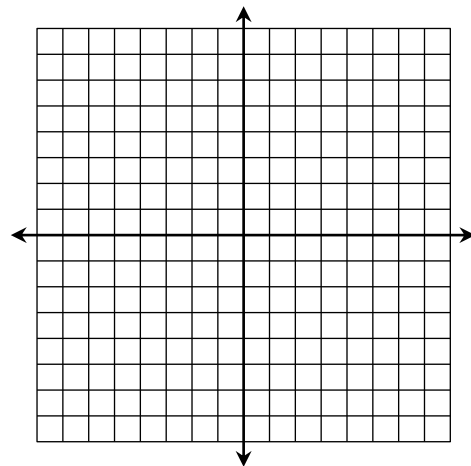
4.
$$\begin{cases} 3x - 2y = 10 \\ x = 2 \end{cases}$$



5.
$$\begin{cases} -6y = 36 - x \\ 4x - 3y = -3 \end{cases}$$



6.
$$\begin{cases} -12x = 30y + 60 \\ 2x + 5y = 20 \end{cases}$$



Solve each system of equations by substitution. Clearly identify your solution.

7.
$$\begin{cases} y = -2x + 14 \\ y = 5x - 7 \end{cases}$$

8.
$$\begin{cases} y = -4x - 9 \\ 7x + 4y = -27 \end{cases}$$

9.
$$\begin{cases} x - y = -9 \\ 8x - 3y = -62 \end{cases}$$

10.
$$\begin{cases} 7x - y = 34 \\ 2x + 3y = -10 \end{cases}$$

11.
$$\begin{cases} 2x = 16 - 8y \\ x + 4y = 25 \end{cases}$$

12.
$$\begin{cases} -\frac{2}{3}x - 7y = 6 \\ x + 3y = -9 \end{cases}$$

13.
$$\begin{cases} 2x - 9y = 26 \\ 6x - 2y = -22 \end{cases}$$

14.
$$\begin{cases} 3x - 2y = 10 \\ 9x + 4y = -50 \end{cases}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
<i>Solving by</i> ELIMINATION	❶	LINE UP the equations.
	❷	MULTIPLY one or both equations by a number to result in a variable with the same coefficient.
	❸	ADD or SUBTRACT the equations to eliminate this variable.
	❹	SOLVE for the remaining variable.
	❺	SUBSTITUTE your answer from step 4 into either original equation to find the other variable.

Solve each system of equations below by elimination. Identify the solution.			
1. $\begin{cases} x + 7y = 17 \\ x - y = -7 \end{cases}$		2. $\begin{cases} 3x + 2y = 22 \\ 5x - 2y = 42 \end{cases}$	
3. $\begin{cases} 4x - 9y = -42 \\ x + 5y = 4 \end{cases}$		4. $\begin{cases} 7x - 6y = -53 \\ 2x - 3y = -13 \end{cases}$	
5. $\begin{cases} 5x + 3y = -7 \\ 2x + 7y = 3 \end{cases}$		6. $\begin{cases} 3x - 9y = 9 \\ 4x - 12y = 36 \end{cases}$	

7. $\begin{cases} 5x - 6y = 3 \\ 7y = 2x + 8 \end{cases}$	8. $\begin{cases} 10x + 18 = -8y \\ 4 + 9x = 5y \end{cases}$
---	--

APPLICATIONS	❶	DEFINE YOUR VARIABLES – What are you solving for?
	❷	SET UP TWO EQUATIONS using the information given.
	❸	SOLVE the system using your method of choice.

9. Marcie bought a total of 20 used books and CDs during a yard sale for a total of \$54.50. If books cost \$1.50 each and CDs cost \$5 each, how many of each did she buy?
10. Landon babysits and works part time at the water park over the summer. One week, he babysat for 3 hours and worked at the water park for 10 hours and made \$109. The next week he babysat for 8 hours and worked at the water park for 12 hours and made \$177. How much does Landon make per hour at each job?
11. Kent has a collection of pennies and nickels with a value of \$1.98. The number of pennies he has is five less than twice the number of nickels. How many of each coin does Kent have?

MORE APPLICATIONS PRACTICE *with* SYSTEMS

Define your variables, set up a system of equations, and solve each problem.

1. At the fast food restaurant, four cheeseburgers and five small fries have a total of 2,310 calories. Three cheeseburgers and two small fries have a total of 1,330 calories. How many calories does each item contain?
2. One month, a homeowner used 150 units of gas and 520 units of electric for a total cost of \$84.20. The next month, 210 units of gas and 405 units of electric were used for a total cost of \$82.35. Find the cost per unit of gas and electric.
3. Kym is renovating the first floor of her home. She bought 750 square feet of laminate flooring and 525 square feet of carpet and paid \$2717.25. She went back to the store and bought an additional 100 square feet of laminate flooring and 75 square feet of carpet and paid \$374.75. Find the cost per square foot of each type of flooring.
4. Sean bought 1.8 pounds of gummy bears and 0.6 pounds of jelly beans and paid \$10.26. He went back to the store the following week and bought 1.2 pounds of gummy bears and 1.5 pounds of jellybeans and paid \$15.09. What is the price per pound of each type of candy?

5. Courtney has a total of 88 stamps worth \$15.56. Some are 25¢ stamps and some are 2¢ stamps. How many of each does she have?
6. Tom put 18 gallons of midgrade gas in his truck and filled up his empty five-gallon gas can with regular gas for his lawnmower at home. He spent \$59.91. The following week, he put 14 gallons of midgrade gas in his truck and topped off his five-gallon can with just one gallon of regular gas. If he paid \$39.75 and the prices remained the same, find the price per gallon of midgrade and regular gas.
7. A total of \$12,000 was invested in two types of bonds. One pays 8% simple interest while the other pays 10.5%. Last year, the annual interest earned on the two investments was \$1,145. How much was invested at each rate?
8. A collection of dimes and quarters is worth \$9.55. If the quarters were dimes and the dimes were quarters, the total value would be \$7.60. Find the number of each coin.

Name: _____

Unit 2: Linear Functions & Systems



Date: _____ Per: _____

Homework 6: Solving Systems of Equations
by Elimination & Applications

**** This is a 2-page document! ****

Solve each system of equations by elimination. Clearly identify your solution.

1.
$$\begin{cases} 3x + y = 11 \\ 5x - y = 21 \end{cases}$$

2.
$$\begin{cases} 2x - 9y = -51 \\ 2x - 3y = -9 \end{cases}$$

3.
$$\begin{cases} 5x + 12y = -34 \\ 2x + 4y = -12 \end{cases}$$

4.
$$\begin{cases} -3x + 8y = 73 \\ x - 7y = -46 \end{cases}$$

5.
$$\begin{cases} 9x + 4y = -72 \\ 2x - 3y = -16 \end{cases}$$

6.
$$\begin{cases} 4x - 11y = 68 \\ 6x + 5y = -27 \end{cases}$$

7.
$$\begin{cases} 5x = 15 - 5y \\ 8y + 26 = 2x \end{cases}$$

8.
$$\begin{cases} 14x + 7y = -7 \\ y = -2x - 1 \end{cases}$$

Define your variables and set up a system of equations, then solve.

9. A storeowner mixed 8 pounds of peanuts and 5 pounds of M&M's. This 13-pound mixture sold for \$55.27. The second mixture included 6 pounds of peanuts and 4 pounds of M&M's. This 10-pound mixture sold for \$42.70. Find the cost per pound of the peanuts and M&M's.
10. Katy's favorite rides at the amusement park are the roller coaster and water slide. The wait time for the roller coaster is 25 minutes and the wait time for the water slide is 10 minutes. If she went on 12 rides total and waited three hours in line, how many times did she go on each ride?
11. A collection of nickels and dimes is worth \$9.45. If the number of dimes is doubled, the value is \$16.65. Find the number of each coin.
12. Zack has two savings accounts with a total of \$9,000. He withdrew 10% from one and 60% from the other to buy his girlfriend an engagement ring. If the ring cost \$2,175, find the remaining balance in each account.

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
THREE- VARIABLE <i>Systems</i>	① CHOOSE TWO EQUATIONS in which you can easily eliminate a variable. Let's call this new equation after the elimination Equation A .
	② ELIMINATE THE SAME VARIABLE from a combination of the equation you didn't use and one of the ones you did. Let's call this equation after the elimination Equation B .
	③ PUT EQUATION A AND EQUATION B TOGETHER and SOLVE THAT SYSTEM . You now have two of your variables!
	④ SUBSTITUTE your answers from step 3 into either original equation to find the remaining variable. Write your solution as (x, y, z) .

EXAMPLES

1.
$$\begin{cases} 2x - 6y + 9z = -8 \\ 5x + y + 2z = 10 \\ 3x + y - 8z = -28 \end{cases}$$

2.
$$\begin{cases} 4x + y - 3z = 45 \\ 6x - 4y + 3z = 41 \\ 2x - 7y - 11z = 85 \end{cases}$$

3.
$$\begin{cases} 5x - 3y + 8z = -59 \\ 9x - 4y - z = -60 \\ 5x + 2y + 4z = -47 \end{cases}$$

4.
$$\begin{cases} x - 5y + 3z = 39 \\ 3x + 8y - z = -19 \\ 2x - 6y - 5z = -29 \end{cases}$$

5.
$$\begin{cases} x + y + 3z = 12 \\ -3x + 7y - 2z = -35 \\ 7x - 6y - 5z = 32 \end{cases}$$

APPLICATIONS

6. The theater sells three types of tickets: general admission, senior citizen, and child. Mark bought 9 general admission, 2 senior citizen, and 3 child tickets and paid \$170. Sarah bought 5 general admission, 4 senior citizen, and 7 child tickets and paid \$173. Kyle bought 4 general admission, 1 senior citizen, and 6 child tickets and paid \$116.50. Find the cost of each type of ticket.

7. Jamie has a collection of 78 nickels, dimes, and quarters worth \$12.40. If the number of quarters is doubled, the value becomes \$22.15. How many of each type of coin does Jamie have?

NOTE: Substitution might be better method for the examples below.

8. Mason has a collection of 49¢ stamps, 20¢ stamps, and 3¢ stamps worth \$23.55. He has 56 total 49¢ and 20¢ stamps and the number of 3¢ stamps is nine more than the number of 20¢ stamps. How many of each does he have?
9. The three medals earned in the Olympics are the gold, silver, and bronze medals. In the 2014 Winter Olympics, the United States had 28 total medal winners. There were two more gold medals than silver medals and the number of bronze medals was four less than the sum of the number of gold and silver medals. How many winners were there of each type of medal?
10. Enrique has a pay-per-use plan on his cell phone. It costs \$0.10 per minute talk, \$0.20 per text message, and \$0.25 per picture message. His total bill last month was \$30.95. The total number of text and picture messages sent was 87. The cost for the use of minutes was \$1.60 less than the cost of the text messages. Find the number of talk minutes, text messages, and picture messages he used.

Name: _____

Unit 2: Linear Functions & Systems



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Homework 7: Systems with Three Variables

**** This is a 2-page document! ****

Solve each system of equations. Clearly identify your solution.

1.
$$\begin{cases} 9x + 3y - 4z = 37 \\ 4x + 3y + 7z = 16 \\ x - 5y + 8z = -31 \end{cases}$$

2.
$$\begin{cases} 3x + 4y - 6z = -60 \\ 7x - 5y + z = -37 \\ 2x + 3y - z = -29 \end{cases}$$

3.
$$\begin{cases} 10x + 7y - 2z = 46 \\ 3x - 2y + 9z = 22 \\ 5x + y - 3z = 28 \end{cases}$$

4. One week, the music store sold 2 trumpets, 3 clarinets, and 5 violins for \$1,240. The next week, they sold 3 trumpets, 1 clarinet, and 4 violins for \$1,027. The following week, they sold 5 trumpets, 7 clarinets, and 2 violins for \$2,091. Find the cost of a trumpet, clarinet, and violin.

5. Caleb earns points on his credit card that he can use towards future purchases. He earns four points per dollar spent on flights, two points per dollar spent at hotels, and one point per dollar spent on all other purchases. Last year, he charged a total of \$9,480 and earned 14,660 points. The amount of money spent on flights was \$140 more than twice the amount of money spent on hotels. Find the amount of money spent on each type of purchase.

Name: _____

Algebra 2

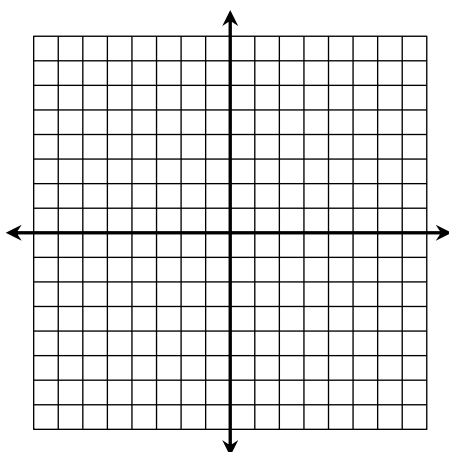
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Unit 2: Linear Functions & Systems

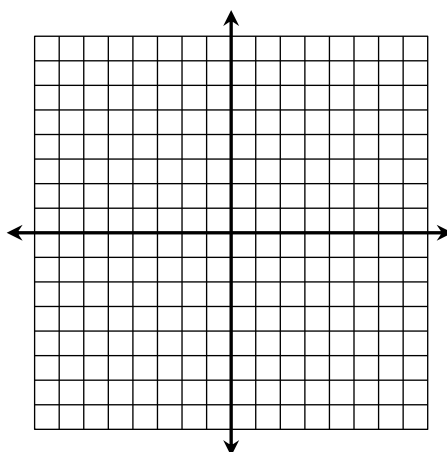
Quiz 2-3: Systems of Equations

For questions 1-2, solve the system by graphing.

1.
$$\begin{cases} 2x + 3y = 9 \\ x - 2y = 8 \end{cases}$$



2.
$$\begin{cases} 6x = 10y - 10 \\ x + y + 7 = 0 \end{cases}$$



1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

For questions 3-4, solve the system using the substitution method.

3.
$$\begin{cases} 9x - y = 17 \\ 5x + 4y = -27 \end{cases}$$

4.
$$\begin{cases} 2x - 4y = 8 \\ 7x - 3y = 61 \end{cases}$$

For questions 5-8, solve the system using the elimination method.

5.
$$\begin{cases} 8x + y = -39 \\ 2x - 5y = 27 \end{cases}$$

6.
$$\begin{cases} 3x - 7y = -18 \\ 4x - 2y = -24 \end{cases}$$

7.
$$\begin{cases} 12y = 17 - 9x \\ -4y - 3x = 31 \end{cases}$$

8.
$$\begin{cases} 11y + 1 = -6x \\ 0 = 43 - 7x - 4y \end{cases}$$

For questions 9-11, solve using a system of equations with your method of choice.

9. Casey bought a 15.4-pound turkey and an 11.6-pound ham for Thanksgiving and paid \$38.51. Her friend Jane bought a 10.2-pound turkey and a 7.3-pound ham from the same store and paid \$24.84. Find the cost per pound of turkey.

9. _____

10. _____

11. _____

10. Owen has a collection of nickels and quarters worth \$8.10. If the nickels were quarters and the quarters were nickels, the value would be \$17.70. How many of each type of coin does Owen have?

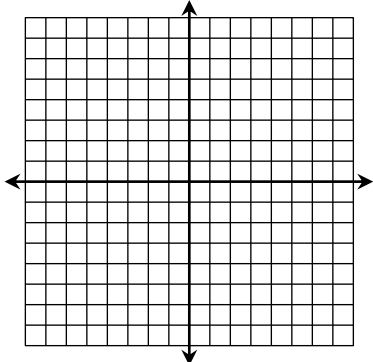
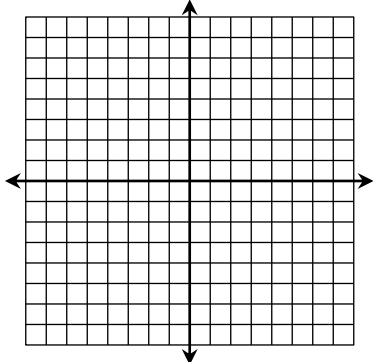
11. Michelle invested \$12,000 in stocks and bonds, paying 4.8% and 7% annual interest, respectively. If the total interest earned for the year was \$763, how much was invested in stocks?

Solve the system of three variables below.

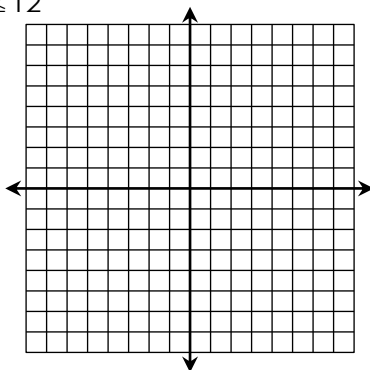
12.
$$\begin{cases} x - 7y + 3z = 17 \\ 5x + 2y - 2z = -57 \\ 3x - 10y - z = -11 \end{cases}$$

12. _____

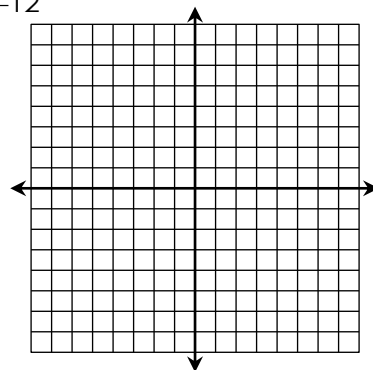
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
LINEAR <i>Inequalities</i>	A linear inequality is similar to a linear equation, however with an inequality symbol instead. Example: Given $y \geq -2x + 9$, determine which points satisfy the inequality: a) (7, 4) b) (-1, 3) c) (10, -5)	
<i>Real World</i> EXAMPLE	John sells t-shirts. Plain shirts cost \$5 and graphic shirts cost \$7. In order to match his product costs, he must make at least \$1,200. a) Write an inequality to represent this situation. b) If he sells 150 plain shirts and 60 graphic shirts, will he be able to cover the product costs? c) If he sells 90 graphic shirts, what's the minimum number of plain shirts he must sell to break even?	
GRAPHING <i>Linear Inequalities</i> Graphing linear inequalities is a visual way of showing the ordered pairs that satisfy the inequality.	1 Put the inequality in SLOPE-INTERCEPT FORM . Be sure to flip the inequality if you multiply or divide by a negative.	
	2 GRAPH the line. Use a dashed line for $<$ and $>$ symbols and a solid line for $\leq$ and $\geq$ symbols.	
	3 SHADE above the line for $>$ or $\geq$ symbols and below the line for $<$ or $\leq$ symbols.	
	Graph each linear inequality.	
1. $y < \frac{2}{3}x - 1$		2. $y \geq -4x$
		

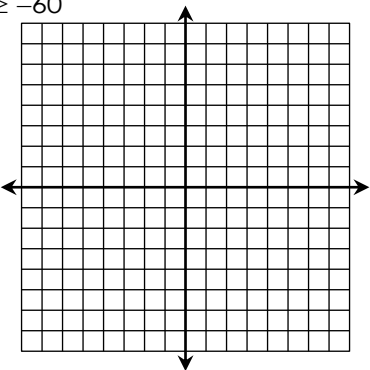
3. $5x + 3y \leq 12$



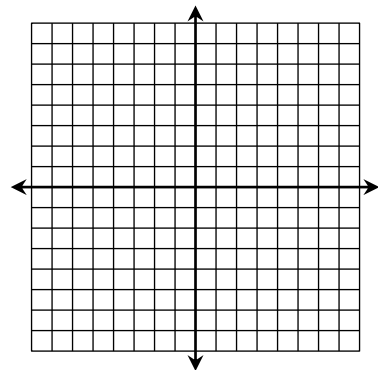
4. $x - 6y < -12$



5. $9x + 12y \geq -60$



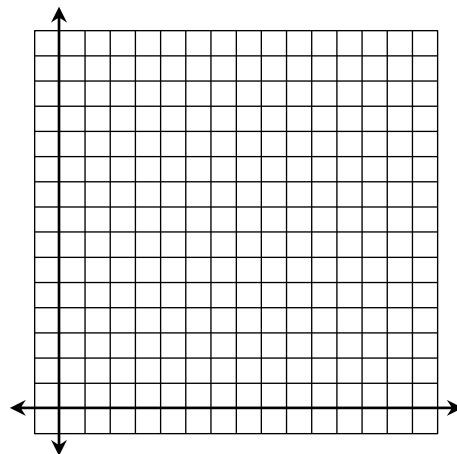
6. $x < -4$



7. Stephanie spends \$0.10 per text message sent and \$0.25 per picture message sent on her cell phone. Her monthly budget for messaging is \$30.

a) Write a linear inequality to represent this situation, then graph.

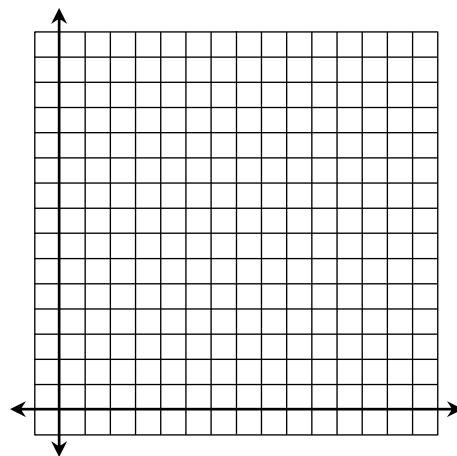
b) Using your graph, give two possible combinations of text and picture messages she can send and stay within her budget.



8. Scott makes \$12 per hour babysitting and \$8 per hour working at the grocery store. He will need to make at least \$600 to buy a new computer.

a) Write a linear inequality to represent this situation, then graph.

b) Using your graph, give two possible combinations of the number of hours Scott can babysit and work in the grocery store to ensure he can afford his new computer.



Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

SYSTEMS

of Linear Inequalities

The **solution** to a system of linear inequalities is the **set of ordered pairs that satisfy ALL the linear inequalities**.
The solution can be shown by graphing.

①

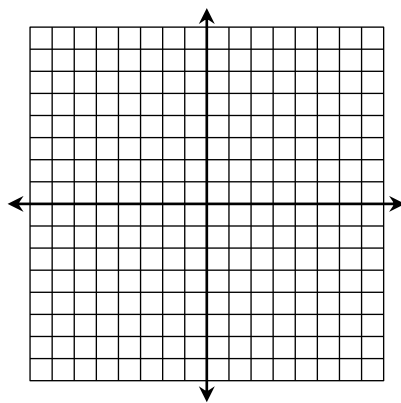
GRAPH and SHADE each linear inequality.

②

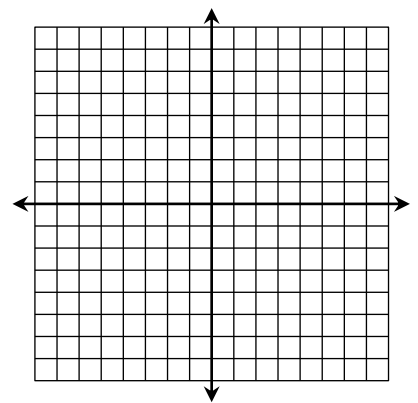
The **SOLUTION** to the system is where the **shading overlaps**.

Graph each system of linear inequalities. Clearly indicate the solution region.

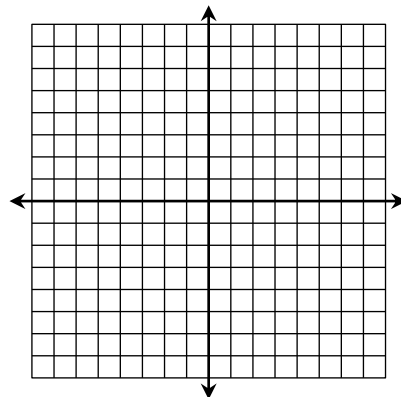
1.
$$\begin{cases} y \leq -\frac{2}{3}x - 1 \\ y \geq \frac{1}{2}x - 8 \end{cases}$$



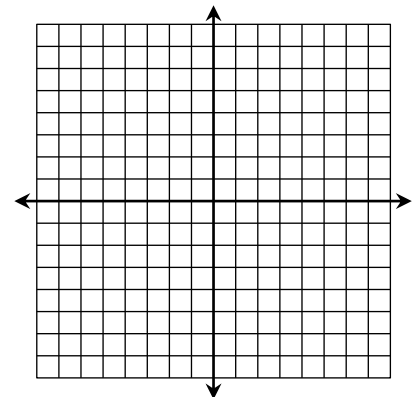
2.
$$\begin{cases} y \leq \frac{3}{4}x - 4 \\ y > -7 \end{cases}$$



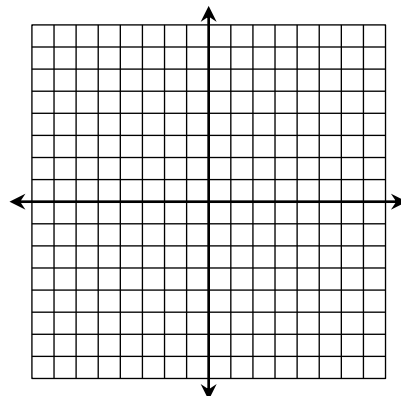
3.
$$\begin{cases} x - y > 1 \\ x + 4y > 20 \end{cases}$$



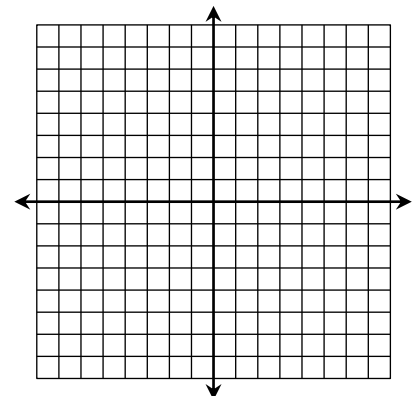
4.
$$\begin{cases} 7x + 3y \leq 18 \\ 2x - 3y \leq 9 \end{cases}$$



5.
$$\begin{cases} x + 2y < -6 \\ -4x - 2y < -10 \end{cases}$$



6.
$$\begin{cases} 4x - 5y > -5 \\ x < -1 \end{cases}$$



7. Which points are solutions to the system of inequalities below? Check all that apply.

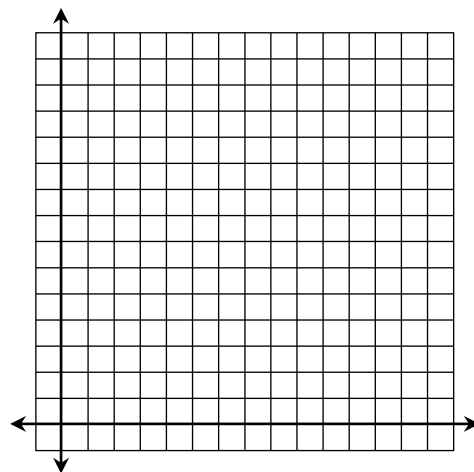
$$\begin{cases} x - 6y > -48 \\ 2x + 3y < 9 \end{cases}$$

☐ (1, 1)	☐ (-2, 6)
☐ (9, -2)	☐ (-8, -3)
☐ (3, 4)	☐ (4, -7)

8. To raise money to attend Boy Scout camp, Alec is selling popcorn. Small bags of popcorn sell for \$4 each and large bags sell for \$6 each. He needs to earn at least \$300 to attend camp. He will also earn a badge if he sells at least 60 bags total of any combination.

a) Write a system of linear inequalities to represent this situation, then graph.

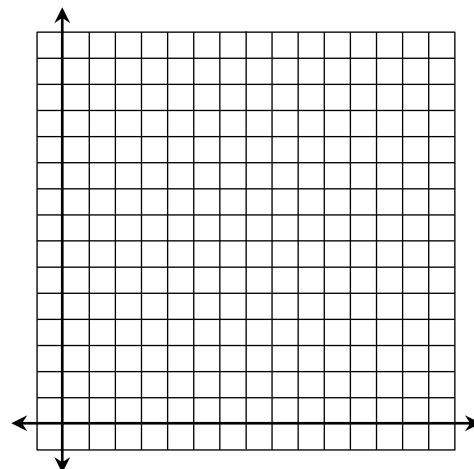
b) Using your graph, give two possible combinations of bags of popcorn that Alec can sell to attend camp and earn his badge.



9. Greg needs at least \$1.60 in stamps to mail a package. He has 28¢ and 4¢ stamps. He can use no more than twenty 4¢ stamps as he only has one book left.

a) Write a system of linear inequalities to represent this situation, then graph.

b) Using your graph, give two possible combinations of stamps that Greg can use to ensure he mails his package with the correct postage.



Name: _____

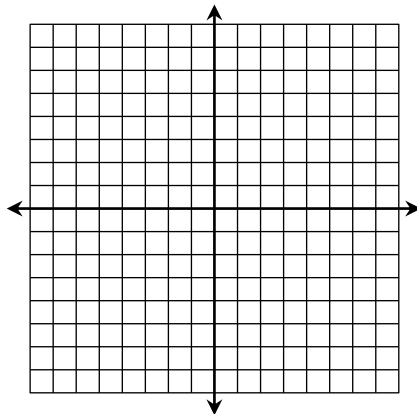
Unit 2: Linear Functions & Systems

Date: _____ Per: _____

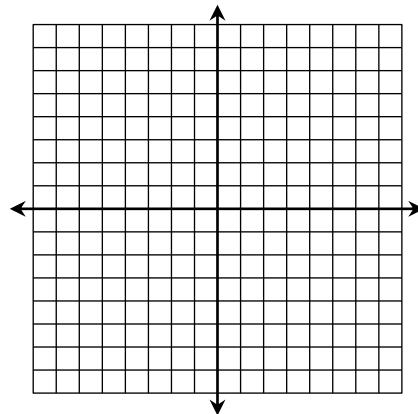
Homework 8: Linear Inequalities

**** This is a 2-page document! ******Graph each linear inequality.**

1. $y \leq -\frac{5}{4}x + 2$



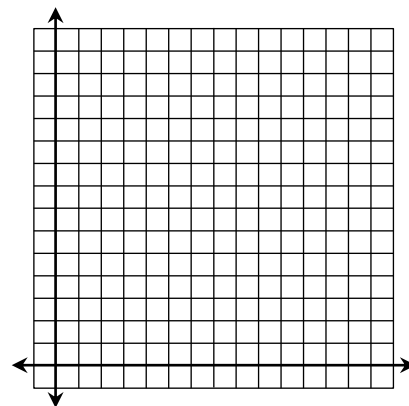
2. $4x - y < -1$



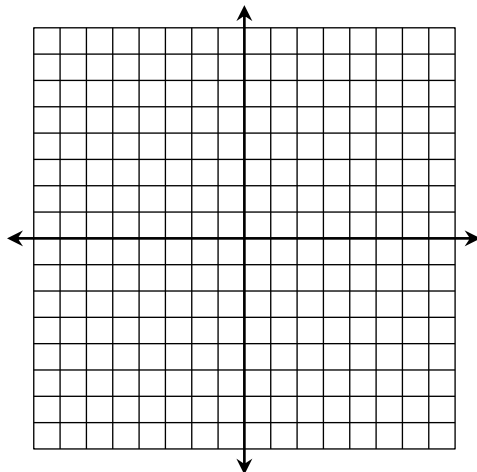
3. In football, a kicker can score one point for the extra-point on a touchdown or three points for a field goal. Logan is a kicker for his high school football team. The record number of points scored by a kicker in a single game is 24. Logan is hoping to beat this record in his next game.

a) Write a linear inequality to represent this situation, then graph.

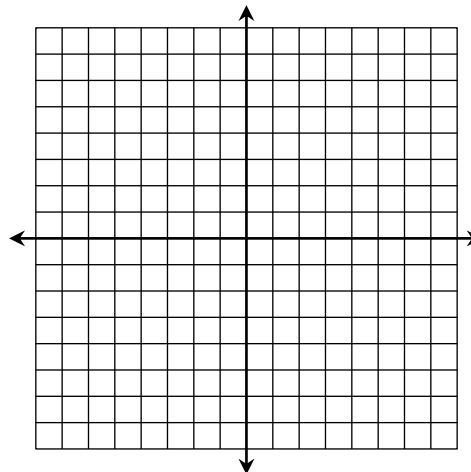
b) Using your graph, give two possible combinations extra-point kicks and field goals Logan could score to beat the current record.

**Graph each system of linear inequalities. Clearly indicate the solution region.**

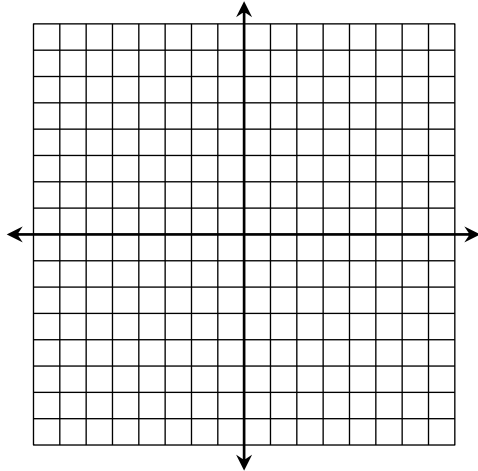
4.
$$\begin{cases} x + 2y \geq -12 \\ y < 6x + 7 \end{cases}$$



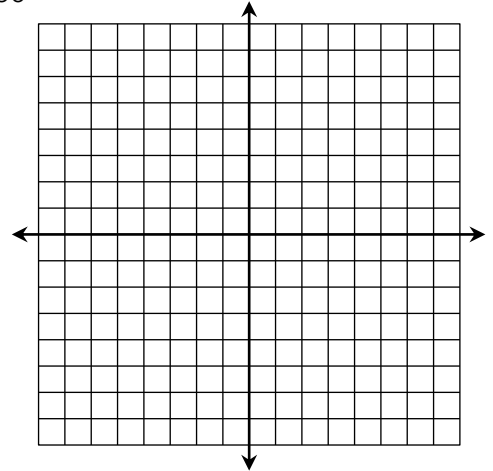
5.
$$\begin{cases} 2x - 5y < -20 \\ x > -3 \end{cases}$$



6.
$$\begin{cases} x - y \geq 4 \\ x + 5y \leq 0 \end{cases}$$



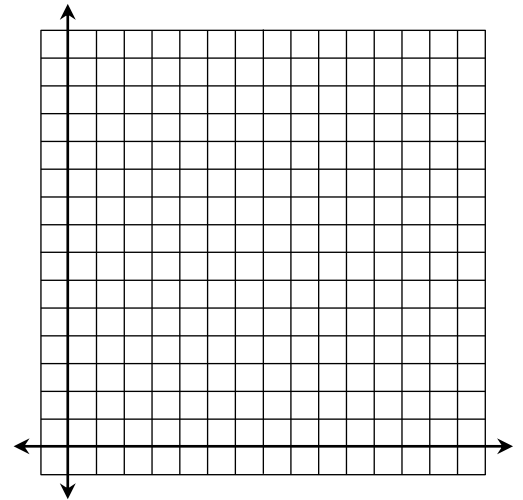
7.
$$\begin{cases} 8x - 4y > -28 \\ 6y \geq 12x - 36 \end{cases}$$



8. Brandon is shopping at Old Navy during a sale. All shorts are \$16 each and all t-shirts are \$10 each. He has \$100 to spend and would like to purchase at least 2 pairs of shorts.

a) Write a system of linear inequalities to represent this situation, then graph.

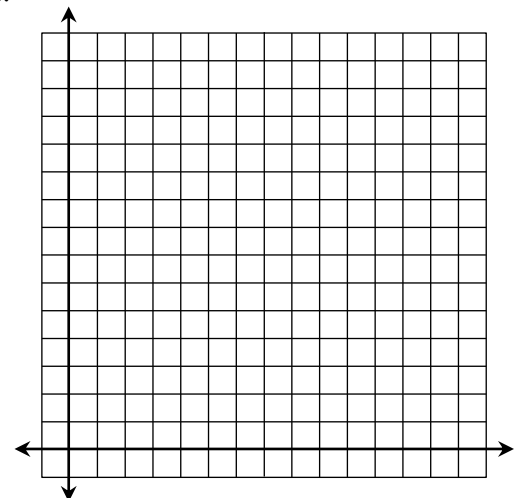
b) Using your graph, give two possible combinations of shorts and t-shirts that Brandon can buy.



9. Darcy is buying apples and oranges for a large fruit basket to give away as a door prize at a charity event. Apples cost \$0.24 each and oranges cost \$0.80 each. She has \$12 to spend and would like to purchase at least 20 pieces of fruit in total.

a) Write a system of linear inequalities to represent this situation, then graph.

b) Using your graph, give two possible combinations of apples and oranges Darcy can buy.



Unit 2 Test Study Guide

(Linear Functions & Systems)

Name: _____

Date: _____ Per: _____

Topic 1: Relations & Functions

Find the domain and range of each relation. Then determine if the relation is a function.

1.	2.	3.	4.										
<table><tr><th>x</th><th>y</th></tr><tr><td>5</td><td>5</td></tr><tr><td>6</td><td>6</td></tr><tr><td>7</td><td>7</td></tr><tr><td>8</td><td>8</td></tr></table>	x	y	5	5	6	6	7	7	8	8			
x	y												
5	5												
6	6												
7	7												
8	8												
D:	D:	D:	D:										
R:	R:	R:	R:										
Function?	Function?	Function?	Function?										

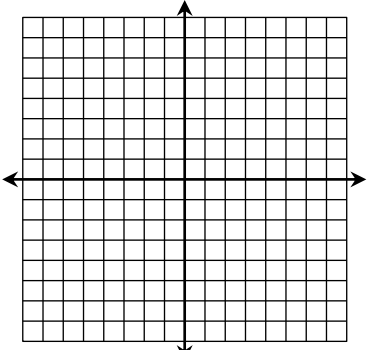
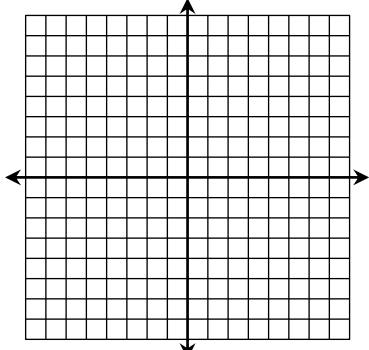
Use $f(x) = -4x + 1$, $g(x) = x^2 - 8x + 21$, and $h(x) = |9 - 3x|$ for questions 5 – 10.

5. Find $g(-5)$	6. Find $f\left(\frac{11}{12}\right)$	7. Find $h(8) - f(-7)$
8. Find $g(x + 4)$	9. If $f(x) = 53$, find x .	10. If $h(x) = 39$, find x .

Topic 2: Linear Equations: Slope-Intercept & Standard Form

Write each equation in slope-intercept form, when necessary, then graph the line.

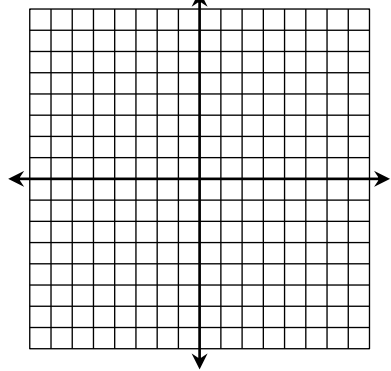
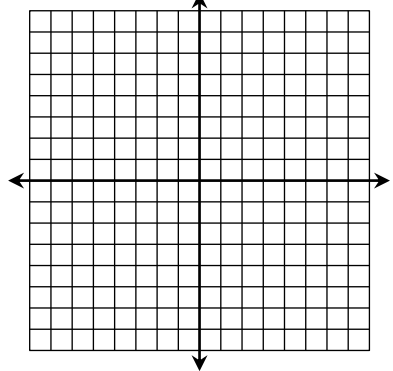
11. $y = -\frac{2}{3}x + 3$	12. $\frac{5}{2}x - 4y = -8$

13. $4x - 5y = -10$ 	14. $12x = 4y - 28$ 
--	--

Give an example of a line that is parallel and a line that is perpendicular to each given line.

15. $9x + 6y = -6$ Parallel: _____ Perpendicular: _____	16. $y = -5$ Parallel: _____ Perpendicular: _____
---	---

Find the x- and y-intercepts of each line, then graph the line.

17. $y = -5x - 3$ 	18. $4y = 10x - 24$ 
---	---

Topic 3: Writing Linear Equations & Applications

Write the equation in SLOPE-INTERCEPT FORM with the given information.

19. Passes through $(-8, 3)$ with a slope of -2	20. Passes through $(-4, -2)$ and $(8, 7)$
--	---

21. Passes through $(-6, 4)$ and parallel to $x + y = 8$.

22. Passes through $(-4, 1)$ and perpendicular to the line passing through $(-5, 2)$ and $(10, 8)$.

DEFINE VARIABLES and WRITE EQUATIONS to represent the following situations, then solve.

23. For commission as a realtor, Michelle earns \$349 plus 3% of the purchase price for each home she helps buy or sell. If she earned \$8,965 in commission on a certain home, find its purchase price.

24. On Ryan's last social studies test, there were two types of questions: true/false worth 2 points each and multiple choice questions worth 4 points each. If Ryan earned 86 points on the test and answered 18 multiple choice questions correctly, how many true/false questions did he answer correctly?

25. The table below shows the altitude of a plane once it begins its descent to a runway.

Time (minutes)	Altitude (feet)
0	28,500
1	26,378
2	24,105
3	21,774
4	19,452
5	17,991

a) Write an equation to model the data using linear regression.

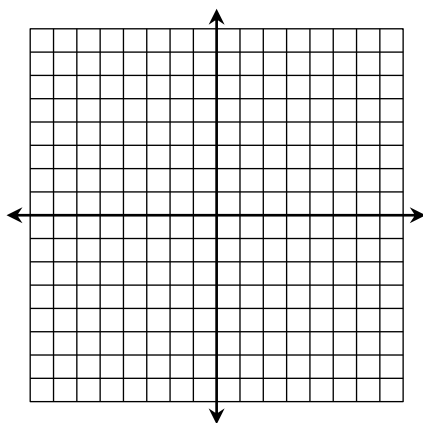
b) Find the height of the plane after 8 minutes.

c) How long will it take the plane to land?

Topic 4: Systems of Equations

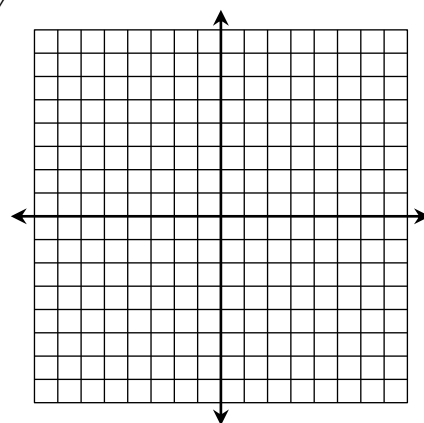
26.
$$\begin{cases} x - 2y = 8 \\ 6x - y = -7 \end{cases}$$

(Solve by Graphing)



27.
$$\begin{cases} 3y - 6 = 4x \\ 16x + 24 = 12y \end{cases}$$

(Solve by Graphing)

**For questions 28 and 29, solve using the substitution method.**

28.
$$\begin{cases} 5x - 4y = 9 \\ x + 7y = -6 \end{cases}$$

29.
$$\begin{cases} 2x + 3y = -35 \\ 8x - y = -23 \end{cases}$$

For questions 30 and 31, solve using the elimination method.

30.
$$\begin{cases} 3x + 10 = 14y \\ 8x - 7y = 34 \end{cases}$$

31.
$$\begin{cases} 18x = 12y + 7 \\ -8y + 21 = -12x \end{cases}$$

32. Ben has a collection of quarters and nickels worth \$5.35. If the number of nickels is five less than twice the number of quarters, find the number of each coin.

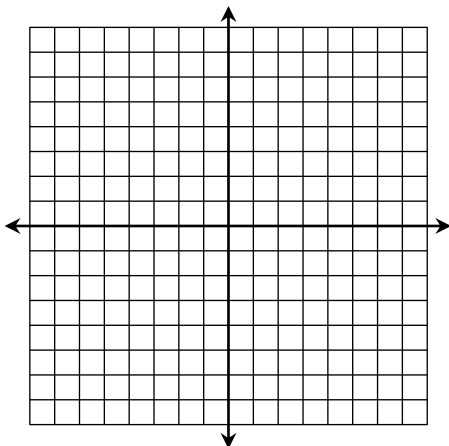
33. Aliyah bought four composition notebooks and three packs of pencils from the school bookstore and paid \$10.93. Laura bought seven composition notebooks and two packs of pencils and paid \$13.31. If each pencil pack contains ten pencils, what is the unit price per pencil?

34. Solve the system below using your method of choice:

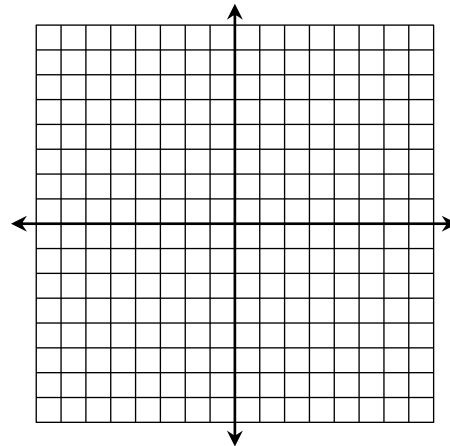
$$\begin{cases} 4x + 2y - 5z = 47 \\ x - 2y + 6z = -10 \\ 9x - 7y - z = 75 \end{cases}$$

Topic 5: Linear Inequalities & Systems of Linear Inequalities**Show the solution to each linear inequality and system of linear inequalities by graphing.**

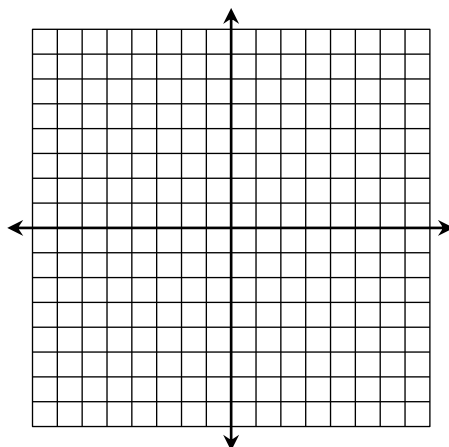
35. $y > -\frac{7}{2}x + 3$



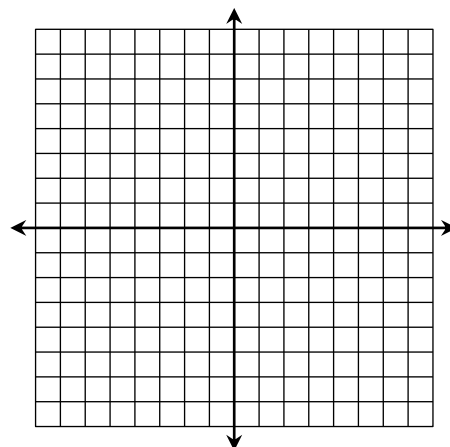
36. $2x - 6y \geq 36$



37.
$$\begin{cases} 5x - 2y > -8 \\ y \leq -3 \end{cases}$$



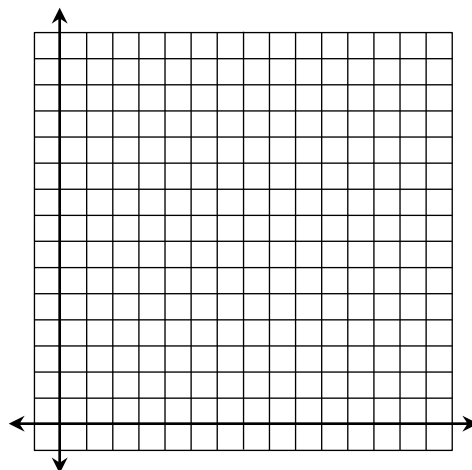
38.
$$\begin{cases} x - y \leq 4 \\ 3y \geq 3x + 9 \end{cases}$$



- 39.** As part of Kayla's exercise program, she either runs 6 miles/day or rides her bike 10 miles/day. Her new goal is to cover a minimum distance of 200 miles, with at least 15 of the days running. She would like to determine the number of days it would take to accomplish this.

a) Write a system of linear inequality to represent this situation, then graph.

b) Using your graph, give two possible combinations of days Kayla can run and ride her bike to meet her goal.



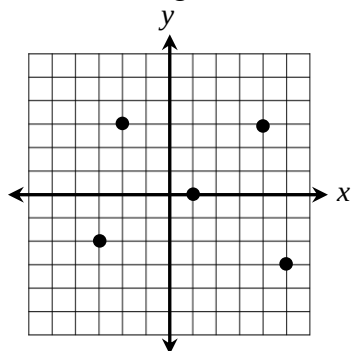
Name: _____

Date: _____ Per: _____

Unit 2 Test

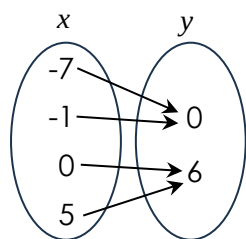
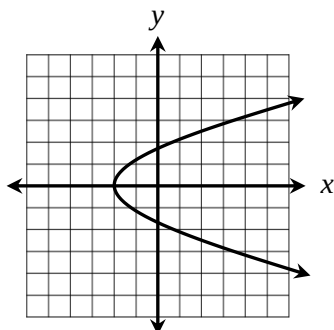
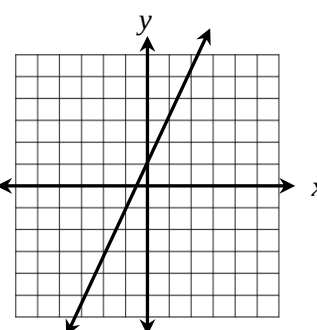
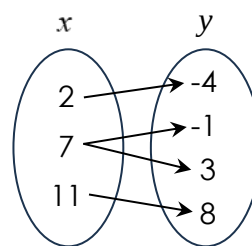
Linear Functions & Systems

1. Find the range of the relation below, then determine whether the relation is a function.

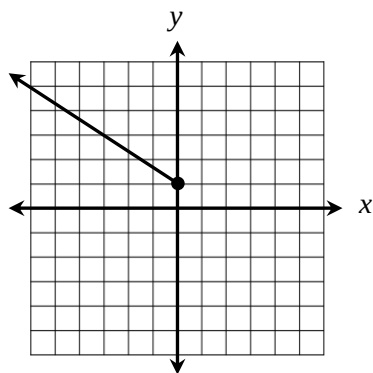


- A. $\{-3, -2, 1, 4, 5\}$; Function
 B. $\{-3, -2, 1, 4, 5\}$; Not a function
 C. $\{-3, -2, 0, 3\}$; Function
 D. $\{-3, -2, 0, 3\}$; Not a function

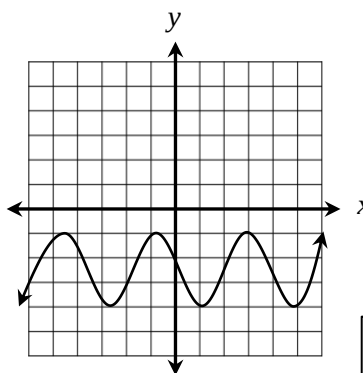
2. Which relations represent a function? Check all that apply.

☐

☐

☐

☐


3. What is the domain of the graph below?



4. What is the range of the graph below?



5. Given the function below, find $f(7)$.

$$f(x) = -x^2 + 9x$$

6. Given the functions below, find $b(-2) - a(12)$.

$$a(x) = |13 - 2x|; b(x) = 8x - 14$$

7. Given the function below, find $p(x - 1)$.

$$p(x) = x^2 - 7x$$

8. Given the function below, if $h(x) = -1$, find x .

$$h(x) = -6x + 20$$

$x =$

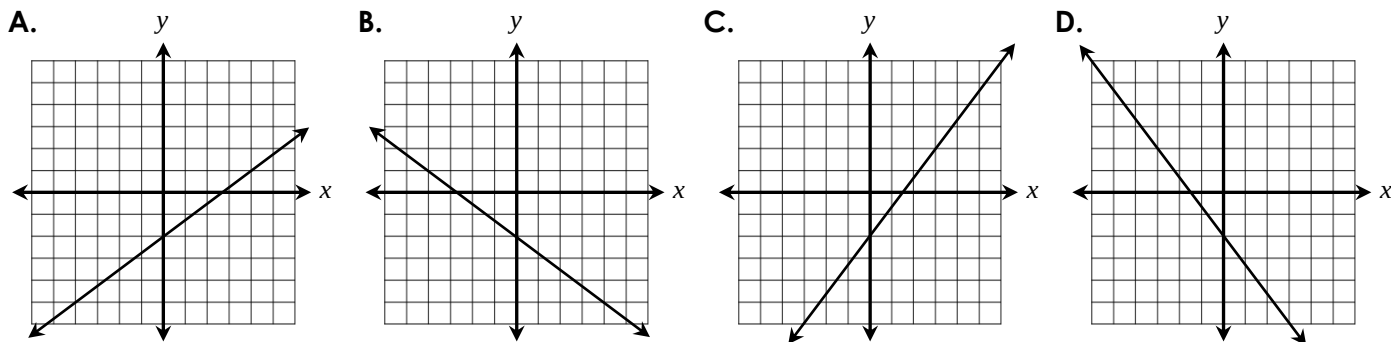
9. Write the equation below in **standard form**.

$$\frac{3}{4}x - 3y = -21$$

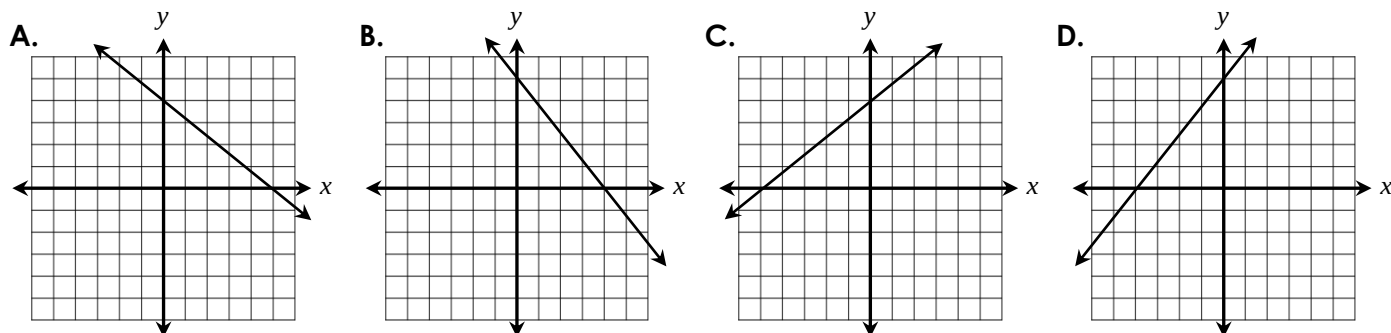
10. Write the equation below in **slope-intercept form**.

$$14x = 6y - 12$$

11. Which graph best represents the equation $12x - 9y = 18$?



12. Which graph best represents the equation $-15y - 12x = -60$?



13. Find the x - and y -intercepts of the equation below. Write your answers as ordered pairs.

$$8y = -9x - 12$$

x -int.:

y -int.:

14. Which of the following lines is **perpendicular** to the equation given below?

$$-9x = 15 - 3y$$

A. $6x + 2y - 8 = 0$

B. $x - 3y = -3$

C. $12x - 24 = 4y$

D. $6y = 42 - 2x$

15. Give an example of a linear equation that is **parallel** to the y -axis.

16. Write a linear equation in **slope-intercept form** that passes through the point $(-1, -8)$ and is **parallel** to the line $15x + 3y = 21$.

17. Write a linear equation in **slope-intercept form** that passes through the points $(-11, -5)$ and $(1, -2)$.

18. For each boat he sells, Mick earns \$149 in addition to 2% of the purchase price of the boat as commission. If p represents the purchase price of the boat, which equation represents Mick's commission, c ?

A. $c = 149 + .2p$

B. $c = 149 + .02p$

C. $p = 149 + .2c$

D. $p = 149 + .02c$

19. Angela fully charged her laptop, then unplugged it and tracked the battery level at certain times under constant use. The table below shows her findings. Find an equation for the line of best fit, then estimate the battery level after two hours of constant use unplugged.

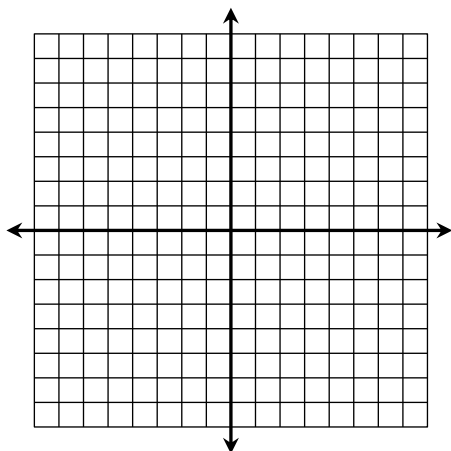
Time	Battery Level (%)
2:00	100%
2:10	94%
2:25	87%
2:40	83%
3:00	79%
3:10	76%

Equation:

Solution:

20. Solve the system below by graphing.

$$\begin{cases} y = 3 - 2x \\ -4x - 4y = -20 \end{cases}$$



21. Solve the system below by substitution.

$$\begin{cases} 6x - y = -23 \\ 2x + 5y = -13 \end{cases}$$

22. Solve the system below by elimination.

$$\begin{cases} x = y - 4 \\ 7x - 2y = 17 \end{cases}$$

23. Solve the system below by elimination.

$$\begin{cases} 2y = -24 - 8x \\ 6x + 18 = 5y \end{cases}$$

24. Which system of equations has an infinite solution?

A. $\begin{cases} y = 2x - 7 \\ 2x + y = 5 \end{cases}$

B. $\begin{cases} x = 3y + 15 \\ 12y = 4x - 60 \end{cases}$

C. $\begin{cases} 10x + 8y = -16 \\ -5x = 4y + 28 \end{cases}$

D. $\begin{cases} y = 8 \\ x = 8 \end{cases}$

25. Jasmine bought 1.3 pounds of ham and 0.8 pounds of cheese from the deli and paid \$11.28. She went back the following week and bought 1.5 pounds of ham and 1.2 pounds of cheese and paid \$14.76. If the prices remained the same, find the price per pound of ham and cheese.

Ham:

Cheese:

26. Solve the system below.

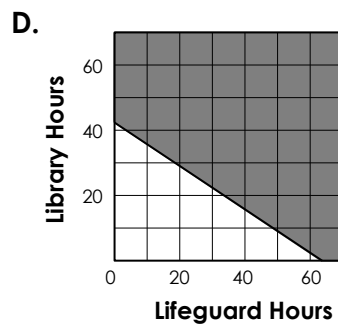
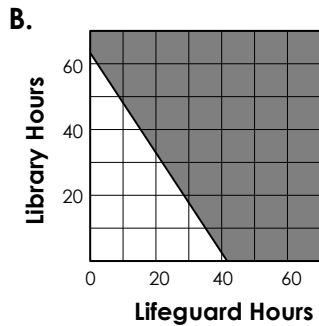
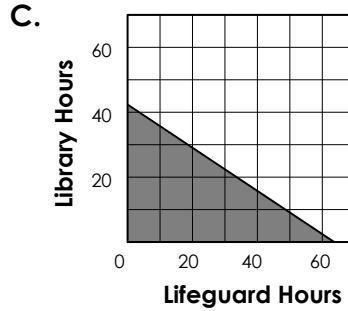
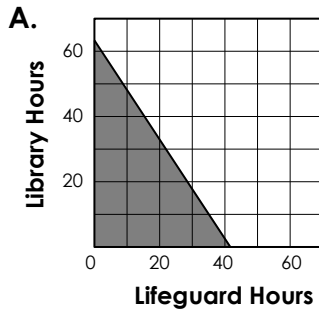
$$\begin{cases} 3x - 5y + 2z = -53 \\ x - 7y - 4z = -37 \\ 5x + 9y + 2z = 57 \end{cases}$$

27. Which ordered pairs represent solutions to the linear inequality below? Check all that apply.

$$x - 4y < -12$$

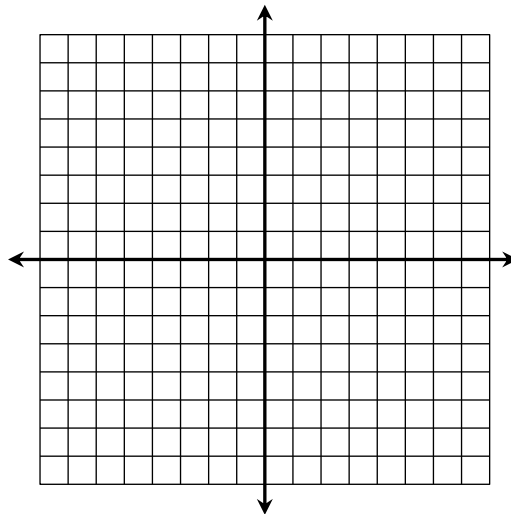
☐ (1, -8)	☐ (-8, -3)	☐ (-12, 0)
☐ (3, 5)	☐ (-7, 9)	☐ (-4, 2)

28. Jon makes \$12 per hour as a lifeguard and \$8 per hour working at the library. This summer, he would like to make at least \$500 working his two jobs. Which graph represents the number of hours he must work at each job to reach his goal?



29. Graph the system of linear inequalities. Then, choose the point that represents a solution to the system.

$$\begin{cases} 5x - 2y \geq -14 \\ x < -1 \end{cases}$$



- A. $(-6, 1)$
 B. $(4, 0)$
 C. $(1, 8)$
 D. $(-3, -7)$

30. To save money for prom this weekend, Tom is going to walk his neighbor's dog for \$6 per hour and wash cars for \$7.50 per hour. His mother said he can work no more than 15 hours to ensure he keeps up with his homework. Tom needs to make at least \$75 to cover prom expenses. If d represents the number of hours walking his neighbor's dog and c represents the number of hours washing cars, which system of inequalities represents this situation?

A. $\begin{cases} 6d + 7.5c \leq 15 \\ d + c \geq 75 \end{cases}$

B. $\begin{cases} 6d + 7.5c \geq 15 \\ d + c \leq 75 \end{cases}$

C. $\begin{cases} 6d + 7.5c \leq 75 \\ d + c \geq 15 \end{cases}$

D. $\begin{cases} 6d + 7.5c \geq 75 \\ d + c \leq 15 \end{cases}$

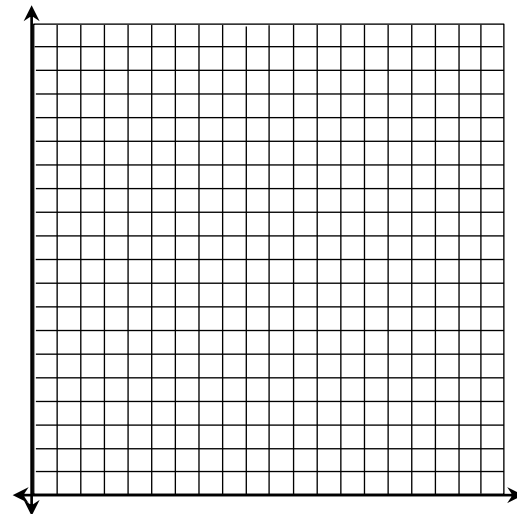
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
LINEAR PROGRAMMING		
Finding the OPTIMAL SOLUTION	❶	Use the CONSTRAINTS to write a SYSTEM OF INEQUALITIES . Constraints are the limitations on a problem.
	❷	GRAPH the system, clearly showing the FEASIBLE REGION .
	❸	Write an OBJECTIVE FUNCTION . This is the quantity you would like to minimize or maximize.
	❹	TEST THE VERTICES of the feasible region to find the OPTIMAL SOLUTION .
Example 1: The smallest plane in the Coast2Coast Airlines fleet offers first class and tourist class seats, and can hold a maximum of 50 passengers. In order to charter a flight, the airline must sell at least 8 first class tickets and at least 20 tourist class tickets. The company makes \$40 profit for each first class ticket and \$30 profit for each tourist class ticket sold. In order for the company to maximize its profits on a single flight, how many first class and tourist class tickets should they sell?		
CONSTRAINTS: GRAPH:		
OBJECTIVE FUNCTION: TEST VERTICES:		SOLUTION:

Example 2: Natalie is a hair stylist. When making appointments for her clients, she allots 30 minutes for each cut and two hours for each cut and color. She has ten hours available for appointments each day and prefers to see no more than 8 total clients. Each routine cut costs \$40 and each cut and color costs \$100. How many of each type of appointment should she schedule each day to maximize her income?

CONSTRAINTS:

GRAPH:

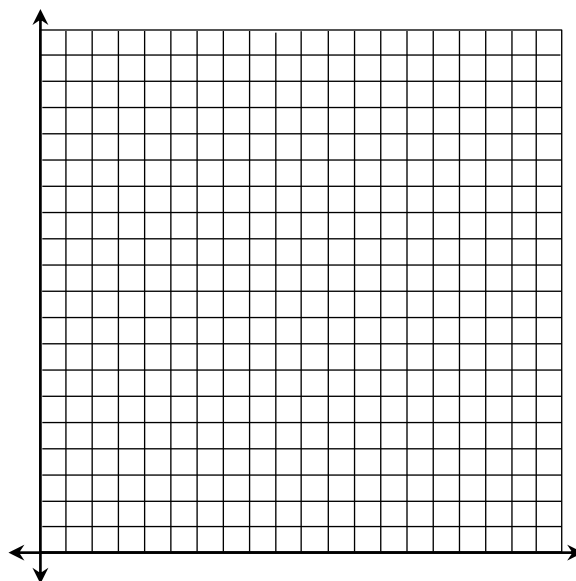


OBJECTIVE FUNCTION:

TEST VERTICES:

SOLUTION:

Example 3: Baking a tray of chocolate chip cookies takes three cups of milk and two cups of flour. Baking a tray of oatmeal raisin cookies takes one cup of milk and four cups of flour. The baker has 9 cups of milk and 16 cups of flour available. He makes \$2.50 profit per tray of chocolate chip cookies and \$2 profit per tray of oatmeal raisin. How many trays of each type of cookies should he make to maximize his profit?



LINEAR PROGRAMMING

CHOICE BOARD

1

Steamline Bicycles produces two kinds of bikes, mountain bikes and road bikes. It takes 4 hours to build a mountain bike and 3 hours to build a road bike. The company has 21 employees, each with a 40-hour work week. Due to demand, they must produce at least 250 total bikes each week. If they profit \$75 on each mountain bike and \$60 on each road bike, how many of each type should they produce in a week to maximize their profit?

2

Beach Cruises prints brochures and fliers to advertise their dolphin watching tours. To print, the brochure requires three pages while the flier requires two. They are limited to 500 sheets of paper but need to print at least 80 brochures and 100 fliers. Each brochure costs 10¢ to print and each flier costs 6¢ to print. How many of each should they print to minimize their cost?

3

The amphitheater has two types of tickets available, reserved seats and lawn seats. The maximum capacity of the venue is 20,000 people; however, they must sell at least 5,000 tickets for a concert to be held. As an additional constraint, the number of lawn seats cannot exceed the number of reserved seats. If reserved seats profit \$65 each and lawn tickets profit \$40 each, how many of each must they sell to maximize their profits?

4

JP Computers produces both laptop and desktop computers. Laptop computers take up 2 cubic feet of space and desktop computers take up 5 cubic feet of space. The maximum capacity on the delivery truck is 800 cubic feet. Due to demand, they need to load at least 300 laptop computers. They profit \$225 on each laptop and \$280 on each desktop computer. How many of each should they load on the truck to maximize profit?

5

The Polar Sports company manufactures both snowboards and skis. It takes 8 hours to manufacture a snowboard and 10 hours to manufacture a set of skis. There are 400 hours of labor available each week for production. Due to demand, they must produce at least 15 snowboards. Snowboards profit \$55 each and skis profit \$80 each. How many of each should they produce in a week to maximize their profit?

6

Ben works for a software company in Cleveland. To train new employees, Ben travels to their two other locations, Buffalo and Philadelphia. A Buffalo trip takes two days while a Philadelphia trip takes three days. He can travel no more than 105 days each year. Due to many new employees this year, he must travel to Philadelphia at least 5 times and Buffalo at least 15 times. It costs the company \$180 for each Buffalo trip and \$300 for each Philadelphia trip. How many times should Ben travel to Buffalo and Philadelphia to minimize costs to the company?

LINEAR PROGRAMMING

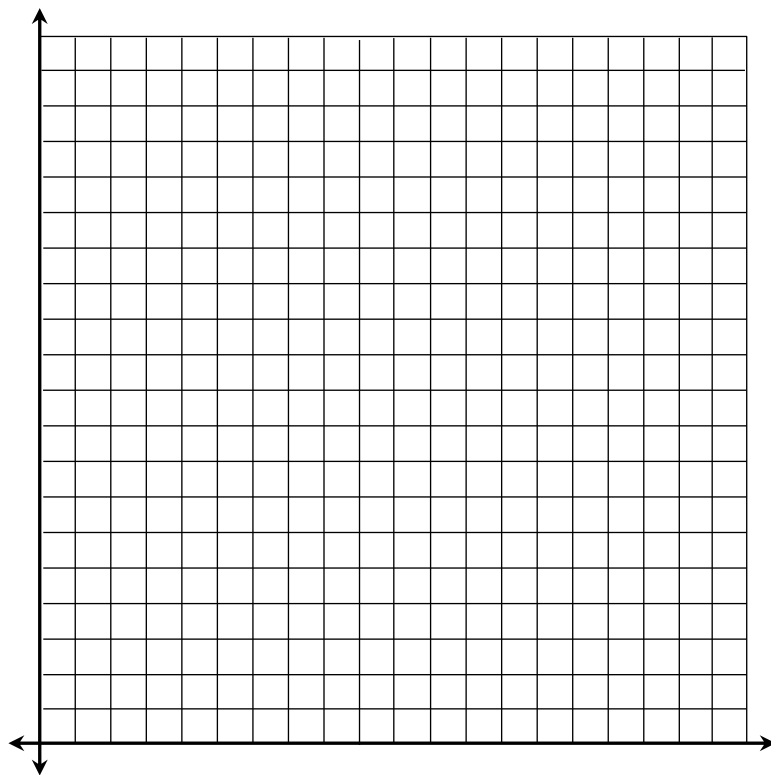
Choice Board Assignment

Name: _____

Date: _____ Per: _____

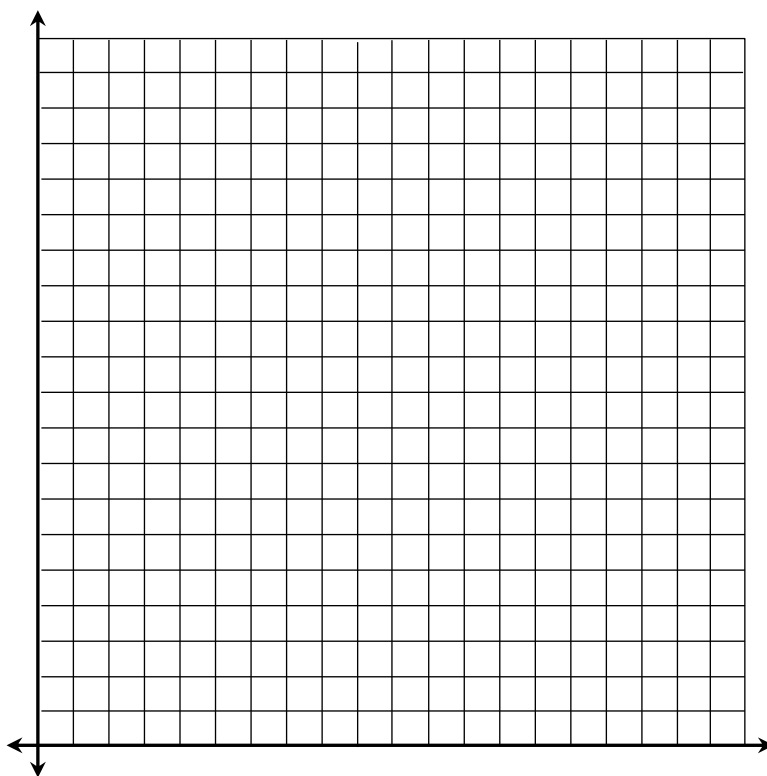
Directions: Choose any **two** problems from the linear programming choice board. Organize and show all work to support your final answer. Give a written summary at the end with your recommendation.

Choice # _____



Summary: _____

Choice # _____

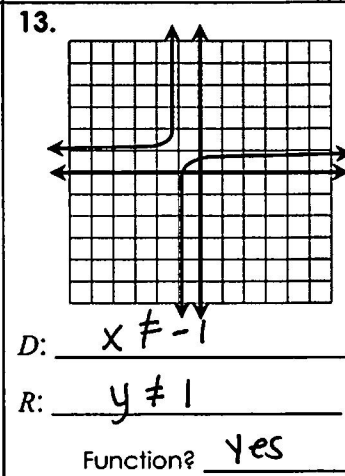
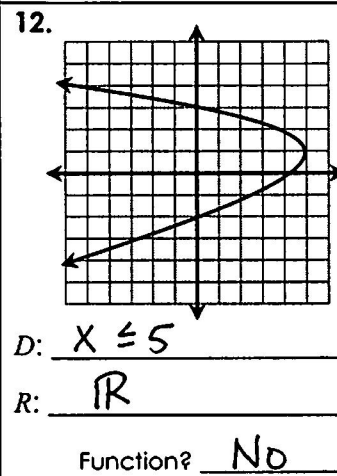
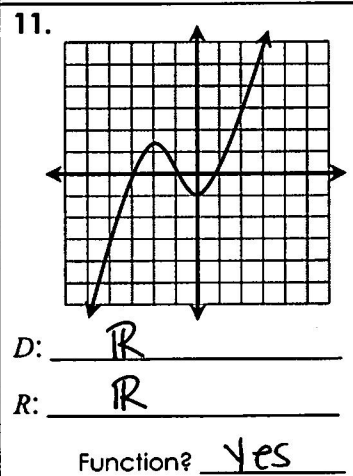
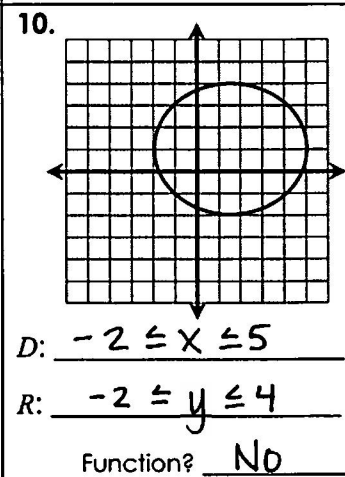
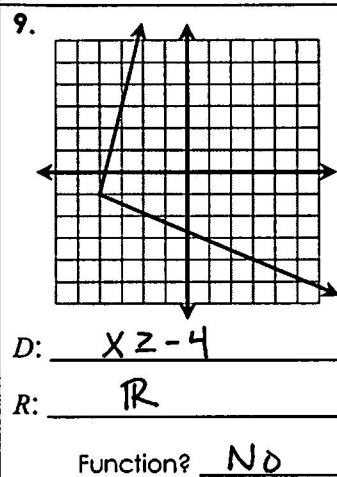
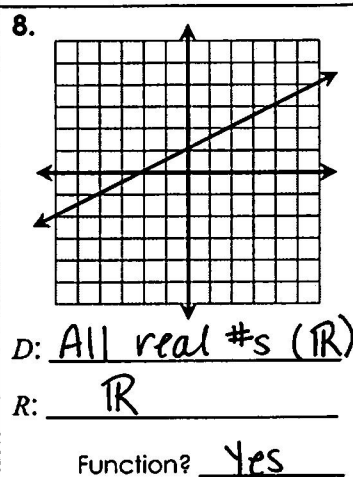
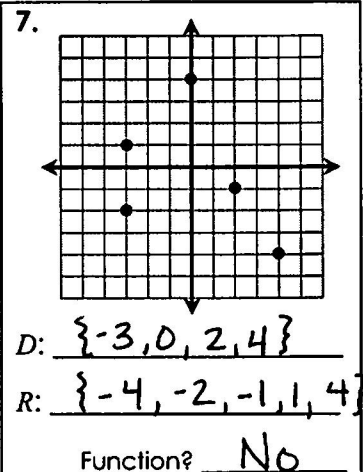
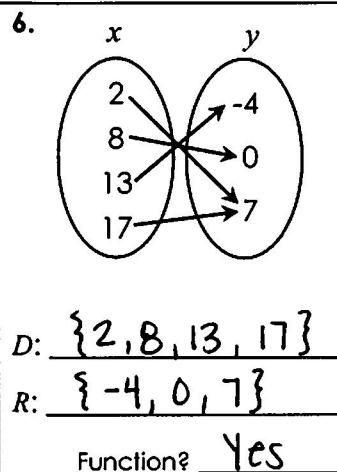
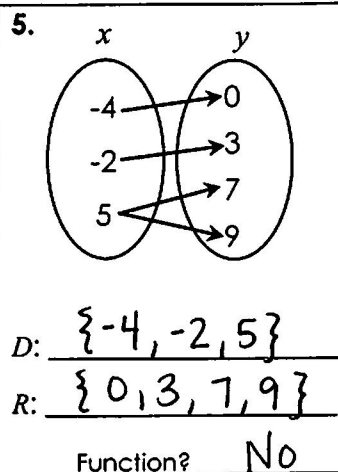


Summary: _____

Name:	Date:
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Topic:	Class:
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Main Ideas/Questions	Notes/Examples																								
RELATIONS	- A relation is a set of ordered pairs. Relations can be shown in a <u>table</u>, <u>mapping</u>, <u>graph</u>, or expressed as an <u>equation</u>. - The set of x-values is called the <u>domain</u>. - The set of y-values is called the <u>range</u>. - A relation with distinct points is called <u>discrete</u>. - A relation with connected lines or curves is called <u>Continuous</u>.																								
What is a FUNCTION?	A function is a relation in which each element of the domain is paired with exactly one element of the range.																								
	<table><tr><td>FUNCTION: $\{(0,1), (-5,2), (6,-2), (5,7)\}$</td><td>NOT A FUNCTION: $\{(-7,2), (6,-1), (6,2), (-3,9)\}$ </td></tr></table> This means x-values do not repeat!	FUNCTION: $\{(0,1), (-5,2), (6,-2), (5,7)\}$	NOT A FUNCTION: $\{(-7,2), (6,-1), (6,2), (-3,9)\}$ 																						
FUNCTION: $\{(0,1), (-5,2), (6,-2), (5,7)\}$	NOT A FUNCTION: $\{(-7,2), (6,-1), (6,2), (-3,9)\}$ 																								
VERTICAL LINE TEST In order to be a function, a vertical line can not pass through more than one point in the relation.	<table><tr><td>FUNCTION: </td><td>NOT A FUNCTION: </td></tr></table>	FUNCTION: 	NOT A FUNCTION: 																						
FUNCTION: 	NOT A FUNCTION: 																								
EXAMPLES	Give the domain, range, and determine whether the relation is a function. <table><tr><td> 1. $\{(-1, -9), (0, -5), (4, 3), (6, -2)\}$ D: <u>$\{-1, 0, 4, 6\}$</u> R: <u>$\{-9, -5, -2, 3\}$</u> Function? <u>Yes</u> </td><td> 2. $\{(5, 2), (-3, 4), (5, -6), (8, 0)\}$ D: <u>$\{-3, 5, 8\}$</u> R: <u>$\{-6, 0, 2, 4\}$</u> Function? <u>No</u> </td></tr><tr><td> 3. <table><tr><td>x</td><td>7</td><td>2</td><td>0</td><td>-4</td></tr><tr><td>y</td><td>-8</td><td>3</td><td>-1</td><td>3</td></tr></table> D: <u>$\{-4, 0, 2, 7\}$</u> R: <u>$\{-8, -1, 3\}$</u> Function? <u>Yes</u> </td><td> 4. <table><tr><td>x</td><td>10</td><td>15</td><td>18</td><td>24</td></tr><tr><td>y</td><td>0</td><td>3</td><td>7</td><td>11</td></tr></table> D: <u>$\{10, 15, 18, 24\}$</u> R: <u>$\{0, 3, 7, 11\}$</u> Function? <u>Yes</u> </td></tr></table>	1. $\{(-1, -9), (0, -5), (4, 3), (6, -2)\}$ D: <u>$\{-1, 0, 4, 6\}$</u> R: <u>$\{-9, -5, -2, 3\}$</u> Function? <u>Yes</u>	2. $\{(5, 2), (-3, 4), (5, -6), (8, 0)\}$ D: <u>$\{-3, 5, 8\}$</u> R: <u>$\{-6, 0, 2, 4\}$</u> Function? <u>No</u>	3. <table><tr><td>x</td><td>7</td><td>2</td><td>0</td><td>-4</td></tr><tr><td>y</td><td>-8</td><td>3</td><td>-1</td><td>3</td></tr></table> D: <u>$\{-4, 0, 2, 7\}$</u> R: <u>$\{-8, -1, 3\}$</u> Function? <u>Yes</u>	x	7	2	0	-4	y	-8	3	-1	3	4. <table><tr><td>x</td><td>10</td><td>15</td><td>18</td><td>24</td></tr><tr><td>y</td><td>0</td><td>3</td><td>7</td><td>11</td></tr></table> D: <u>$\{10, 15, 18, 24\}$</u> R: <u>$\{0, 3, 7, 11\}$</u> Function? <u>Yes</u>	x	10	15	18	24	y	0	3	7	11
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x	7	2	0	-4																					
y	-8	3	-1	3																					
x	10	15	18	24																					
y	0	3	7	11																					



NOTE:
 Use a graphing
 device for #14-17.

14. $y = -x^2 + 3$
 D: $\mathbb{R}$
 R: $y \leq 3$
 Function? Yes

15. $y = -5x - 1$
 D: $\mathbb{R}$
 R: $\mathbb{R}$
 Function? Yes

16. $y = -7$
 D: $\mathbb{R}$
 R: $y = -7$
 Function? Yes

17. $x = -1$
 D: $x = -1$
 R: $\mathbb{R}$
 Function? No

Name:		Date:	
Topic:		Class:	

Main Ideas/Questions	Notes/Examples
FUNCTION Notation	Equations that represent functions are often written in function notation. Example: $y = 2x + 7 \rightarrow f(x) = 2x + 7$ <u> </u> x is the input and <u> </u> $f(x)$ is the output.
EVALUATING Functions Given: • $f(x) = -5x - 14$ • $g(x) = -x^2 + 9x - 1$	Example: If $f(x) = 2x + 7$, find $f(-3)$ $f(-3) = 2(-3) + 7$ $= -6 + 7$ $= 1$ For questions 1-6, evaluate using $f(x)$ and $g(x)$ given to the left. 1. $f(-9)$ $f(-9) = -5(-9) - 14$ $= 45 - 14$ $= 31$ 2. $f\left(\frac{3}{10}\right)$ $f\left(\frac{3}{10}\right) = -5\left(\frac{3}{10}\right) - 14$ $= -\frac{3}{2} - 14$ $= -\frac{31}{2}$ 3. $g(0.4)$ $g(0.4) = -(0.4)^2 + 9(0.4) - 1$ $= -0.16 + 3.6 - 1$ $= 2.44$ 4. $g(-7)$ $g(-7) = -(-7)^2 + 9(-7) - 1$ $= -49 - 63 - 1$ $= -113$ 5. $f(1) - f(-4)$ $f(1) = -5(1) - 14$ $= -19$ $f(-4) = -5(-4) - 14$ $= 6$ $-19 - 6 = -25$ 6. $2[g(12)]$ $g(12) = -(12)^2 + 9(12) - 1$ $= -144 + 108 - 1$ $= -37$ $2(-37) = -74$
Given: • $p(x) = \frac{1}{2} - \frac{4}{3}x$ • $q(x) = 3x - x^3$	For questions 7-12, evaluate using $p(x)$ and $q(x)$ given to the left. 7. $p\left(-\frac{9}{16}\right)$ $p\left(-\frac{9}{16}\right) = \frac{1}{2} - \frac{4}{3}\left(-\frac{9}{16}\right)$ $= \frac{1}{2} + \frac{3}{4}$ $= \frac{5}{4}$ 8. $q(5)$ $q(5) = 3(5) - 5^3$ $= 15 - 125$ $= -110 = 110$ 9. $q(-2)$ $q(-2) = 3(-2) - (-2)^3$ $= -6 - (-8)$ $= 2 = 2$ 10. $p(18)$ $p(18) = \frac{1}{2} - \frac{4}{3}(18)$ $= \frac{1}{2} - 24$ $= -\frac{47}{2}$

	11. $7 - q(4.5)$ $q(4.5) = 3(4.5) - 4.5^3 $ $= 13.5 - 91.125 $ $= 77.625$ $7 - 77.625 = \boxed{-70.625}$	12. $3[p(-1)] + 5$ $p(-1) = \frac{1}{2} - \frac{4}{3}(-1)$ $= \frac{1}{2} + \frac{4}{3}$ $= \frac{11}{6}$ $3\left(\frac{11}{6}\right) + 5 = \boxed{\frac{21}{2}}$
Given: • $f(x) = 3x + 19$	For questions 13 and 14, find the value of x given $f(x)$ to the left.	
	13. If $f(x) = 34$, find x . $3x + 19 = 34$ $3x = 15$ $\boxed{x = 5}$	14. If $f(x) = -62$, find x . $3x + 19 = -62$ $3x = -81$ $\boxed{x = -27}$
Given: • $f(x) = -4x + 9$ • $g(x) = x^2 - 5$	For questions 15-18, evaluate using $f(x)$ and $g(x)$ given to the left.	
	15. $f(c+2)$ $f(c+2) = -4(c+2) + 9$ $= -4c - 8 + 9$ $\boxed{= -4c + 1}$	16. $f(7w-5)$ $f(7w-5) = -4(7w-5) + 9$ $= -28w + 20 + 9$ $\boxed{= -28w + 29}$
	17. $g(a-3)$ $g(a-3) = (a-3)^2 - 5$ $= (a-3)(a-3) - 5$ $= a^2 - 6a + 9 - 5$ $\boxed{= a^2 - 6a + 4}$	
	18. $g(2m+1) - f(3m-7)$ $g(2m+1) = (2m+1)^2 - 5$ $= (2m+1)(2m+1) - 5$ $= 4m^2 + 4m + 1 - 5$ $= 4m^2 + 4m - 4$ $f(3m-7) = -4(3m-7) + 9$ $= -12m + 28 + 9$ $= -12m + 37$ $(4m^2 + 4m - 4) - (-12m + 37) = \boxed{4m^2 + 16m - 41}$	
APPLICATION: Owen paid a lawn company \$108 for an initial lawn treatment, then \$74 per month thereafter. The function $p(t) = 108 + 74t$ represents the total amount of money he will have paid after t months.	Use the information to the left for questions 19 and 20.	
	19. Find and interpret $p(24)$. $p(24) = 108 + 74(24)$ $= 108 + 1776$ $= 1884$ For 24 months of service, Owen will pay \$1,884.	20. How long has Owen been receiving lawn service if he has paid a total of \$2,994 to date? $2994 = 108 + 74t$ $2886 = 74t$ $39 = t$ 39 months

Name: _____

Unit 2: Linear Functions & Systems

Date: _____ Per: _____

Homework 1: Relations & Functions

**** This is a 2-page document! ******Find the domain and range of each relation. Then determine if the relation is a function.**

1. $\{(-1, 4), (-1, 9), (-1, 15), (-1, 0)\}$

D: $\{-1\}$

R: $\{0, 4, 9, 15\}$

Function? No

2.

x	5	6	7	8
y	0	-2	10	-3

D: $\{5, 6, 7, 8\}$

R: $\{-3, -2, 0, 10\}$

Function? Yes

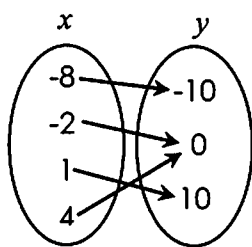
3. $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$

D: $\{1, 2, 3, 4\}$

R: $\{1, 2, 3, 4\}$

Function? Yes

4.

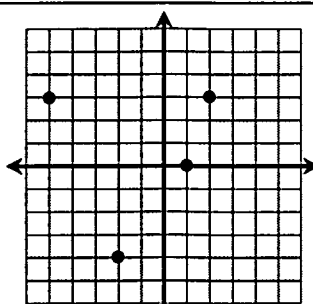


D: $\{-8, -2, 1, 4\}$

R: $\{-10, 0, 10\}$

Function? Yes

5.

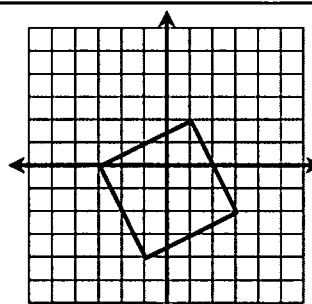


D: $\{-5, -2, 1, 2\}$

R: $\{-4, 0, 3\}$

Function? Yes

6.

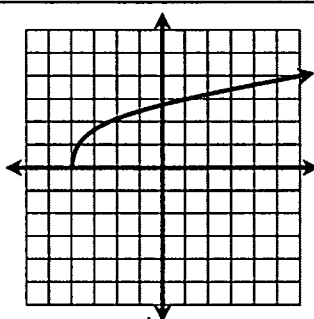


D: $-3 \leq x \leq 3$

R: $-4 \leq y \leq 2$

Function? No

7.

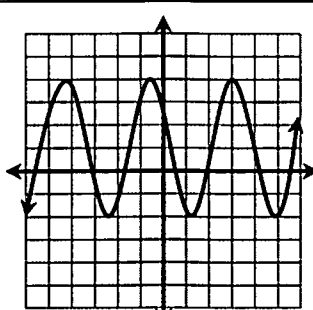


D: $x \geq -4$

R: $y \geq 0$

Function? Yes

8.

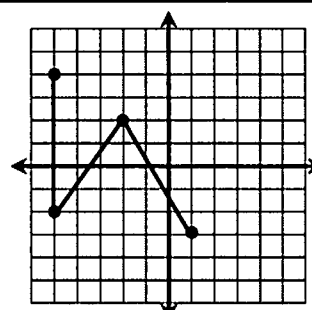


D: $-2 \leq x \leq 4$

R: $\mathbb{R}$

Function? Yes

9.

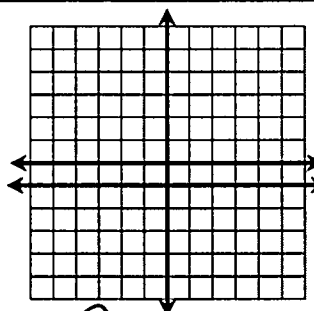


D: $-5 \leq x \leq 1$

R: $-3 \leq y \leq 4$

Function? No

10.

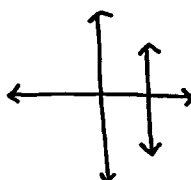


D: $\mathbb{R}$

R: $y = 1$

Function? Yes

11. $x = 4$

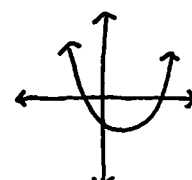


D: $x = 4$

R: $\mathbb{R}$

Function? No

12. $y = x^2 - 4x - 5$



D: $\mathbb{R}$

R: $y \geq -9$

Function? Yes

Use $f(x) = -\frac{3}{2}x + 11$, $g(x) = x^2 - 7x$, and $h(x) = |15 - 9x|$ to evaluate each function.

13. $f(-8)$

$$= -\frac{3}{2}(-8) + 11$$

$$= 12 + 11 = \boxed{23}$$

14. $g(-2)$

$$= (-2)^2 - 7(-2)$$

$$= 4 + 14 = \boxed{18}$$

15. $h\left(\frac{2}{3}\right)$

$$= \left| 15 - 9\left(\frac{2}{3}\right) \right|$$

$$= |15 - 6| = \boxed{9}$$

16. $h(5) + g(9)$

$$h(5) = |15 - 9(5)|$$

$$= |15 - 45|$$

$$= |-30|$$

$$= 30$$

$$g(9) = (9)^2 - 7(9)$$

$$= 81 - 63$$

$$= 18$$

$$30 + 18 = \boxed{48}$$

17. $1 - 2\left[f\left(\frac{8}{9}\right)\right]$

$$f\left(\frac{8}{9}\right) = -\frac{3}{2}\left(\frac{8}{9}\right) + 11$$

$$= -\frac{4}{3} + 11$$

$$= \frac{29}{3}$$

$$1 - 2\left(\frac{29}{3}\right)$$

$$= 1 - \frac{58}{3}$$

$$= \boxed{-\frac{55}{3}}$$

18. If $f(x) = -7$, find x .

$$-\frac{3}{2}x + 11 = -7$$

$$-\frac{3}{2}x = -18$$

$$\boxed{x = 12}$$

19. If $h(x) = 48$, find x .

$$|15 - 9x| = 48$$

$$15 - 9x = 48$$

$$-9x = 33$$

$$x = -\frac{11}{3}$$

$$15 - 9x = -48$$

$$-9x = -63$$

$$x = 7$$

$$\boxed{x = \left\{-\frac{11}{3}, 7\right\}}$$

20. Find $f(-8c + 2)$

$$= -\frac{3}{2}(-8c + 2) + 11$$

$$= 12c - 3 + 11$$

$$\boxed{= 12c + 8}$$

21. Find $g(y - 8)$

$$= (y - 8)^2 - 7(y - 8)$$

$$= y^2 - 16y + 64 - 7y + 56$$

$$\boxed{= y^2 - 23y + 120}$$

22. The commission earned on the sale of a car can be represented by the equation $c(p) = 250 + 0.02p$ where c represents the commission and p represents the purchase price of the car.

a) Jack sold two cars last weekend, one for \$15,075 and the other for \$21,640. Find his total commission.

$$c(15075) = 250 + .02(15075)$$

$$= 250 + 301.5$$

$$= 551.50$$

$$c(21640) = 250 + .02(21640)$$

$$= 250 + 432.8$$

$$= 682.80$$

Total:

$$551.50$$

$$+ 682.80$$

$$= \boxed{\$1234.30}$$

b) If $c(p) = 745.80$, find the purchase price of the car.

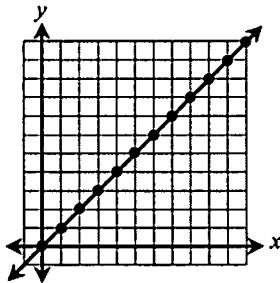
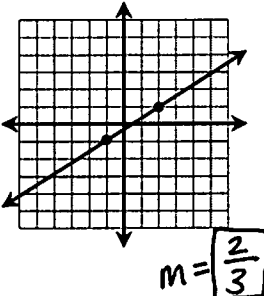
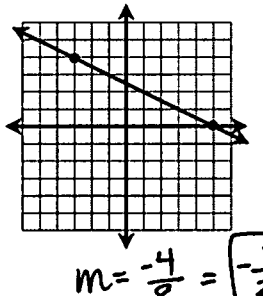
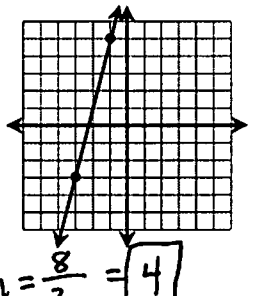
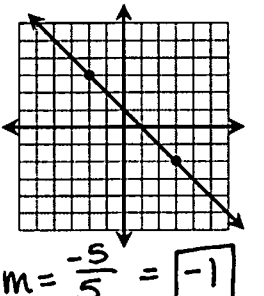
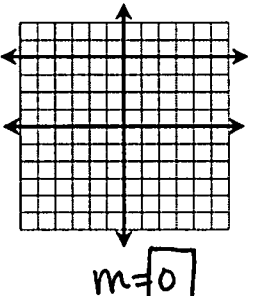
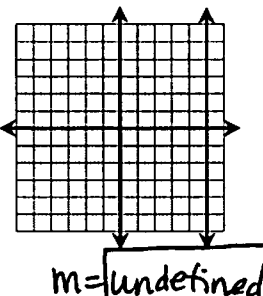
$$745.80 = 250 + .02p$$

$$495.80 = .02p$$

$$\boxed{p = \$24,790}$$

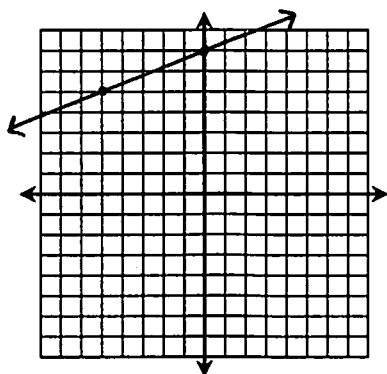
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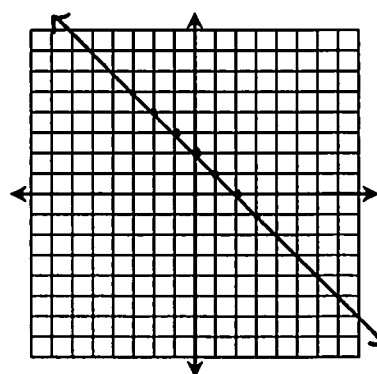
Main Ideas/Questions	Notes/Examples
LINEAR <i>Relations</i>	- When points on a graph lie along a straight line, it is called a linear relation. - Linear relations can be represented by an equation, for example, the equation of the graph to the right can be written as $y = x$. 
SLOPE Slope is the constant rate of change between the points on a line. (Remember rise/run!)	Find the slope of the line graphed. 1.  2.  3.  4.  5.  6.  Find the slope of the line passing through the given points. 7. (-8, -6) and (2, 14) $m = \frac{14 - (-6)}{2 - (-8)} = \frac{20}{10} = 2$ 8. (8, -7) and (16, -13) $m = \frac{-13 - (-7)}{16 - 8} = \frac{-6}{8} = -\frac{3}{4}$ 9. (5, -6) and (5, 2) $m = \frac{2 - (-6)}{5 - 5} = \frac{8}{0} = \text{undefined}$ 10. (-7, -1) and (4, -1) $m = \frac{-1 - (-1)}{4 - (-7)} = \frac{0}{11} = 0$
SLOPE-INTERCEPT <i>Form</i>	Linear equations are often written in slope-intercept form: $y = mx + b \quad \text{(or)} \quad f(x) = mx + b$ (where m is the <u>slope</u> and b is the <u>y-intercept</u>)

Identify the slope and y-intercept, then graph the line.

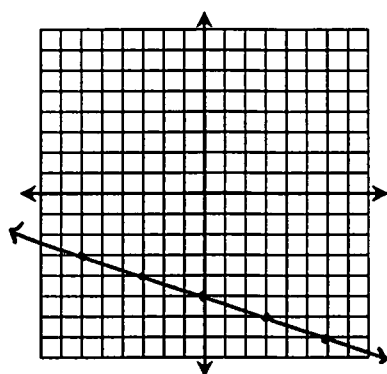
11. $y = \frac{2}{5}x + 7$ $m = \frac{2}{5}$
 $b = 7$



12. $y = -x + 2$ $m = -1$
 $b = 2$

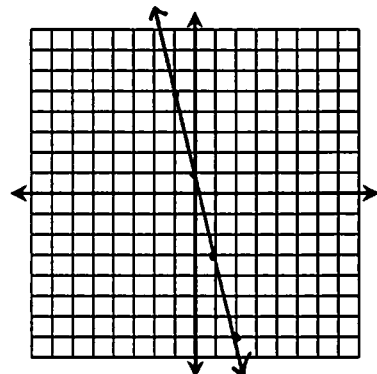


13. $y = -\frac{1}{3}x - 5$ $m = -\frac{1}{3}$
 $b = -5$



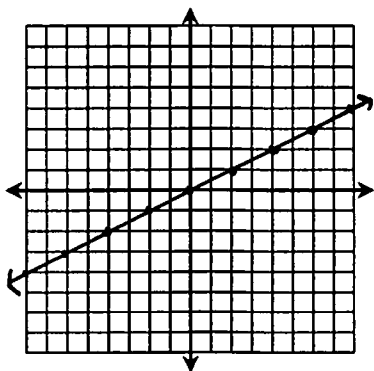
14. $f(x) = -4x + 1$

$m = -4$
 $b = 1$



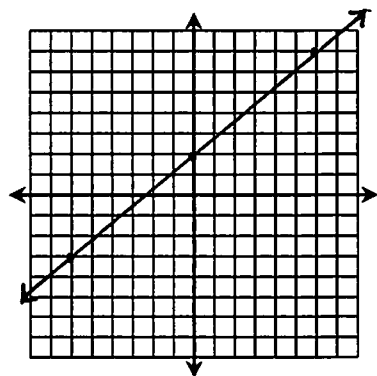
15. $f(x) = \frac{1}{2}x$

$m = \frac{1}{2}$
 $b = 0$



16. $f(x) = \frac{5}{6}x + 2$

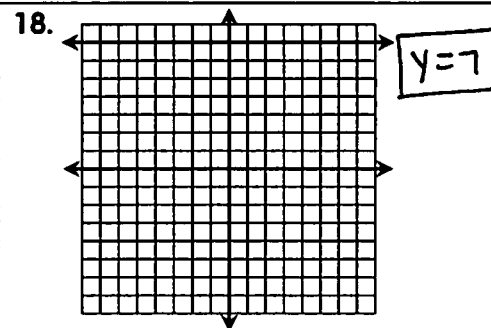
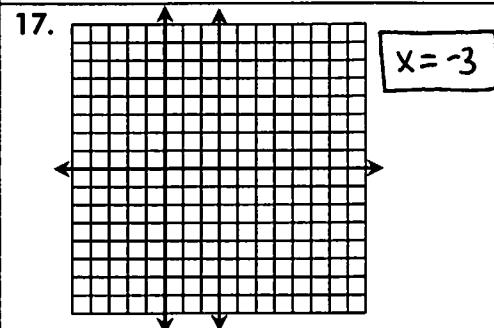
$m = \frac{5}{6}$
 $b = 2$



**VERTICAL &
HORIZONTAL**

Lines

Write each equation to represent the line shown on the graph.



Name:

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Main Ideas/Questions

Notes/Examples

STANDARD*Form*

Directions: Write each equation in both standard form and slope-intercept form, then graph the line.

GCF = 3

GCF = 7

Linear equations are also often written in **standard form**:

$$Ax + By = C$$

(where A , B , and C are integers, $A \geq 1$, and the GCF of A , B , and C is 1)

Standard Form

Slope-Intercept Form

Graph

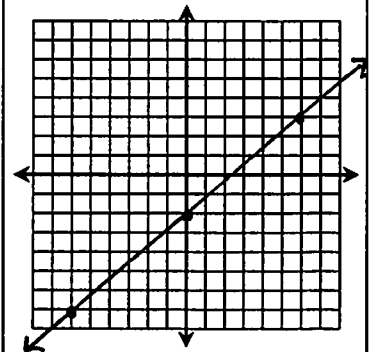
$$1. \frac{-5x}{-1} + \frac{6y}{-1} = \frac{-12}{-1}$$

$$5x - 6y = 12$$

$$-5x + 6y = -12$$

$$6y = 5x - 12$$

$$y = \frac{5}{6}x - 2$$



$$2. 9x + 6y - 24 = 0$$

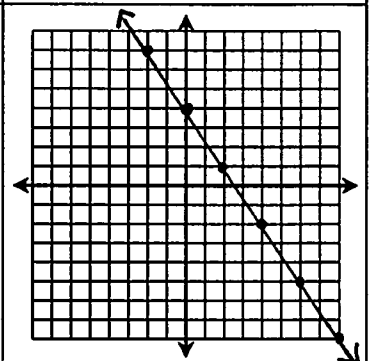
$$\frac{9x}{3} + \frac{6y}{3} = \frac{24}{3}$$

$$3x + 2y = 8$$

$$9x + 6y - 24 = 0$$

$$6y = -9x + 24$$

$$y = -\frac{3}{2}x + 4$$



$$3. 7y = 7 - 7x$$

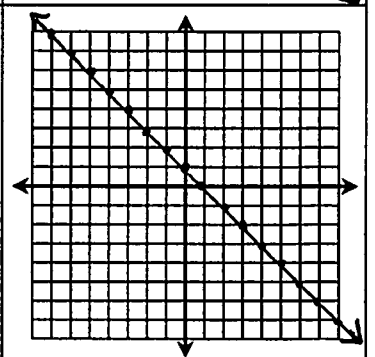
$$\frac{7x}{7} + \frac{7y}{7} = \frac{7}{7}$$

$$x + y = 1$$

$$7y = 7 - 7x$$

$$7y = -7x + 7$$

$$y = -x + 1$$



$$4. \frac{3}{4}x - 3y = 1$$

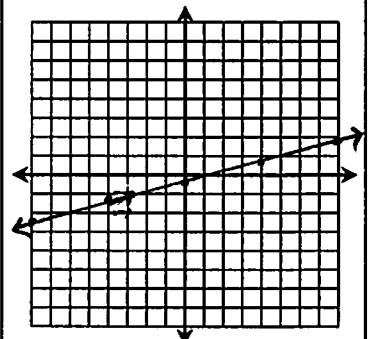
$$3x - 12y = 4$$

$$\frac{3}{4}x - 3y = 1$$

$$3x - 12y = 4$$

$$-12y = -3x + 4$$

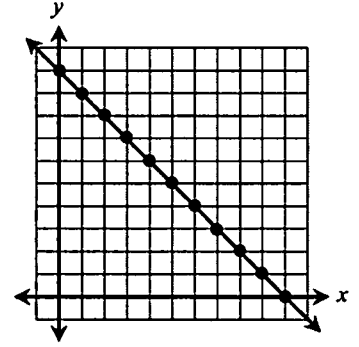
$$y = \frac{1}{4}x - \frac{1}{3}$$



X- and Y-INTERCEPTS

- The point at which the graph intersects the x-axis is called the **x-intercept**. Likewise, the point at which the graph intersects the y-axis is called the **y-intercept**.
- Identify the x- and y-intercept of the graph shown on the right.

x-intercept: $(10, 0)$
y-intercept: $(0, 10)$



The x-intercept is also referred to as a root, zero, or solution to the equation.

FINDING INTERCEPTS Algebraically

To find the x-intercept of an equation, set y equal to 0 and solve for x . To find the y-intercept of an equation, set x equal to 0 and solve for y .

Example: Find the x- and y-intercepts of the equation $y = -2x + 5$.

$$\begin{aligned} \text{x-int: } 0 &= -2x + 5 \\ -5 &= -2x \\ \frac{5}{2} &= x \end{aligned}$$

$(\frac{5}{2}, 0)$

$$\begin{aligned} \text{y-int: } y &= -2(0) + 5 \\ y &= 0 + 5 \\ y &= 5 \end{aligned}$$

$(0, 5)$

EXAMPLES

Directions: Find the x- and y-intercept of each equation. Then, graph the equation using the intercepts.

5. $y = 5x - 4$

x-int:

$$0 = 5x - 4$$

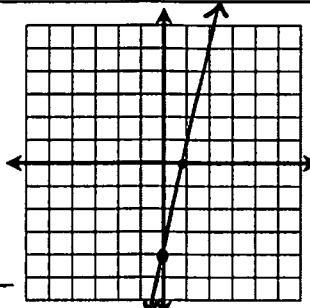
$$4 = 5x$$

$$\frac{4}{5} = x$$

y-int:

$$\begin{aligned} y &= 5(0) - 4 \\ y &= -4 \end{aligned}$$

x-int: $(\frac{4}{5}, 0)$ y-int: $(0, -4)$



6. $y = -\frac{6}{5}x + 2$

x-int:

$$0 = -\frac{6}{5}x + 2$$

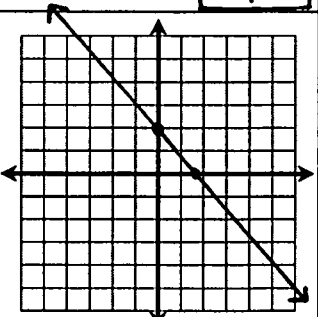
$$-2 = -\frac{6}{5}x$$

$$\frac{5}{3} = x$$

y-int:

$$\begin{aligned} y &= -\frac{6}{5}(0) + 2 \\ y &= 2 \end{aligned}$$

x-int: $(\frac{5}{3}, 0)$ y-int: $(0, 2)$



7. $4x - y = 6$

x-int:

$$4x - 0 = 6$$

$$4x = 6$$

$$x = \frac{3}{2}$$

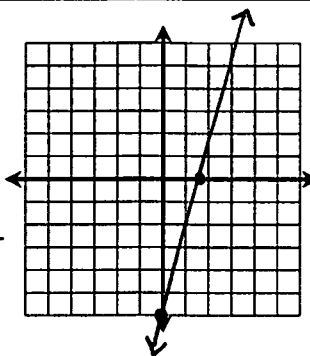
y-int:

$$4(0) - y = 6$$

$$-y = 6$$

$$y = -6$$

x-int: $(\frac{3}{2}, 0)$ y-int: $(0, -6)$



8. $-5x - 8y = -25$

x-int:

$$-5x - 8(0) = -25$$

$$-5x = -25$$

$$x = 5$$

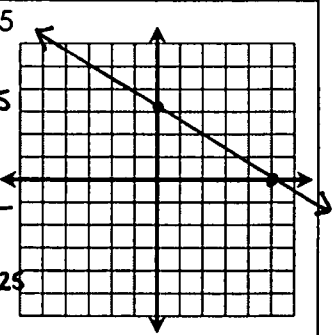
y-int:

$$-5(0) - 8y = -25$$

$$-8y = -25$$

$$y = \frac{25}{8}$$

x-int: $(5, 0)$ y-int: $(0, \frac{25}{8})$



Name: _____

Unit 2: Linear Functions & Systems

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Homework 2: Slope-Intercept & Standard Form

**** This is a 2-page document! ****

Find the slope of the line passing through the given points.

1. (5, -2) and (-7, -8)

$$m = \frac{-8 - (-2)}{-7 - 5} = \frac{-6}{-12}$$

$$= \frac{1}{2}$$

2. (-3, 4) and (-3, -5)

$$m = \frac{-5 - 4}{-3 - (-3)} = \frac{-9}{0}$$

$$= \text{undefined}$$

3. (-2, 1) and (-10, 3)

$$m = \frac{3 - 1}{-10 - (-2)} = \frac{2}{-8}$$

$$= -\frac{1}{4}$$

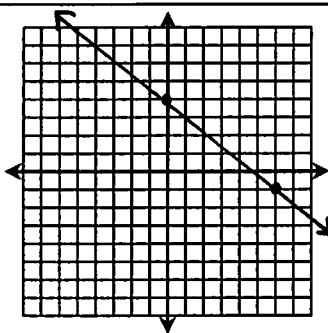
4. (0, 8) and (-6, 8)

$$m = \frac{8 - 8}{-6 - 0} = \frac{0}{-6}$$

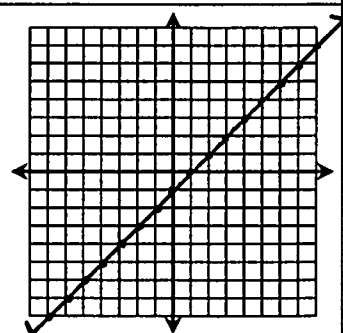
$$= 0$$

Graph each line. Write the equation in slope-intercept form when necessary.

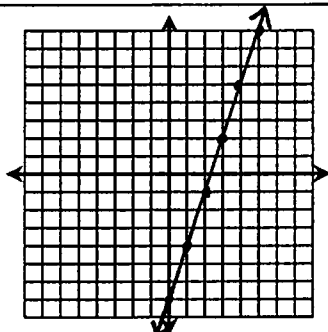
5. $y = -\frac{5}{6}x + 4$



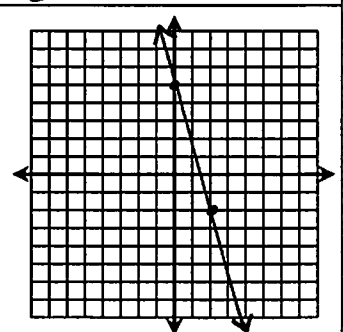
6. $y = x - 1$



7. $f(x) = 3x - 7$



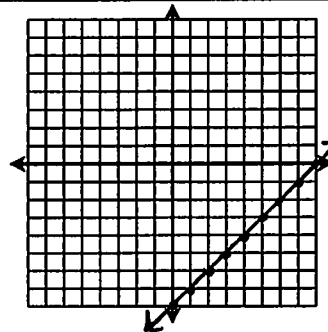
8. $f(x) = -\frac{7}{2}x + 5$



9. $x - y = 8$

$-y = -x + 8$

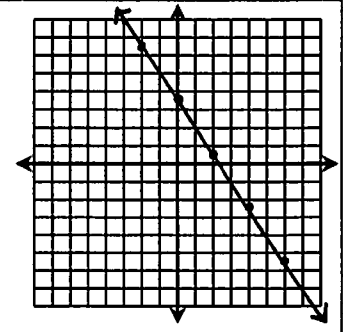
$y = x - 8$



10. $12x + 8y = 28$

$8y = -12x + 28$

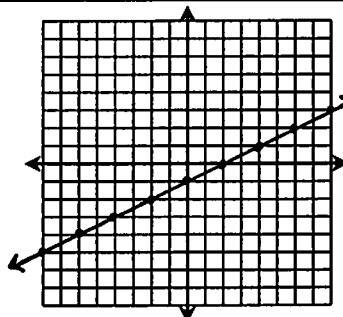
$y = -\frac{3}{2}x + \frac{7}{2}$



11. $x - 2y - 2 = 0$

$-2y = -x + 2$

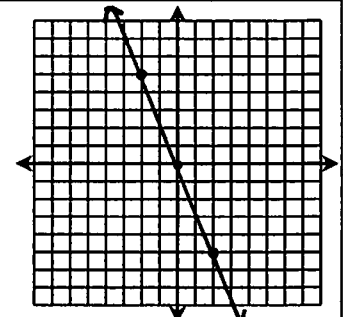
$y = \frac{1}{2}x - 1$



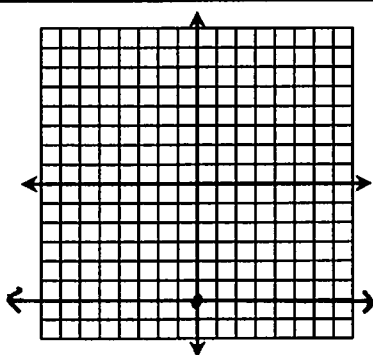
12. $-5x = 2y$

$2y = -5x$

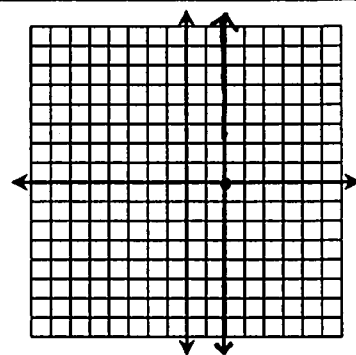
$y = -\frac{5}{2}x$



13. $y = -6$

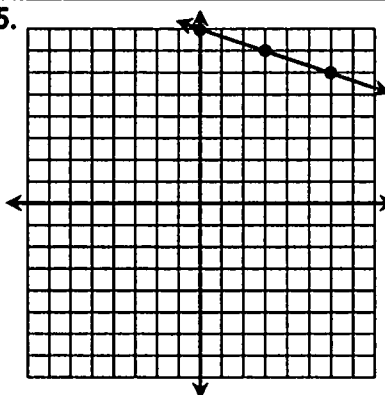


14. $x = 2$



Write the equation of the line graphed in both slope-intercept form and standard form.

15.



$$m = -\frac{1}{3}$$

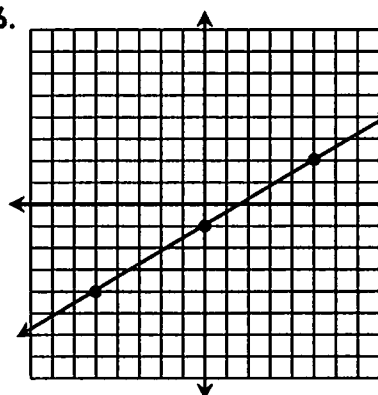
$$b = 8$$

$$y = -\frac{1}{3}x + 8$$

$$3y = -x + 24$$

$$x + 3y = 24$$

16.



$$m = \frac{3}{5}$$

$$b = -1$$

$$y = \frac{3}{5}x - 1$$

$$5y = 3x - 5$$

$$-3x + 5y = -5$$

$$3x - 5y = 5$$

Slope-Intercept Form: $y = -\frac{1}{3}x + 8$

Slope-Intercept Form: $y = \frac{3}{5}x - 1$

Standard Form: $x + 3y = 24$

Standard Form: $3x - 5y = 5$

Find the x- and y-intercepts of each equation. Use the intercepts to graph the line.

17. $y = 2x - 2$

x-int: $0 = 2x - 2$

$$2 = 2x$$

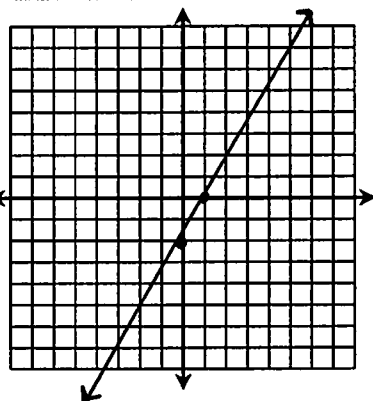
$$1 = x$$

$$(1, 0)$$

y-int: $y = 2(0) - 2$

$$y = -2$$

$$(0, -2)$$



18. $6x + 5y = -15$

x-int: $6x + 5(0) = -15$

$$6x = -15$$

$$x = -\frac{5}{2}$$

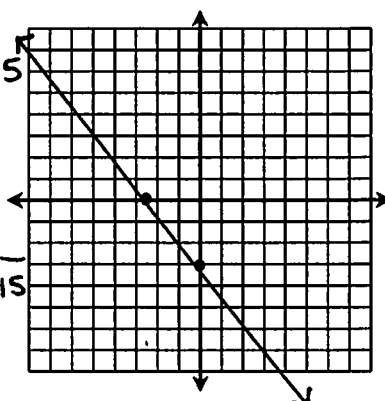
$$(-\frac{5}{2}, 0)$$

y-int: $6(0) + 5y = -15$

$$5y = -15$$

$$y = -3$$

$$(0, -3)$$



19. $20 + 12x = 4y$

x-int: $20 + 12x = 4(0)$

$$20 + 12x = 0$$

$$12x = -20$$

$$x = -\frac{5}{3}$$

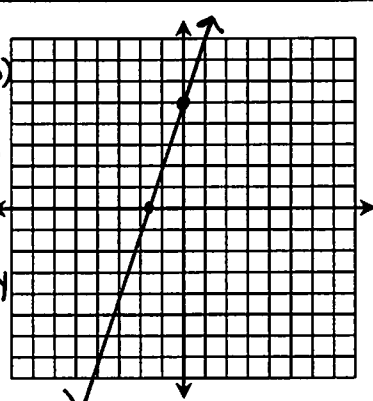
$$(-\frac{5}{3}, 0)$$

y-int: $20 + 12(0) = 4y$

$$20 = 4y$$

$$5 = y$$

$$(0, 5)$$



20. $7x - 10y = 35$

x-int: $7x - 10(0) = 35$

$$7x = 35$$

$$x = 5$$

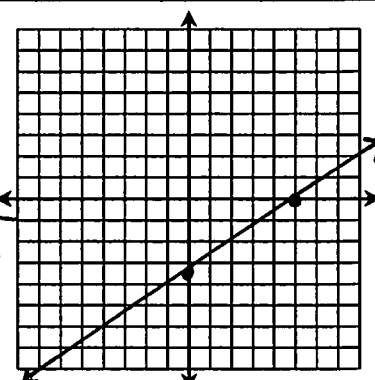
$$(5, 0)$$

y-int: $7(0) - 10y = 35$

$$-10y = 35$$

$$y = -\frac{7}{2}$$

$$(0, -\frac{7}{2})$$



Name: _____

Algebra 2

Date: _____ Per: _____

Unit 2: Linear Functions & Systems

Quiz 2-1: Relations, Functions, & Graphing Linear Functions

1. Which relations represent a function? Check all that apply.

☒

x	y
-8	1
-7	1
-6	1
-5	1

☒

☐

☒

2. Which graph does not represent a function?

A.

B.

C.

D.

Give the domain and range of each relation below.

3.

4.

3. $D = \mathbb{R}$

$R = y \leq 1$

4. $D = -2 \leq x \leq 3$

$R = -3 \leq y \leq 3$

5. $\frac{23}{2}$

Use the functions $f(x) = 10 - \frac{9}{2}x$, $g(x) = |-3x + 16|$, and $h(x) = -x^2 + 8x$ for the questions 5 and 6.

5. Find $f\left(-\frac{1}{3}\right)$

$$10 - \frac{9}{2}\left(-\frac{1}{3}\right)$$

$$= 10 + \frac{3}{2} = \frac{23}{2}$$

6. Find $g(20) + h(7)$

$$g(20) = |-3(20) + 16|$$

$$= |-60 + 16|$$

$$= |-44| = 44$$

$$h(7) = -(7)^2 + 8(7)$$

$$= -49 + 56$$

$$= 7$$

6. 51

$$\left. \begin{array}{l} 44 \\ 7 \end{array} \right\} 44 + 7 = 51$$

Use the functions $f(x) = 10 - \frac{9}{2}x$, $g(x) = |-3x + 16|$, and $h(x) = -x^2 + 8x$ for the questions 7 and 8.

7. If $f(x) = -53$, find x .

$$-53 = 10 - \frac{9}{2}x$$

$$-63 = -\frac{9}{2}x$$

$$x = 14$$

8. If $g(x) = 17$, find x .

$$|-3x + 16| = 17$$

$$-3x + 16 = 17$$

$$-3x = 1$$

$$x = -\frac{1}{3}$$

$$-3x + 16 = -17$$

$$-3x = -33$$

$$x = 11$$

7. $x = 14$

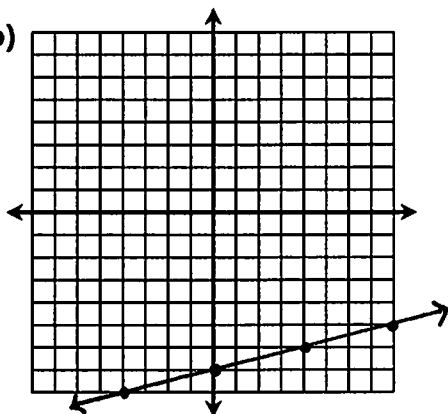
8. $x = \{-\frac{1}{3}, 11\}$

For questions 9 and 10, (a) write the equation in slope-intercept form, then (b) graph the line.

9. $x - 4y = 28$ a) $y = \frac{1}{4}x - 7$

$$-4y = -x + 28$$

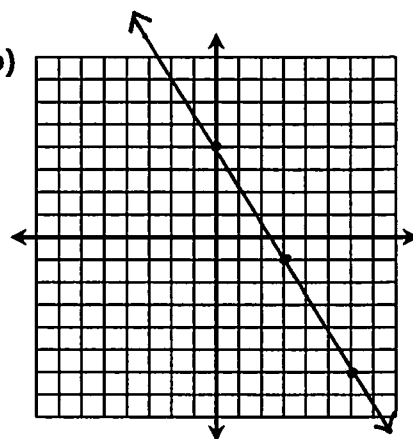
$$y = \frac{1}{4}x - 7$$



10. $9y - 36 = -15x$ a) $y = -\frac{5}{3}x + 4$

$$9y = -15x + 36$$

$$y = -\frac{5}{3}x + 4$$



For questions 11 and 12, (a) give the x- and y-intercepts of the function, then (b) graph the line.

11. $y = -6x - 5$ a) x-intercept: $(-\frac{5}{6}, 0)$

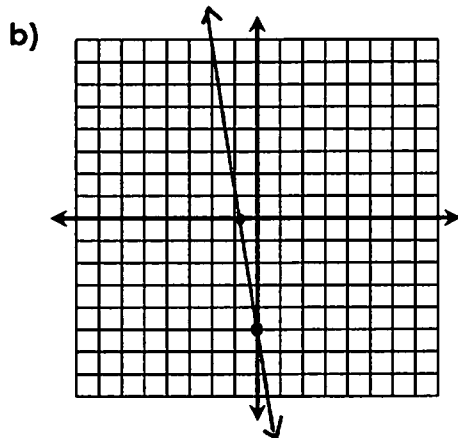
x-int:

$$0 = -6x - 5$$

$$5 = -6x$$

$$-\frac{5}{6} = x$$

y-intercept: $(0, -5)$



12. $\frac{9}{4}x - \frac{1}{2}y = -3$ a) x-intercept: $(-\frac{4}{3}, 0)$

x-int:

$$\frac{9}{4}x - \frac{1}{2}(0) = -3$$

$$\frac{9}{4}x = -3$$

$$x = -\frac{4}{3}$$

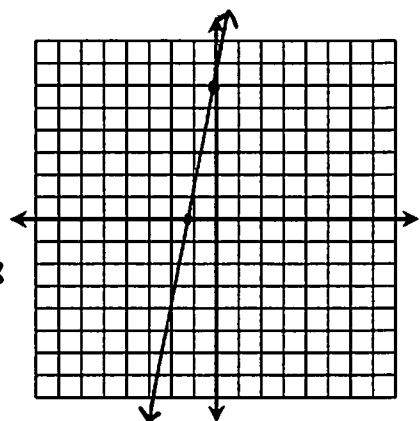
y-intercept: $(0, 6)$

y-int:

$$\frac{9}{4}(0) - \frac{1}{2}y = -3$$

$$-\frac{1}{2}y = -3$$

$$y = 6$$



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Main Ideas/Questions

Notes/Examples

WRITING LINEAR EQUATIONS Given a Point & Slope

To write a linear equation given the slope of the line (m), and a point that it passes through (x_1, y_1) , use the **point-slope formula**:

$$y - y_1 = m(x - x_1)$$

Write the equation of the line in slope-intercept form given a point on the line and the slope of the line.

1. $(-1, 7); m = -4$

$$y - 7 = -4(x + 1)$$

$$y - 7 = -4x - 4$$

$$y = -4x + 3$$

2. $(-6, -5); m = \frac{2}{3}$

$$y + 5 = \frac{2}{3}(x + 6)$$

$$y + 5 = \frac{2}{3}x + 4$$

$$y = \frac{2}{3}x - 1$$

3. $(4, -3); m = 1$

$$y + 3 = 1(x - 4)$$

$$y + 3 = x - 4$$

$$y = x - 7$$

4. $(-8, 12); m = -\frac{5}{4}$

$$y - 12 = -\frac{5}{4}(x + 8)$$

$$y - 12 = -\frac{5}{4}x - 10$$

$$y = -\frac{5}{4}x + 2$$

WRITING LINEAR EQUATIONS Given Two Points

To write a linear equation given two points (x_1, y_1) and (x_2, y_2) use the **slope formula** followed by the **point-slope formula**:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \rightarrow y - y_1 = m(x - x_1)$$

5. $(-6, 1)$ and $(-3, 7)$

$$m = \frac{7 - 1}{-3 - (-6)} = \frac{6}{3} = 2$$

$$y - 1 = 2(x + 6)$$

$$y - 1 = 2x + 12$$

$$y = 2x + 13$$

6. $(-2, -2)$ and $(-6, 8)$

$$m = \frac{8 - (-2)}{-6 - (-2)} = \frac{10}{-4} = -\frac{5}{2}$$

$$y + 2 = -\frac{5}{2}(x + 2)$$

$$y + 2 = -\frac{5}{2}x - 5$$

$$y = -\frac{5}{2}x - 7$$

7. $(10, -5)$ and $(1, -8)$

$$m = \frac{-8 - (-5)}{1 - 10} = \frac{-3}{-9} = \frac{1}{3}$$

$$y + 8 = \frac{1}{3}(x - 1)$$

$$y + 8 = \frac{1}{3}x - \frac{1}{3}$$

$$y = \frac{1}{3}x - \frac{25}{3}$$

8. $(-8, -3)$ and $(-2, -3)$

$$m = \frac{-3 - (-3)}{-2 - (-8)} = \frac{0}{6} = 0$$

$$y + 3 = 0(x + 8)$$

$$y + 3 = 0$$

$$y = -3$$

PARALLEL VS. PERPENDICULAR LINES

- The slopes of parallel lines are equal.
- The slopes of perpendicular lines are negative reciprocals.

Determine whether the pair of lines are parallel, perpendicular, or neither.

9. $10x + 5y = 5$ and $-x = 2y + 16$
 $5y = -10x + 5$ $-x - 16 = 2y$
 $y = -2x + 1$ $-\frac{1}{2}x - 8 = y$
 $\hookrightarrow m = -2$ $\hookrightarrow m = -\frac{1}{2}$

Neither

10. $8x - 6y = 42$ and $-3y = 9 - 4x$
 $-6y = -8x + 42$ $-3y = -4x + 9$
 $y = \frac{4}{3}x - 7$ $y = \frac{4}{3}x - 3$
 $\hookrightarrow m = \frac{4}{3}$ $\hookrightarrow m = \frac{4}{3}$

Parallel

11. $-8x + 8y = 0$ and $x - y = 4$
 $8y = 8x$ $-y = -x + 4$
 $y = x$ $y = x - 4$
 $\hookrightarrow m = 1$ $\hookrightarrow m = 1$

Parallel

12. $x = 2$ and $y = 2$
 $m = \text{undef.}$ $m = 0$

Perpendicular

Write an equation in slope-intercept form with the given information.

13. Passing through the point $(-6, 2)$ and parallel to the line $x - 3y = 21$.
 $y - 2 = \frac{1}{3}(x + 6)$ $-3y = -x + 21$
 $y - 2 = \frac{1}{3}x + 2$ $y = \frac{1}{3}x - 7$
 $\hookrightarrow m = \frac{1}{3}$

$y = \frac{1}{3}x + 4$

14. Passing through the point $(-1, -3)$ and parallel to the line passing through the points $(-4, 11)$ and $(2, -1)$.
 $m = \frac{-1 - 11}{2 - (-4)} = \frac{-12}{6} = -2$

$y + 3 = -2(x + 1)$
 $y + 3 = -2x - 2$
 $y = -2x - 5$

15. Passing through the point $(-9, 1)$ and perpendicular to the line $6x + 4y = -4$.

$y - 1 = \frac{2}{3}(x + 9)$
 $y - 1 = \frac{2}{3}x + 6$

$y = \frac{2}{3}x + 7$

$4y = -6x - 4$
 $y = -\frac{3}{2}x - 1$
 $\hookrightarrow m = -\frac{3}{2}$
 $\downarrow$
 $m = \frac{2}{3}$

16. Passing through the point $(7, -16)$ and perpendicular to the line passing through the points $(4, 2)$ and $(-8, -10)$.

$y + 16 = -1(x - 7)$
 $y + 16 = -x + 7$

$y = -x - 9$

$m = \frac{-10 - 2}{-8 - 4} = \frac{-12}{-12} = 1$
 $\downarrow$
 $m = -1$

Name: _____

Unit 2: Linear Functions & Systems

Date: _____ Per: _____

Homework 3: Writing Linear Equations

Write a linear equation in slope-intercept form with the given information.

1. slope = -6; passes through (-4, 1)

$$y - 1 = -6(x + 4)$$

$$y - 1 = -6x - 24$$

$$y = -6x - 23$$

2. slope = $\frac{4}{3}$; passes through (-5, -6)

$$y + 6 = \frac{4}{3}(x + 5)$$

$$y + 6 = \frac{4}{3}x + \frac{20}{3}$$

$$y = \frac{4}{3}x + \frac{2}{3}$$

3. passes through (-4, 11) and (2, 8)

$$m = \frac{8 - 11}{2 - (-4)} = \frac{-3}{6} = -\frac{1}{2}$$

$$y - 8 = -\frac{1}{2}(x - 2)$$

$$y - 8 = -\frac{1}{2}x + 1$$

$$y = -\frac{1}{2}x + 9$$

4. passes through (6, 3) and (14, -5)

$$m = \frac{-5 - 3}{14 - 6} = \frac{-8}{8} = -1$$

$$y - 3 = -1(x - 6)$$

$$y - 3 = -x + 6$$

$$y = -x + 9$$

5. passes through (-8, 3) and parallel to the line

$$3x - 4y = 4$$

$$-4y = -3x + 4$$

$$y = \frac{3}{4}x - 1$$

↓

$$m = \frac{3}{4}$$

$$y - 3 = \frac{3}{4}(x + 8)$$

$$y - 3 = \frac{3}{4}x + 6$$

$$y = \frac{3}{4}x + 9$$

6. passes through (-2, 9) and parallel to the line passing through (1, -1) and (-2, 14)

$$m = \frac{-1 - 1}{-2 - 1} = \frac{-2}{-3} = \frac{2}{3}$$

$$m = \frac{15}{-3} = -5$$

$$m = -5$$

$$y - 9 = -5(x + 2)$$

$$y - 9 = -5x - 10$$

$$y = -5x - 1$$

7. passes through (4, -1) and perpendicular to the line $x - 2y = 6$

$$-2y = -x + 6$$

$$y = \frac{1}{2}x - 3$$

↓

$$m = -2$$

$$y + 1 = -2(x - 4)$$

$$y + 1 = -2x + 8$$

$$y = -2x + 7$$

8. passes through (8, 9) and perpendicular to the line passing through (-6, 2) and (-1, -2)

$$m = \frac{-2 - 2}{-1 - 6} = \frac{-4}{-7} = \frac{4}{7}$$

$$m = \frac{-4}{5}$$

$$m = \frac{5}{4}$$

$$y - 9 = \frac{5}{4}(x - 8)$$

$$y - 9 = \frac{5}{4}x - 10$$

$$y = \frac{5}{4}x - 1$$

Determine if the given lines are parallel, perpendicular, or neither.

9. $x + y = 8$ and $y = -x - 1$

$$y = -x + 8$$

$$m = -1$$

$$m = -1$$

Parallel

10. $9x - 6y = -20$ and $-3y = 2x - 11$

$$-6y = -9x - 20$$

$$y = \frac{3}{2}x + \frac{10}{3}$$

$$m = \frac{3}{2}$$

$$y = -\frac{2}{3}x + \frac{11}{3}$$

$$m = -\frac{2}{3}$$

Perpendicular

11. $2x = 14 + y$ and $4x + 2y = 10$

$$2x - 14 = y$$

$$m = 2$$

$$2y = -4x + 10$$

$$y = -2x + 5$$

$$m = -2$$

Neither

12. $y = -4$ and $y = 7$

$$m = 0$$

$$m = 0$$

Parallel

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

Linear Equation APPLICATIONS

TYPE I:

Slope-Intercept Form

$$y = mx + b$$

1. A cab ride costs \$4 plus \$1.75 per mile driven.

a) Write a linear equation to represent the cab fare. Define your variables.

 $C = \text{cab fare}$ $m = \text{miles}$

$$C = 1.75m + 4$$

b) If Clay's destination is 17 miles away and he has \$30, will he make it?

$$C = 1.75(17) + 4$$

$$= 29.75 + 4$$

$$= 33.75$$

No, Clay will not have enough money.

2. The Jones' are purchasing a new home. The closing costs are \$249 plus 3% of the purchase price of the home.

a) Write a linear equation to represent the total closing costs. Define your variables.

 $C = \text{closing cost}$ $p = \text{purchase price}$

$$C = 0.03p + 249$$

b) The purchase price of their home is \$195,000. The sellers offered \$5,000 in closing cost assistance. Will this cover all their closing costs? If not, how much will the Jones' need to pay out of pocket?

$$C = 0.03(195000) + 249$$

$$= 5850 + 249$$

$$= 6099$$

No, the Jones' will pay \$1,099.

3. A 22.5-ounce candle burns at the rate of one ounce every five hours.

a) Write an equation to represent the amount of candle left. Define your variables.

 $C = \text{candle left}$ $h = \text{hours}$

$$C = -\frac{1}{5}h + 22.5$$

b) How many hours will the candle last?

$$0 = -\frac{1}{5}h + 22.5$$

$$\frac{1}{5}h = 22.5$$

$$h = 112.5$$

The candle will last 112.5 hours.

4. The water level of a lake is at 80 feet. The lake will be drained at a rate of 3 inches per day each day in September, beginning on September 1.

a) Write an equation to represent the level of the lake each day since it was drained.

 $W = \text{water level}$ $d = \text{days}$

$$W = -.25d + 80$$

b) Find the level of the lake at the end of the day on September 30.

$$W = -.25(30) + 80$$

$$= -7.5 + 80$$

$$= 72.5$$

The lake is at 72.5 ft.

TYPE 2:

Standard Form

$$Ax + By = C$$

5. Josh bought 3 bags of potting soil and 10 bags of mulch from Home Depot.

a) Write an equation to represent the total cost. Define your variables.

p = potting soil

m = mulch

t = total

$$3p + 10m = t$$

b) If Josh spent \$40.91 and mulch cost \$2.75 per bag, how much was a bag of potting soil?

$$3p + 10(2.75) = 40.91$$

$$3p + 27.5 = 40.91$$

$$3p = 13.41$$

$$p = 4.47$$

$$\boxed{\$4.47/\text{bag}}$$

6. Kayleigh has a collection of quarters and nickels.

a) Write an equation to represent the total value of the coins. Define your variables.

q = quarters

n = nickels

t = total

$$.25q + .05n = t$$

b) If the total value of the coins is \$5.50 and Kayleigh has 17 quarters, how many nickels does she have?

$$.25(17) + .05n = 5.50$$

$$4.25 + .05n = 5.50$$

$$.05n = 1.25$$

$$n = 25$$

$$\boxed{25 \text{ nickels}}$$

7. The baseball concession stand sells soft pretzels for \$4.50 and bags of popcorn for \$3.25.

a) Write an equation to represent the total sales from pretzels and popcorn on a given day. Define your variables.

s = soft pretzels

p = popcorn

t = total

$$4.50s + 3.25p = t$$

b) If the concession stand sold 137 soft pretzels, how many bags of popcorn would they need to sell to make at least \$1,000?

$$4.50(137) + 3.25p \geq 1000$$

$$616.50 + 3.25p \geq 1000$$

$$3.25p \geq 383.50$$

$$p \geq 118$$

$$\boxed{\text{At least 118 bags of popcorn.}}$$

8. A math test has two sections: Section A consists of true/false questions worth 2 points each, and Section B consists of multiple choice worth 5 points each.

a) Write an equation to represent the total number of points possible a student could earn on the test. Define your variables.

A = section A

B = section B

t = total

$$2A + 5B = t$$

b) If the highest possible score is 125 points and there are 5 questions in Section A, how many total questions are on the test?

$$2(5) + 5B = 125$$

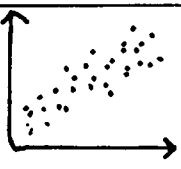
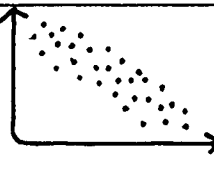
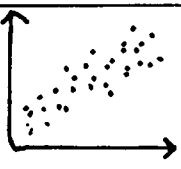
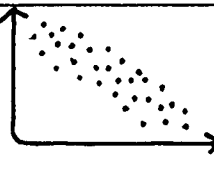
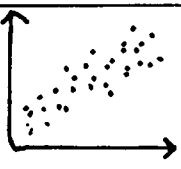
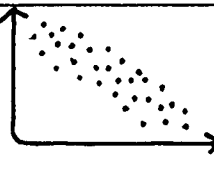
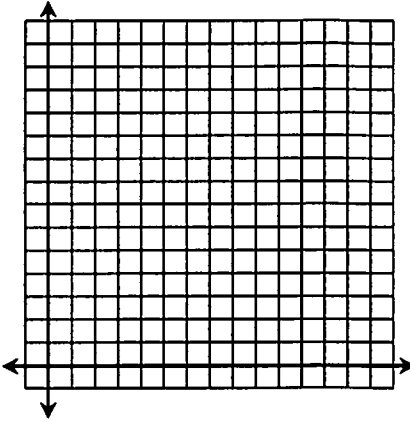
$$10 + 5B = 125$$

$$5B = 115$$

$$B = 23$$

$$\left. \begin{array}{l} A = 5 \text{ questions} \\ B = 23 \text{ questions} \end{array} \right\} \boxed{28 \text{ questions}}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples														
SCATTER PLOTS	A scatter plot shows the relationship between two variables. Sketch the three possible relationships below. <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 33%; padding: 5px;">Positive Correlation</th> <th style="width: 33%; padding: 5px;">Negative Correlation</th> <th style="width: 33%; padding: 5px;">No Correlation</th> </tr> <tr> <td style="text-align: center; padding: 20px;">  </td> <td style="text-align: center; padding: 20px;">  </td> <td style="text-align: center; padding: 20px;">  </td> </tr> </table>	Positive Correlation	Negative Correlation	No Correlation											
	Positive Correlation	Negative Correlation	No Correlation												
															
Line of BEST FIT	Given a positive or negative correlation, it's possible to write a prediction equation, called a line of best fit. It can be used to predict the value of a variable given the value of the other variable.														
EXAMPLE <table border="1" style="margin: 10px auto; border-collapse: collapse;"> <thead> <tr> <th style="padding: 5px;"><i>d</i></th> <th style="padding: 5px;"><i>t</i></th> </tr> </thead> <tbody> <tr><td style="text-align: center;">0</td><td style="text-align: center;">85</td></tr> <tr><td style="text-align: center;">4</td><td style="text-align: center;">82</td></tr> <tr><td style="text-align: center;">7</td><td style="text-align: center;">81</td></tr> <tr><td style="text-align: center;">10</td><td style="text-align: center;">79</td></tr> <tr><td style="text-align: center;">12</td><td style="text-align: center;">77</td></tr> <tr><td style="text-align: center;">15</td><td style="text-align: center;">76</td></tr> </tbody> </table>	<i>d</i>	<i>t</i>	0	85	4	82	7	81	10	79	12	77	15	76	The table to the left shows the relationship between the depth, <i>d</i>, of the lake in feet and the water temperature, <i>t</i>, in degrees Fahrenheit. a) Make a scatter plot.  b) Use two ordered pairs to write a line of best fit. $(0, 85)$ $m = \frac{79 - 85}{10 - 0} = \frac{-6}{10} = -\frac{3}{5}$ $(10, 79)$ $y - 85 = -\frac{3}{5}(x - 0)$ $y - 85 = -\frac{3}{5}x$ $y = -\frac{3}{5}x + 85$ c) Predict the temperature of the lake at 25 feet. $y = -\frac{3}{5}(25) + 85$ $y = -15 + 85$ $y = 70$ 70°
<i>d</i>	<i>t</i>														
0	85														
4	82														
7	81														
10	79														
12	77														
15	76														
LINEAR Regression Calculator Steps →	Another method to find the line of best fit is called linear regression. This can be done through your graphing calculator and will create the most precise equation possible. <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 50%; padding: 5px;">Enter your data.</th> <th style="width: 50%; padding: 5px;">Find the line of best fit.</th> </tr> <tr> <td style="padding: 10px;"> - Hit STAT, ENTER - Enter x-values in L1 - Enter y-values in L2 </td> <td style="padding: 10px;"> - Hit STAT - Arrow over to CALC - Choose #4-LinReg(ax+b) - Hit ENTER (possibly twice) </td> </tr> </table> 	Enter your data.	Find the line of best fit.	- Hit STAT, ENTER - Enter x-values in L1 - Enter y-values in L2	- Hit STAT - Arrow over to CALC - Choose #4-LinReg(ax+b) - Hit ENTER (possibly twice)										
Enter your data.	Find the line of best fit.														
- Hit STAT, ENTER - Enter x-values in L1 - Enter y-values in L2	- Hit STAT - Arrow over to CALC - Choose #4-LinReg(ax+b) - Hit ENTER (possibly twice)														

LINEAR REGRESSION EXAMPLES

1. The table below shows the number of people, p , and suitcases, s , on 8 airplanes.

p	63	115	153	192	224	297	331	375
s	112	209	278	302	379	500	586	712

a) Find the regression equation.

$$y = 1.83x - 16.50$$

b) Predict the number of suitcases on an airplane with 500 people.

$$y = 1.83(500) - 16.50$$

$$y = 898.5$$

About 899 suitcases

2. The table below shows the population, in thousands, during certain years since 2000.

year	0	3	6	9	12	15
population	10.7	11.1	11.9	12.5	13.2	13.6

a) Find the regression equation.

$$y = 0.20x + 10.64$$

b) Predict the population in 2025. $\rightarrow x = 25$

$$y = 0.20(25) + 10.64$$

$$y = 15.64$$

Approx. 15,640 people

3. The table below shows the salary, in millions, made by the highest paid major league baseball player during certain years.

	0	6	8	11	13	16
year	1985	1991	1993	1996	1998	2001
salary	2.1	3.8	6.2	9.3	14.9	22

a) Find the regression equation.

$$y = 1.22x - 1.29$$

b) Predict the salary of the highest paid player in 2020. $\rightarrow x = 35$

$$y = 1.22(35) - 1.29$$

$$y = 41.41$$

Approx. \$41,410,000

4. The table below shows the mileage, m , on six cars that were recently traded in, along with their value, v .

m	5,824	25,824	58,126	73,005	105,092	159,062
v	\$16,920	\$10,300	\$7,225	\$5,300	\$3,125	\$1,800

a) Find the regression equation.

$$y = -0.09x + 13888.67$$

b) Predict the value of a car with 13,258 miles.

$$y = -0.09(13258) + 13888.67$$

$$y = 12695.45$$

About \$12,695.45

Name: _____

Unit 2: Linear Functions & Systems

Date: _____ Per: _____

Homework 4: Applications & Linear Regression

**** This is a 2-page document! ******Linear Equation Applications**

1. Troy's truck has a 30-gallon gas tank and gets an average of 21 miles per gallon.

a) Write an equation to represent the amount of gas in Troy's truck after driving a certain number of miles (assuming he starts with a full tank). Define your variables.

 t = total gas left m = miles

$$t = -\frac{m}{21} + 30$$

b) If the tank is half full, approximately how many miles has Troy driven since last filling up his tank?

 $t = 15$

$$15 = -\frac{m}{21} + 30$$

$$-15 = -\frac{m}{21}$$

$$m = 315$$

315 mi

2. Mikayla is training for a half-marathon. At the beginning of her training, she can run 2 miles without stopping. Her goal is to increase her distance by three quarters of a mile each week.

a) Write an equation to represent the number of miles Mikayla can run after each week. Define your variables.

 t = total miles w = weeks

$$t = \frac{3}{4}w + 2$$

b) If a half-marathon (13.1 miles) is 12 weeks after the start of Mikayla's training program, will she be ready?

$$t = \frac{3}{4}(12) + 2$$

$$t = 9 + 2$$

$$t = 11$$

No, she will only be at 11 mi.

3. The music store sold 12 trumpets and 19 clarinets last month.

a) Write an equation to show the amount of money earned on trumpet and clarinet sales last month. Define your variables.

 S = total sales t = trumpets c = clarinets

$$12t + 19c = S$$

b) If clarinets sell for \$139.95 and the store made \$4,578.45 from sales on the two instruments, find the price of a trumpet.

$$12t + 19(139.95) = 4578.45$$

$$12t + 2659.05 = 4578.45$$

$$12t = 1919.40$$

$$t = 159.95$$

\$159.95

4. Alysha has a collection of dime and nickel rolls that she'd like to take to the bank. Each dime roll has 50 dimes and each nickel roll has 40 nickels.

a) Write an equation to represent the total value of Alysha's dime and nickel collection. Define your variables.

 t = total d = dimes n = nickels

$$5d + 2n = t$$

* 50 dimes = \$5
40 nickels = \$2

b) If the total value of the coins is \$71 and she has 9 rolls of dimes, how many rolls of nickels does she have?

$$5(9) + 2n = 71$$

$$45 + 2n = 71$$

$$2n = 26$$

$$n = 13$$

13 rolls

Linear Regression

5. The table below shows the average price of a movie ticket during certain years.

<i>year</i>	2002	2003	2004	2005	2006	2007	2008	2009
<i>price</i>	5.80	6.03	6.21	6.41	6.55	6.88	7.18	7.35

a) Find the regression equation.

$$y = 0.22x + 5.77$$

b) During what approximate year will the price of a movie ticket reach \$15?

$$15 = 0.22x + 5.77 \rightarrow y = 15$$

$$9.23 = 0.22x$$

$$41.95 = x$$

In the year 2043

6. The table below shows the recorded temperature on a certain day starting at 6:00 a.m.

	0	0.75	1.50	2.50	3	3.75	4.50	5
<i>time</i>	6:00	6:45	7:30	8:30	9:00	9:45	10:30	11:00
<i>temp</i>	65°	67°	68°	72°	74°	75°	77°	80°

a) Find the regression equation.

$$y = 2.91x + 64.6$$

b) Predict the temperature at 2:00 p.m.

$$y = 2.91(8) + 64.6 \rightarrow x = 8$$

$$y = 87.88$$

Approx. 88°

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 2: Linear Functions & Systems

Quiz 2-2: Parallel & Perpendicular Lines, Writing Equations, & Applications

1. Which equations are
- parallel**
- to the line
- $-3x - 6y = 30$
- ?

Check all that apply.

$$-6y = 3x + 30$$

$$y = -\frac{1}{2}x - 5$$

$$m = -\frac{1}{2}$$

☐ $2x - y = 15$	☐ $y = -3x + 1$
☒ $4y = 20 - 2x$	☒ $5x + 10y = 10$
☒ $x = 18 - 2y$	☐ $6 - 2y = 4x$

2. Give an example of a linear equation that is
- perpendicular**
- to
- $y = 5$
- .

2. $x = 5$

↑ * Answers may vary.

Write the equation of the line in slope-intercept form with the given information.

3. Passes through
- $(-8, 7)$
- with a slope of
- $-\frac{1}{4}$

$$y - 7 = -\frac{1}{4}(x + 8)$$

$$y - 7 = -\frac{1}{4}x - 2$$

$$y = -\frac{1}{4}x + 5$$

3. $y = -\frac{1}{4}x + 5$

4. $y = \frac{7}{2}x - 1$

5. $y = 3x - 8$

6. $y = -\frac{2}{5}x + 4$

7. $y = \frac{1}{2}x - 3$

4. Passes through
- $(-2, -8)$
- and parallel to
- $7x - 2y = -8$
- .

$$y + 8 = \frac{7}{2}(x + 2)$$

$$y + 8 = \frac{7}{2}x + 7$$

$$y = \frac{7}{2}x - 1$$

$$-2y = -7x - 8$$

$$y = \frac{7}{2}x + 4$$

5. Passes through
- $(2, -2)$
- and
- $(-3, -17)$

$$m = \frac{-17 + 2}{-3 - 2} = \frac{-15}{-5} = 3$$

$$y + 2 = 3(x - 2)$$

$$y + 2 = 3x - 6$$

$$y = 3x - 8$$

6. Passes through
- $(5, 2)$
- and perpendicular to
- $5x - 2y = -6$
- .

$$y - 2 = -\frac{2}{5}(x - 5)$$

$$-2y = -5x - 6$$

$$y = \frac{5}{2}x + 3$$

$$y - 2 = -\frac{2}{5}x + 2$$

$$y = -\frac{2}{5}x + 4$$

$$\downarrow$$

$$m = -\frac{2}{5}$$

7. Passes through
- $(-8, -7)$
- and perpendicular to the line passing through the points
- $(-6, 3)$
- and
- $(-1, -7)$
- .

$$m = \frac{-7 - 3}{-1 - 6} = \frac{-10}{-5} = 2$$

$$y + 7 = \frac{1}{2}(x + 8)$$

$$y + 7 = \frac{1}{2}x + 4$$

$$y = \frac{1}{2}x - 3$$

8. To manage a rental property, a property manager charges \$59 per month plus 8% of the rent.

- a) Write an equation to represent the total monthly fees charged by the property manager.

$$f = \text{fees}$$

$$r = \text{rent}$$

- b) If Jack pays the property manager \$165 each month, what is the monthly rent on his property?

$$165 = 0.08r + 59$$

$$106 = 0.08r$$

$$r = 1325$$

- c) If Jack raises the rent on the house to \$1,495, how much more will he pay the property manager each month?

$$f = 0.08(1495) + 59$$

$$= 178.6$$

$$178.6 - 165$$

$$= 13.6$$

8a) $f = 0.08r + 59$

b) $\$1,325$

c) $\$13.60$

9. Katya tutors and works in the college bookstore to pay for her college tuition. She makes \$20 per hour tutoring and \$8.50 per hour working in the bookstore.

- a) Write an equation to represent the total amount of money Katya earns tutoring and working in the bookstore.

$$t = \text{total}$$

$$e = \text{tutoring}$$

$$b = \text{bookstore}$$

9a) $20e + 8.50b = t$

b) 9 hours/week

- b) The tuition for one semester (16 weeks) is \$5,736. If Katya works 21 hours each week in the bookstore, how many hours per week must she tutor to pay her tuition?

$$5736 = 20e + 8.5(336)$$

$$5736 = 20e + 2856$$

$$2880 = 20e$$

$$144 = e$$

$$\frac{144 \text{ hr}}{16 \text{ wk}} = 9 \text{ hr/wk}$$

10. The table below shows the average price of a personal computer during certain years.

	0	2	6	10	13	15
year	1997	1999	2003	2007	2010	2012
price	\$1,950	\$1,719	\$1,420	\$1,175	\$949	\$725

- a) Write an equation to model the data using linear regression.

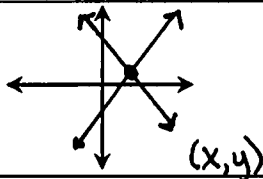
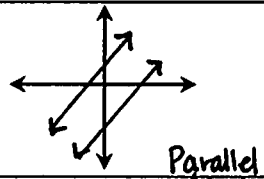
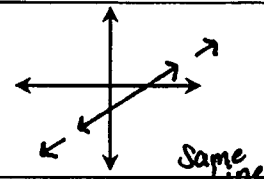
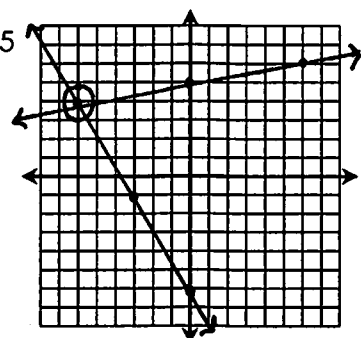
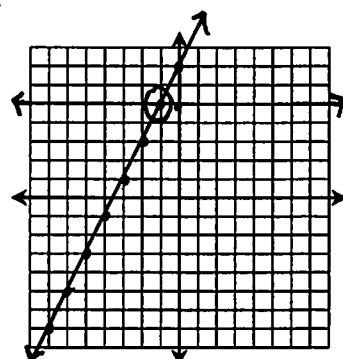
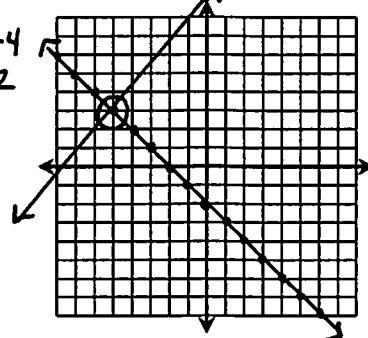
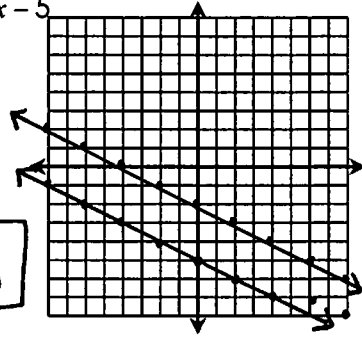
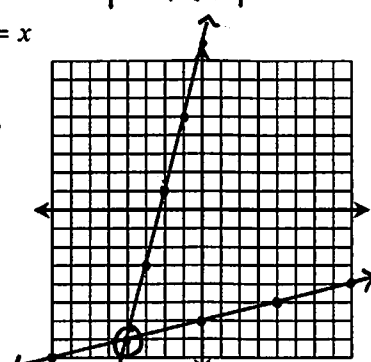
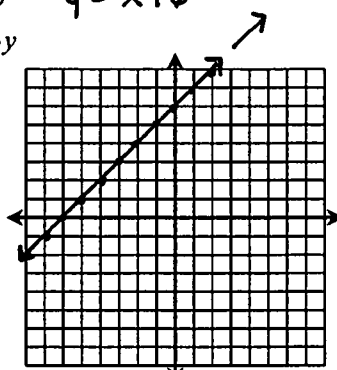
10a) $y = -76.86x + 1912.29$

b) $\$144.51$

- b) Use the equation to predict the average cost of a personal computer in 2020. $\rightarrow x = 23$

$$y = -76.86(23) + 1912.29$$

$$y = 144.51$$

Name:		Date:	
Topic:		Class:	
Main Ideas/Questions	Notes/Examples		
SYSTEMS OF EQUATIONS	Two or more linear equations		
Types of SOLUTIONS			
	One Solution	No Solution	Infinite Solution
Solving by GRAPHING	Solve each system of equations below by graphing. Identify the solution.		
	1. $\begin{cases} y = -\frac{5}{3}x - 6 \\ y = \frac{1}{6}x + 5 \end{cases}$  $(-6, 4)$	2. $\begin{cases} y = 5 \\ y = 2x + 7 \end{cases}$  $(-1, 5)$	
	3. $\begin{cases} 6x - 5y = -45 \\ 2x + 2y = -4 \end{cases}$ $\begin{aligned} -5y &= -6x - 45 \\ y &= \frac{6}{5}x + 9 \end{aligned}$ $\begin{aligned} 2y &= -2x - 4 \\ y &= -x - 2 \end{aligned}$  $(-5, 3)$	4. $\begin{cases} 3x + 6y = -12 \\ y = -\frac{1}{2}x - 5 \end{cases}$ $\begin{aligned} 6y &= -3x - 12 \\ y &= -\frac{1}{2}x - 2 \end{aligned}$  No Solution	
	5. $\begin{cases} 2y = 8x + 18 \\ 24 + 4y = x \end{cases}$ $\begin{aligned} y &= 4x + 9 \\ 4y &= x - 24 \\ y &= \frac{1}{4}x - 6 \end{aligned}$  $(-4, -7)$	6. $\begin{cases} -y = -x - 6 \\ 3x + 18 = 3y \end{cases}$ $\begin{aligned} y &= x + 6 \\ x + 6 &= y \end{aligned}$  Infinite Solutions	

Solving by SUBSTITUTION

- ① **SOLVE** one equation for x or y . (Isolate a variable)
- ② **SUBSTITUTE** the resulting equation from step 1 into the other equation for that variable.
- ③ **SOLVE** for the remaining variable.
- ④ **SUBSTITUTE** your answer from step 3 into either original equation to find the other variable.

Solve each system of equations below by substitution. Identify the solution.

7. $\begin{cases} y = -7x - 1 \\ y = x - 9 \end{cases}$

$$x - 9 = -7x - 1$$

$$8x - 9 = -1$$

$$8x = 8$$

$$x = 1$$

$$y = 1 - 9$$

$$y = -8$$

$$(1, -8)$$

8. $\begin{cases} y = -5x + 30 \\ 7x + 3y = 42 \end{cases}$

$$7x + 3(-5x + 30) = 42$$

$$7x - 15x + 90 = 42$$

$$-8x + 90 = 42$$

$$-8x = -48$$

$$x = 6$$

$$y = -5(6) + 30$$

$$y = 0$$

$$(6, 0)$$

9. $\begin{cases} 6x - 5y = -28 \\ 7x + y = 22 \end{cases} \rightarrow y = -7x + 22$

$$6x - 5(-7x + 22) = -28$$

$$6x + 35x - 110 = -28$$

$$41x = 82$$

$$x = 2$$

$$y = -7(2) + 22$$

$$y = 8$$

$$(2, 8)$$

10. $\begin{cases} x - 7y = 53 \rightarrow x = 7y + 53 \\ -4x - 5y = 19 \end{cases}$

$$-4(7y + 53) - 5y = 19$$

$$-28y - 212 - 5y = 19$$

$$-33y = 231$$

$$y = -7$$

$$x = 7(-7) + 53$$

$$x = 4$$

$$(4, -7)$$

11. $\begin{cases} 2y = 6x + 10 \rightarrow y = 3x + 5 \\ 3x - y = 5 \end{cases}$

$$3x - (3x + 5) = 5$$

$$3x - 3x - 5 = 5$$

$$-5 \neq 5$$

No Solution

12. $\begin{cases} 5x + 7y = -17 \\ 4x - 3y = -5 \end{cases} \rightarrow \begin{aligned} -3y &= -4x - 5 \\ y &= \frac{4}{3}x + \frac{5}{3} \end{aligned}$

$$5x + 7\left(\frac{4}{3}x + \frac{5}{3}\right) = -17$$

$$5x + \frac{28}{3}x + \frac{35}{3} = -17$$

$$\frac{43}{3}x + \frac{35}{3} = -17$$

$$43x + 35 = -51$$

$$43x = -86$$

$$x = -2$$

$$y = \frac{4}{3}(-2) + \frac{5}{3}$$

$$y = -\frac{3}{3}$$

$$y = -1$$

$$(-2, -1)$$

Name: _____

Unit 2: Linear Functions & Systems

Date: _____ Per: _____

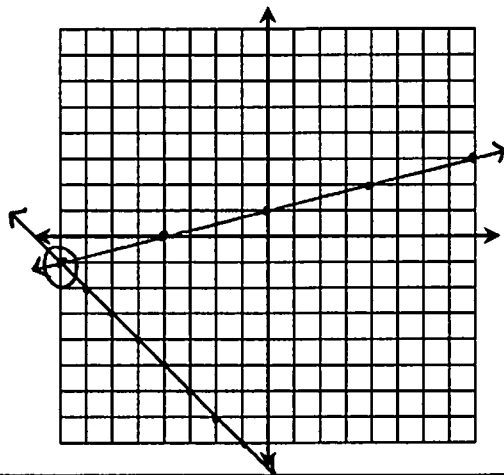
Homework 5: Solving Systems of Equations
by Graphing & Substitution

**** This is a 2-page document! ****

Solve each system of equations by graphing. Clearly identify your solution.

1.
$$\begin{cases} y = \frac{1}{4}x + 1 \\ y = -x - 9 \end{cases}$$

$(-8, -1)$

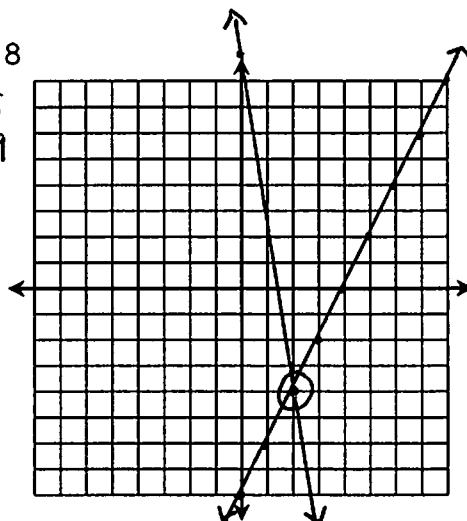


2.
$$\begin{cases} y = 2x - 8 \\ 13x + 2y = 18 \end{cases}$$

$$2y = -13x + 18$$

$$y = -\frac{13}{2}x + 9$$

$(2, -4)$



3.
$$\begin{cases} 3x + 7y = -63 \\ x - y = -1 \end{cases}$$

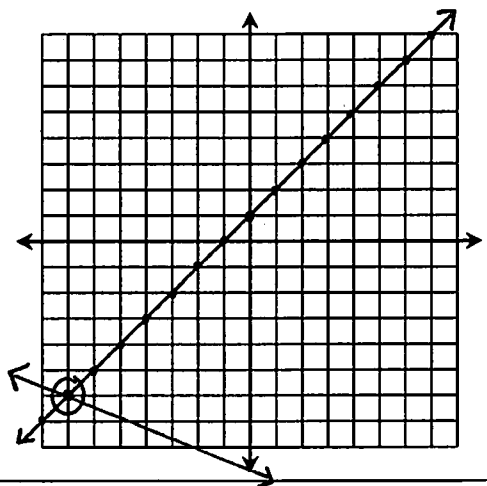
$$7y = -3x - 63$$

$$y = -\frac{3}{7}x - 9$$

$$-y = -x - 1$$

$$y = x + 1$$

$(-7, -6)$

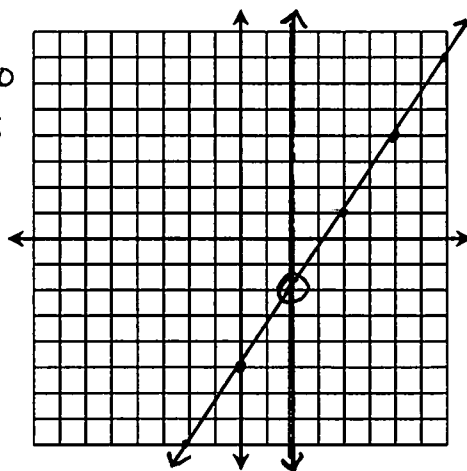


4.
$$\begin{cases} 3x - 2y = 10 \\ x = 2 \end{cases}$$

$$-2y = -3x + 10$$

$$y = \frac{3}{2}x - 5$$

$(2, -2)$



5.
$$\begin{cases} -6y = 36 - x \\ 4x - 3y = -3 \end{cases}$$

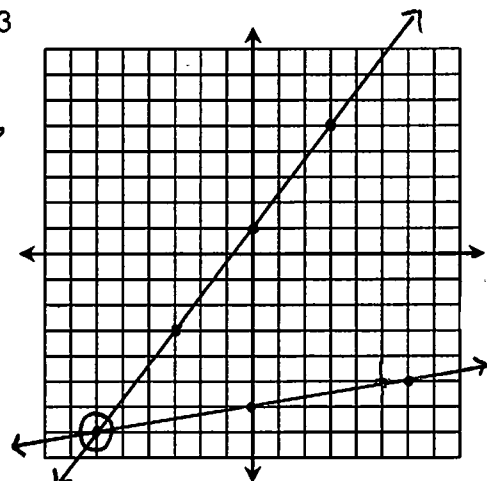
$$-6y = -x + 36$$

$$y = \frac{1}{6}x - 6$$

$$-3y = -4x - 3$$

$$y = \frac{4}{3}x + 1$$

$(-6, -7)$



6.
$$\begin{cases} -12x = 30y + 60 \\ 2x + 5y = 20 \end{cases}$$

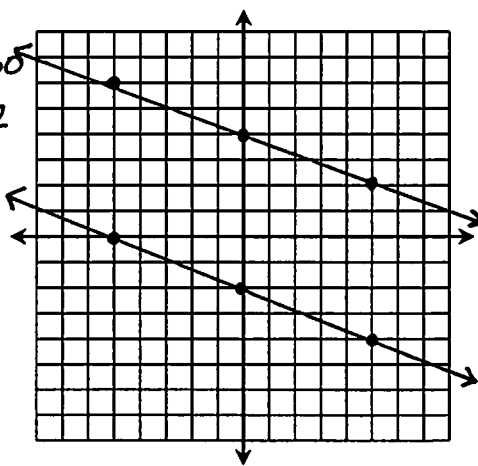
$$30y = -12x - 60$$

$$y = -\frac{2}{5}x - 2$$

$$5y = -2x + 20$$

$$y = -\frac{2}{5}x + 4$$

No
Solution



Solve each system of equations by substitution. Clearly identify your solution.

7. $\begin{cases} y = -2x + 14 \\ y = 5x - 7 \end{cases}$

$$-2x + 14 = 5x - 7$$

$$21 = 7x$$

$$3 = x$$

$$y = 5(3) - 7$$

$$y = 8$$

$$(3, 8)$$

8. $\begin{cases} y = -4x - 9 \\ 7x + 4y = -27 \end{cases}$

$$7x + 4(-4x - 9) = -27$$

$$7x - 16x - 36 = -27$$

$$-9x - 36 = -27$$

$$-9x = 9$$

$$x = -1$$

$$y = -4(-1) - 9$$

$$y = 4 - 9$$

$$y = -5$$

$$(-1, -5)$$

9. $\begin{cases} x - y = -9 \rightarrow x = y - 9 \\ 8x - 3y = -62 \end{cases}$

$$8(y - 9) - 3y = -62$$

$$8y - 72 - 3y = -62$$

$$5y = 10$$

$$y = 2$$

$$x - 2 = -9$$

$$x = -7$$

$$(-7, 2)$$

10. $\begin{cases} 7x - y = 34 \rightarrow y = 7x - 34 \\ 2x + 3y = -10 \end{cases}$

$$2x + 3(7x - 34) = -10$$

$$2x + 21x - 102 = -10$$

$$23x = 92$$

$$x = 4$$

$$7(4) - y = 34$$

$$-y = 6$$

$$y = -6$$

$$(4, -6)$$

11. $\begin{cases} 2x = 16 - 8y \rightarrow x = 8 - 4y \\ x + 4y = 25 \end{cases}$

$$(8 - 4y) + 4y = 25$$

$$8 \neq 25$$

$$\text{No Solution}$$

12. $\begin{cases} -\frac{2}{3}x - 7y = 6 \\ x + 3y = -9 \rightarrow x = -3y - 9 \end{cases}$

$$-\frac{2}{3}(-3y - 9) - 7y = 6$$

$$2y + 6 - 7y = 6$$

$$-5y = 0$$

$$y = 0$$

$$x + 3(0) = -9$$

$$x = -9$$

$$(-9, 0)$$

13. $\begin{cases} 2x - 9y = 26 \\ 6x - 2y = -22 \rightarrow 6x + 22 = 2y \\ 3x + 11 = y \end{cases}$

$$2x - 9(3x + 11) = 26$$

$$2x - 27x - 99 = 26$$

$$-25x = 125$$

$$x = -5$$

$$2(-5) - 9y = 26$$

$$-9y = 36$$

$$y = -4$$

$$(-5, -4)$$

14. $\begin{cases} 3x - 2y = 10 \rightarrow -2y = -3x + 10 \\ 9x + 4y = -50 \end{cases} \quad y = \frac{3}{2}x - 5$

$$9x + 4(\frac{3}{2}x - 5) = -50$$

$$9x + 6x - 20 = -50$$

$$15x = -30$$

$$x = -2$$

$$3(-2) - 2y = 10$$

$$-6 - 2y = 10$$

$$-2y = 16$$

$$y = -8$$

$$(-2, -8)$$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

Solving by
ELIMINATION

- ① **LINE UP** the equations.
- ② **MULTIPLY** one or both equations by a number to result in a variable with the same coefficient.
- ③ **ADD** or **SUBTRACT** the equations to eliminate this variable.
- ④ **SOLVE** for the remaining variable.
- ⑤ **SUBSTITUTE** your answer from step 4 into either original equation to find the other variable.

Solve each system of equations below by elimination. Identify the solution.

$$\begin{array}{rcl}
 1. \begin{cases} x+7y=17 \\ -(x-y=-7) \end{cases} & & \begin{array}{l} x-3=-7 \\ x=-4 \end{array} \\
 \hline
 8y=24 & & \\
 y=3 & &
 \end{array}$$

$(-4, 3)$

$$\begin{array}{rcl}
 2. \begin{cases} 3x+2y=22 \\ + 5x-2y=42 \end{cases} & & \begin{array}{l} 3(8)+2y=22 \\ 24+2y=22 \\ 2y=-2 \\ y=-1 \end{array} \\
 \hline
 8x=64 & & \\
 x=8 & &
 \end{array}$$

$(8, -1)$

$$\begin{array}{rcl}
 3. \begin{cases} 4x-9y=-42 \\ -4(x+5y=4) \end{cases} & & \begin{array}{l} x+5(2)=4 \\ x+10=4 \\ x=-6 \end{array} \\
 \hline
 4x-9y=-42 & & \\
 + -4x-20y=-16 & & \\
 \hline
 -29y=-58 & & \\
 y=2 & &
 \end{array}$$

$(-6, 2)$

$$\begin{array}{rcl}
 4. \begin{cases} 7x-6y=-53 \\ -2(2x-3y=-13) \end{cases} & & \begin{array}{l} 7(-9)-6y=-53 \\ -63-6y=-53 \\ -6y=10 \\ y=-\frac{5}{3} \end{array} \\
 \hline
 7x-6y=-53 & & \\
 + -4x+6y=26 & & \\
 \hline
 3x=-27 & & \\
 x=-9 & &
 \end{array}$$

$(-9, -\frac{5}{3})$

$$\begin{array}{rcl}
 5. \begin{cases} (5x+3y=-7) \cdot 2 \\ (2x+7y=3) \cdot -5 \end{cases} & & \begin{array}{l} 5x+3(1)=-7 \\ 5x+3=-7 \\ 5x=-10 \\ x=-2 \end{array} \\
 \hline
 10x+6y=-14 & & \\
 + -10x-35y=-15 & & \\
 \hline
 -29y=-29 & & \\
 y=1 & &
 \end{array}$$

$(-2, 1)$

$$\begin{array}{rcl}
 6. \begin{cases} (3x-9y=9) \cdot 4 \\ (4x-12y=36) \cdot -3 \end{cases} & & \\
 \hline
 12x-36y=36 & & \\
 + -12x+36y=-108 & & \\
 \hline
 0 \neq -72 & &
 \end{array}$$

No Solution

$$7. \begin{cases} 5x - 6y = 3 \\ 7y = 2x + 8 \end{cases} \quad \begin{cases} (5x - 6y = 3) \cdot 7 \\ (-2x + 7y = 8) \cdot 5 \end{cases}$$

$$\begin{array}{r} 10x - 12y = 6 \\ + -10x + 35y = 40 \\ \hline 23y = 46 \\ y = 2 \\ \hline 7(2) = 2x + 8 \\ 14 = 2x + 8 \\ 6 = 2x \\ 3 = x \end{array} \quad \boxed{(3, 2)}$$

$$8. \begin{cases} 10x + 18 = -8y \\ 4 + 9x = 5y \end{cases} \quad \begin{cases} (10x + 18 = -8y) \cdot 5 \\ (9x - 5y = -4) \cdot 8 \end{cases}$$

$$\begin{array}{r} 50x + 40y = -90 \\ + 72x - 40y = -32 \\ \hline 122x = -122 \\ x = -1 \\ \hline 4 + 9(-1) = 5y \\ 4 - 9 = 5y \\ -5 = 5y \\ y = -1 \end{array} \quad \boxed{(-1, -1)}$$

APPLICATIONS

- ① **DEFINE YOUR VARIABLES** – What are you solving for?
- ② **SET UP TWO EQUATIONS** using the information given.
- ③ **SOLVE** the system using your method of choice.

9. Marcie bought a total of 20 used books and CDs during a yard sale for a total of \$54.50. If books cost \$1.50 each and CDs cost \$5 each, how many of each did she buy?

$$\begin{array}{l} b = \text{books} \\ c = \text{CDs} \end{array} \quad \begin{array}{l} b + c = 20 \\ 1.50b + 5c = 54.50 \end{array} \Rightarrow \begin{array}{r} -5b - 5c = -100 \\ + 1.5b + 5c = 54.5 \\ \hline -3.5b = -45.5 \\ b = 13 \\ \hline 13 + c = 20 \\ c = 7 \end{array}$$

$\boxed{13 \text{ books and } 7 \text{ CDs}}$

10. Landon babysits and works part time at the water park over the summer. One week, he babysat for 3 hours and worked at the water park for 10 hours and made \$109. The next week he babysat for 8 hours and worked at the water park for 12 hours and made \$177. How much does Landon make per hour at each job?

$$\begin{array}{l} b = \text{babysits} \\ w = \text{waterpark} \end{array} \quad \begin{array}{l} 3b + 10w = 109 \\ 8b + 12w = 177 \end{array} \Rightarrow \begin{array}{r} 24b + 80w = 872 \\ + -24b - 36w = -531 \\ \hline 44w = 341 \\ w = 7.75 \\ \hline 3b + 10(7.75) = 109 \\ 3b = 31.5 \\ b = 10.5 \end{array}$$

$\boxed{\$7.75/\text{hr at the waterpark and } \$10.50 \text{ babysitting.}}$

11. Kent has a collection of pennies and nickels with a value of \$1.98. The number of pennies he has is five less than twice the number of nickels. How many of each coin does Kent have?

$$\begin{array}{l} p = \text{pennies} \\ n = \text{nickels} \end{array} \quad \begin{array}{l} .01p + .05n = 1.98 \\ p = 2n - 5 \end{array} \Rightarrow \begin{array}{r} .01(2n - 5) + .05n = 1.98 \\ .02n - .05 + .05n = 1.98 \\ .07n - .05 = 1.98 \\ .07n = 2.03 \\ n = 29 \\ \hline p = 2(29) - 5 \\ p = 53 \end{array}$$

$\boxed{53 \text{ pennies and } 29 \text{ nickels}}$

MORE APPLICATIONS PRACTICE *with* SYSTEMS

Define your variables, set up a system of equations, and solve each problem.

1. At the fast food restaurant, four cheeseburgers and five small fries have a total of 2,310 calories. Three cheeseburgers and two small fries have a total of 1,330 calories. How many calories does each item contain?

$C = \text{cheeseburgers}$
 $f = \text{fries}$

$$\begin{aligned} 4c + 5f &= 2310 \\ 3c + 2f &= 1330 \end{aligned} \Rightarrow \begin{aligned} 12c + 15f &= 6930 \\ + -12c - 8f &= -5320 \\ \hline 7f &= 1610 \\ f &= 230 \end{aligned}$$

Cheeseburger = 290 cal.
and fries = 230 cal.

$$\begin{aligned} 3c + 2(230) &= 1330 \\ 3c &= 870 \\ c &= 290 \end{aligned}$$

2. One month, a homeowner used 150 units of gas and 520 units of electric for a total cost of \$84.20. The next month, 210 units of gas and 405 units of electric were used for a total cost of \$82.35. Find the cost per unit of gas and electric.

$g = \text{gas}$
 $e = \text{electric}$

$$\begin{aligned} 150g + 520e &= 84.20 \\ 210g + 405e &= 82.35 \end{aligned} \Rightarrow \begin{aligned} 1050g + 3640e &= 589.4 \\ + -1050g - 2025e &= -411.75 \\ \hline 1615e &= 177.65 \\ e &= 0.11 \end{aligned}$$

Electric costs \$0.11/unit +
gas costs \$0.18/unit

$$\begin{aligned} 150g + 520(0.11) &= 84.20 \\ 150g &= 27 \\ g &= 0.18 \end{aligned}$$

3. Kym is renovating the first floor of her home. She bought 750 square feet of laminate flooring and 525 square feet of carpet and paid \$2717.25. She went back to the store and bought an additional 100 square feet of laminate flooring and 75 square feet of carpet and paid \$374.75. Find the cost per square foot of each type of flooring.

$L = \text{laminate}$
 $C = \text{carpet}$

$$\begin{aligned} 750L + 525C &= 2717.25 \\ 100L + 75C &= 374.75 \end{aligned} \Rightarrow \begin{aligned} 750L + 525C &= 2717.25 \\ + -700L - 525C &= -2623.25 \\ \hline 50L &= 94 \\ L &= 1.88 \end{aligned}$$

Laminate costs \$1.88/sq.ft.
and Carpet costs \$2.49/sq.ft.

$$\begin{aligned} 100(1.88) + 75C &= 374.75 \\ 75C &= 186.75 \\ C &= 2.49 \end{aligned}$$

4. Sean bought 1.8 pounds of gummy bears and 0.6 pounds of jelly beans and paid \$10.26. He went back to the store the following week and bought 1.2 pounds of gummy bears and 1.5 pounds of jellybeans and paid \$15.09. What is the price per pound of each type of candy?

$g = \text{gummy bears}$
 $j = \text{jelly beans}$

$$\begin{aligned} 1.8g + 0.6j &= 10.26 \\ 1.2g + 1.5j &= 15.09 \end{aligned} \Rightarrow \begin{aligned} 9g + 3j &= 51.30 \\ + -2.4g - 3j &= -30.18 \\ \hline 6.6g &= 21.12 \\ g &= 3.20 \end{aligned}$$

Gummy bears cost \$3.20/lb
and jelly beans cost \$7.50/lb

$$\begin{aligned} 1.8(3.20) + 0.6j &= 10.26 \\ 0.6j &= 4.50 \\ j &= 7.50 \end{aligned}$$

5. Courtney has a total of 88 stamps worth \$15.56. Some are 25¢ stamps and some are 2¢ stamps. How many of each does she have?

$x = 25¢ \text{ stamps}$
 $y = 2¢ \text{ stamps}$

60 - 25¢ stamps
 and
 28 - 2¢ stamps

$$\begin{aligned} x + y &= 88 \rightarrow x = 88 - y \\ .25x + .02y &= 15.56 \\ .25(88 - y) + .02y &= 15.56 \\ 22 - .25y + .02y &= 15.56 \\ -.23y &= -6.44 \\ y &= 28 \end{aligned}$$

$$\begin{aligned} x + 28 &= 88 \\ x &= 60 \end{aligned}$$

6. Tom put 18 gallons of midgrade gas in his truck and filled up his empty five-gallon gas can with regular gas for his lawnmower at home. He spent \$59.91. The following week, he put 14 gallons of midgrade gas in his truck and topped off his five-gallon can with just one gallon of regular gas. If he paid \$39.75 and the prices remained the same, find the price per gallon of midgrade and regular gas.

$m = \text{midgrade}$
 $r = \text{regular}$

Regular costs \$2.37/gal
 and Midgrade costs \$2.67/gal

$$\begin{aligned} 18m + 5r &= 59.91 \\ 14m + r &= 39.75 \end{aligned} \Rightarrow \begin{aligned} 18m + 5r &= 59.91 \\ + -70m - 5r &= -198.75 \\ \hline -52m &= -138.84 \end{aligned}$$

$$\begin{aligned} m &= 2.67 \\ 14(2.67) + r &= 39.75 \\ r &= 2.37 \end{aligned}$$

7. A total of \$12,000 was invested in two types of bonds. One pays 8% simple interest while the other pays 10.5%. Last year, the annual interest earned on the two investments was \$1,145. How much was invested at each rate?

$x = \text{account 1}$
 $y = \text{account 2}$

\$4,600 at 8%
 and \$7400 at 10.5%

$$\begin{aligned} x + y &= 12000 \rightarrow x = 12000 - y \\ .08x + .105y &= 1145 \\ .08(12000 - y) + .105y &= 1145 \\ 960 - .08y + .105y &= 1145 \\ .025y &= 185 \\ y &= 7400 \end{aligned}$$

$$\begin{aligned} x + 7400 &= 12000 \\ x &= 4600 \end{aligned}$$

8. A collection of dimes and quarters is worth \$9.55. If the quarters were dimes and the dimes were quarters, the total value would be \$7.60. Find the number of each coin.

$d = \text{dimes}$
 $q = \text{quarters}$

18 dimes and
 31 quarters

$$\begin{aligned} .10d + .25q &= 9.55 \\ .25d + .10q &= 7.60 \end{aligned} \Rightarrow \begin{aligned} .50d + 1.25q &= 47.75 \\ + - .50d - .20q &= -15.20 \\ \hline 1.05q &= 32.55 \\ q &= 31 \\ .10d + .25(31) &= 9.55 \\ .10d &= 1.80 \\ d &= 18 \end{aligned}$$

Name: _____

Unit 2: Linear Functions & Systems

Date: _____ Per: _____

Homework 6: Solving Systems of Equations
by Elimination & Applications**** This is a 2-page document! ****

Solve each system of equations by elimination. Clearly identify your solution.

1.
$$\begin{cases} 3x + y = 11 \\ 5x - y = 21 \end{cases}$$

$$\begin{array}{r} 8x = 32 \\ x = 4 \end{array}$$

$$\begin{array}{r} 3(4) + y = 11 \\ 12 + y = 11 \\ y = -1 \end{array}$$

 $(4, -1)$

2.
$$\begin{cases} 2x - 9y = -51 \\ 2x - 3y = -9 \end{cases}$$

$$\begin{array}{r} -6y = -42 \\ y = 7 \end{array}$$

$$\begin{array}{r} 2x - 3(7) = -9 \\ 2x - 21 = -9 \\ 2x = 12 \\ x = 6 \end{array}$$

 $(6, 7)$

3.
$$\begin{cases} 5x + 12y = -34 \\ (2x + 4y = -12) \cdot 3 \end{cases}$$

$$\begin{array}{r} 5x + 12y = -34 \\ -(6x + 12y = -36) \\ \hline -x = 2 \\ x = -2 \end{array}$$

$$\begin{array}{r} 2(-2) + 4y = -12 \\ -4 + 4y = -12 \\ 4y = -8 \\ y = -2 \end{array}$$

 $(-2, -2)$

4.
$$\begin{cases} -3x + 8y = 73 \\ (x - 7y = -46) \cdot 3 \end{cases}$$

$$\begin{array}{r} -3x + 8y = 73 \\ + 3x - 21y = -138 \\ \hline -13y = -65 \\ y = 5 \end{array}$$

$$\begin{array}{r} x - 7(5) = -46 \\ x - 35 = -46 \\ x = -11 \end{array}$$

 $(-11, 5)$

5.
$$\begin{cases} (9x + 4y = -72) \cdot 2 \\ (2x - 3y = -16) \cdot 9 \end{cases}$$

$$\begin{array}{r} 18x + 8y = -144 \\ -(18x - 27y = -144) \\ \hline -35y = 0 \\ y = 0 \end{array}$$

$$\begin{array}{r} 2x - 3(0) = -16 \\ 2x = -16 \\ x = -8 \end{array}$$

 $(-8, 0)$

6.
$$\begin{cases} (4x - 11y = 68) \cdot 3 \\ (6x + 5y = -27) \cdot 2 \end{cases}$$

$$\begin{array}{r} 12x - 33y = 204 \\ -(12x + 10y = -54) \\ \hline -43y = 258 \\ y = -6 \end{array}$$

$$\begin{array}{r} 6x + 5(-6) = -27 \\ 6x - 30 = -27 \\ 6x = 3 \\ x = \frac{1}{2} \end{array}$$

 $(\frac{1}{2}, -6)$

7.
$$\begin{cases} 5x = 15 - 5y \rightarrow (5x + 5y = 15) \cdot 2 \\ 8y + 26 = 2x \rightarrow (2x - 8y = 26) \cdot 5 \end{cases}$$

$$\begin{array}{r} 10x + 10y = 30 \\ -(10x - 40y = 130) \\ \hline 50y = -100 \\ y = -2 \end{array}$$

$$\begin{array}{r} 5x = 15 - 5(-2) \\ 5x = 15 + 10 \\ 5x = 25 \\ x = 5 \end{array}$$

 $(5, -2)$

8.
$$\begin{cases} 14x + 7y = -7 \\ y = -2x - 1 \rightarrow (2x + y = -1) \cdot 7 \end{cases}$$

$$\begin{array}{r} 14x + 7y = -7 \\ -(14x + 7y = -7) \\ \hline 0 = 0 \end{array}$$

Infinite Solutions

Define your variables and set up a system of equations, then solve.

9. A storeowner mixed 8 pounds of peanuts and 5 pounds of M&M's. This 13-pound mixture sold for \$55.27. The second mixture included 6 pounds of peanuts and 4 pounds of M&M's. This 10-pound mixture sold for \$42.70. Find the cost per pound of the peanuts and M&M's.

$$\begin{array}{l} x = \text{peanuts} \\ y = \text{m+ms} \end{array} \quad \begin{array}{l} (8x + 5y = 55.27) \cdot 4 \\ (6x + 4y = 42.70) \cdot 5 \end{array} \Rightarrow \begin{array}{r} 32x + 20y = 221.08 \\ -(30x + 20y = 213.50) \end{array}$$

$$2x = 7.58$$

$$x = 3.79$$

$$8(3.79) + 5y = 55.27$$

$$5y = 24.95$$

$$y = 4.99$$

Peanuts are \$3.79/lb.
and M+Ms are \$4.99/lb.

10. Katy's favorite rides at the amusement park are the roller coaster and water slide. The wait time for the roller coaster is 25 minutes and the wait time for the water slide is 10 minutes. If she went on 12 rides total and waited three hours in line, how many times did she go on each ride?

$$\begin{array}{l} x = \text{rollercoaster} \\ y = \text{waterslide} \end{array}$$

$$x + y = 12 \rightarrow y = 12 - x$$

$$25x + 10y = 180$$

$$25x + 10(12 - x) = 180$$

$$25x + 120 - 10x = 180$$

$$15x + 120 = 180$$

$$15x = 60$$

$$x = 4$$

$$\begin{array}{l} 4 + y = 12 \\ y = 8 \end{array}$$

4 times on the
rollercoaster and
8 times on waterslide

11. A collection of nickels and dimes is worth \$9.45. If the number of dimes is doubled, the value is \$16.65. Find the number of each coin.

$$\begin{array}{l} x = \text{nickels} \\ y = \text{dimes} \end{array}$$

$$\begin{array}{r} .05x + .10y = 9.45 \\ -(.05x + .20y = 16.65) \end{array}$$

$$-.10y = -7.20$$

$$y = 72$$

$$.05x + .10(72) = 9.45$$

$$.05x = 2.25$$

$$x = 45$$

45 nickels and
72 dimes

12. Zack has two savings accounts with a total of \$9,000. He withdrew 10% from one and 60% from the other to buy his girlfriend an engagement ring. If the ring cost \$2,175, find the remaining balance in each account.

$$\begin{array}{l} x = \text{account A} \\ y = \text{account B} \end{array}$$

$$x + y = 9000 \rightarrow y = 9000 - x$$

$$.10x + .60y = 2175$$

$$.10x + .60(9000 - x) = 2175$$

$$.10x + 5400 - .60x = 2175$$

$$-.5x = -3225$$

$$x = 6450$$

$$6450 + y = 9000$$

$$y = 2550$$

$$6450(.9) = 5805$$

$$2550(.4) = 1020$$

\$5805 in acct A
and \$1020 in acct B

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
THREE- VARIABLE <i>Systems</i>	1 CHOOSE TWO EQUATIONS in which you can easily eliminate a variable. Let's call this new equation after the elimination Equation A .
	2 ELIMINATE THE SAME VARIABLE from a combination of the equation you didn't use and one of the ones you did. Let's call this equation after the elimination Equation B .
	3 PUT EQUATION A AND EQUATION B TOGETHER and SOLVE THAT SYSTEM . You now have two of your variables!
	4 SUBSTITUTE your answers from step 3 into either original equation to find the remaining variable. Write your solution as (x, y, z) .

EXAMPLES

1. $\left\{ \begin{array}{l} 2x - 6y + 9z = -8 \\ 5x + y + 2z = 10 \\ 3x + y - 8z = -28 \end{array} \right\} \begin{array}{l} A \\ B \end{array}$

A: $2x - 6y + 9z = -8$
 $(5x + y + 2z = 10) \cdot 6$
 $\underline{2x - 6y + 9z = -8}$
 $+ 30x + 6y + 12z = 60$
 $\underline{32x + 21z = 52}$

B: $2x - 6y + 9z = -8$
 $(3x + y - 8z = -28) \cdot 6$
 $\underline{2x - 6y + 9z = -8}$
 $+ 18x + 6y - 48z = -168$
 $\underline{20x - 39z = -176}$

$32x + 21(4) = 52$
 $32x + 84 = 52$
 $32x = -32$
 $x = -1$
 $5(-1) + y + 2(4) = 10$
 $y + 3 = 10$
 $y = 7$

$(32x + 21z = 52) \cdot 5$
 $(20x - 39z = -176) \cdot -8$
 $\underline{160x + 105z = 260}$
 $+ -160x + 312z = 1408$
 $\underline{417z = 1668}$
 $z = 4$

$(-1, 7, 4)$

2. $\left\{ \begin{array}{l} 4x + y - 3z = 45 \\ 6x - 4y + 3z = 41 \\ 2x - 7y - 11z = 85 \end{array} \right\} \begin{array}{l} A \\ B \end{array}$

A: $4x + y - 3z = 45$
 $+ 6x - 4y + 3z = 41$
 $\underline{10x - 3y = 86}$

B: $(6x - 4y + 3z = 41) \cdot 11$
 $(2x - 7y - 11z = 85) \cdot 3$
 $\underline{66x - 44y + 33z = 451}$
 $+ 6x - 21y - 33z = 255$
 $\underline{72x - 65y = 706}$

$10x - 3(-2) = 86$
 $10x + 6 = 86$
 $10x = 80$
 $x = 8$
 $4(8) + (-2) - 3z = 45$
 $30 - 3z = 45$
 $-3z = 15$
 $z = -5$

$36(10x - 3y = 86)$
 $-5(72x - 65y = 706)$
 $\underline{360x - 108y = 3096}$
 $- 360x + 325y = -3530$
 $217y = -434$
 $y = -2$

$(8, -2, -5)$

$$3. \begin{cases} 5x - 3y + 8z = -59 \\ 9x - 4y - z = -60 \\ 5x + 2y + 4z = -47 \end{cases} \begin{matrix} A \\ B \end{matrix}$$

$$\begin{array}{r} A: 5x - 3y + 8z = -59 \\ (9x - 4y - z = -60) \cdot 8 \\ \hline 5x - 3y + 8z = -59 \\ + 72x - 32y - 8z = -480 \\ \hline 77x - 35y = -539 \end{array}$$

$$\begin{array}{r} B: (9x - 4y - z = -60) \cdot 4 \\ \hline 5x + 2y + 4z = -47 \\ 36x - 16y - 4z = -240 \\ + 5x + 2y + 4z = -47 \\ \hline 41x - 14y = -287 \end{array}$$

$$\begin{array}{r} 41(-7) - 14y = -287 \\ -287 - 14y = -287 \\ -14y = 0 \\ y = 0 \end{array}$$

$$\begin{array}{r} 9(-7) - 4(0) - z = -60 \\ -63 - z = -60 \\ -z = 3 \\ z = -3 \end{array}$$

$$\boxed{(-7, 0, -3)}$$

$$\begin{array}{r} (77x - 35y = -539) \cdot 2 \\ (41x - 14y = -287) \cdot 5 \\ \hline 154x - 70y = -1078 \\ - (205x - 70y = -1435) \\ \hline -51x = 357 \\ x = -7 \end{array}$$

$$4. \begin{cases} x - 5y + 3z = 39 \\ 3x + 8y - z = -19 \\ 2x - 6y - 5z = -29 \end{cases} \begin{matrix} A \\ B \end{matrix}$$

$$\begin{array}{r} A: (x - 5y + 3z = 39) \cdot 3 \\ \hline 3x + 8y - z = -19 \\ 3x - 15y + 9z = 117 \\ - (3x + 8y - z = -19) \\ \hline -23y + 10z = 136 \end{array}$$

$$\begin{array}{r} B: (x - 5y + 3z = 39) \cdot 2 \\ \hline 2x - 6y - 5z = -29 \\ 2x - 10y + 6z = 78 \\ - (2x - 6y - 5z = -29) \\ \hline -4y + 11z = 107 \end{array}$$

$$\begin{array}{r} -4y + 11(9) = 107 \\ -4y + 99 = 107 \\ -4y = 8 \\ y = -2 \end{array}$$

$$\begin{array}{r} x - 5(-2) + 3(9) = 39 \\ x + 37 = 39 \\ x = 2 \end{array}$$

$$\boxed{(2, -2, 9)}$$

$$\begin{array}{r} (-23y + 10z = 136) \cdot 4 \\ (-4y + 11z = 107) \cdot 23 \\ \hline -92y + 40z = 544 \\ + 92y - 253z = -2461 \\ \hline -213z = -1917 \\ z = 9 \end{array}$$

$$5. \begin{cases} x + y + 3z = 12 \\ -3x + 7y - 2z = -35 \\ 7x - 6y - 5z = 32 \end{cases} \begin{matrix} A \\ B \end{matrix}$$

$$\begin{array}{r} A: (x + y + 3z = 12) \cdot 3 \\ \hline -3x + 7y - 2z = -35 \\ 3x + 3y + 9z = 36 \\ + -3x + 7y - 2z = -35 \\ \hline 10y + 7z = 1 \end{array}$$

$$\begin{array}{r} B: (x + y + 3z = 12) \cdot 7 \\ \hline 7x - 6y - 5z = 84 \\ -7x - 7y - 21z = -84 \\ + 7x - 6y - 5z = 84 \\ \hline -13y - 26z = -52 \end{array}$$

$$\begin{array}{r} 10y + 7(3) = 1 \\ 10y + 21 = 1 \\ 10y = -20 \\ y = -2 \end{array}$$

$$\begin{array}{r} x + (-2) + 3(3) = 12 \\ x + 7 = 12 \\ x = 5 \end{array}$$

$$\boxed{(5, -2, 3)}$$

$$\begin{array}{r} (10y + 7z = 1) \cdot 13 \\ (-13y - 26z = -52) \cdot 10 \\ \hline 130y + 91z = 13 \\ + -130y - 260z = -520 \\ \hline -169z = -507 \\ z = 3 \end{array}$$

APPLICATIONS

6. The theater sells three types of tickets: general admission, senior citizen, and child. Mark bought 9 general admission, 2 senior citizen, and 3 child tickets and paid \$170. Sarah bought 5 general admission, 4 senior citizen, and 7 child tickets and paid \$173. Kyle bought 4 general admission, 1 senior citizen, and 6 child tickets and paid \$116.50. Find the cost of each type of ticket.

$x = \text{gen. admission}$

$y = \text{Sr. citizen}$

$z = \text{child}$

$$\begin{array}{rcl} 9x + 2y + 3z = 170 & \} & A \\ 5x + 4y + 7z = 173 & \} & B \\ 4x + y + 6z = 116.50 & \} & B \end{array}$$

Gen Ad = \$13.50/ticket
Sr. Cit = \$11.50/ticket
Child = \$8.50/ticket

$$\begin{array}{r} A: (9x + 2y + 3z = 170) \cdot -2 \\ \quad 5x + 4y + 7z = 173 \\ \hline -18x - 4y - 6z = -340 \\ + \quad 5x + 4y + 7z = 173 \\ \hline -13x + z = -167 \end{array}$$

$$\begin{array}{r} B: 5x + 4y + 7z = 173 \\ \quad (4x + y + 6z = 116.50) \cdot -4 \\ \hline 5x + 4y + 7z = 173 \\ + -16x - 4y - 24z = -466 \\ \hline -11x - 17z = -293 \end{array}$$

$$\begin{array}{r} (-13x + z = -167) \cdot 17 \\ \quad -11x - 17z = -293 \\ \hline -221x + 17z = -2839 \\ + -11x - 17z = -293 \\ \hline -232x = -3132 \end{array}$$

$$\begin{array}{r} \quad \quad \quad x = 13.5 \\ -13(13.5) + z = -167 \\ -175.5 + z = -167 \\ \quad \quad \quad z = 8.5 \end{array}$$

$$\begin{array}{r} 9(13.5) + 2y + 3(8.5) = 170 \\ 2y = 23 \quad \quad y = 11.5 \end{array}$$

7. Jamie has a collection of 78 nickels, dimes, and quarters worth \$12.40. If the number of quarters is doubled, the value becomes \$22.15. How many of each type of coin does Jamie have?

$x = \text{nickel}$

$y = \text{dime}$

$z = \text{quarter}$

$$\begin{array}{rcl} x + y + z = 78 & \} & A \\ .05x + .10y + .25z = 12.40 & \} & B \\ .05x + .10y + .50z = 22.15 & \} & B \end{array}$$

$$\begin{array}{r} A: (x + y + z = 78) \cdot -.05 \\ \quad .05x + .10y + .25z = 12.4 \\ \hline -.05x - .05y - .05z = -3.9 \\ + .05x + .10y + .25z = 12.4 \\ \hline .05y + .2z = 8.5 \end{array}$$

$$\begin{array}{r} .05y + .2(39) + 8.5 \\ .05y + 7.8 = 8.5 \\ .05y = .70 \\ \quad y = 14 \end{array}$$

$$\begin{array}{r} B: .05x + .10y + .25z = 12.40 \\ \quad - (.05x + .10y + .50z = 22.15) \\ \hline \quad \quad \quad -.25z = -9.75 \\ \quad \quad \quad z = 39 \end{array}$$

$$\begin{array}{r} x + 14 + 39 = 78 \\ x = 25 \end{array}$$

25 nickels, 14 dimes,
and 39 quarters

NOTE: Substitution might be better method for the examples below.

8. Mason has a collection of 49¢ stamps, 20¢ stamps, and 3¢ stamps worth \$23.55. He has 56 total 49¢ and 20¢ stamps and the number of 3¢ stamps is nine more than the number of 20¢ stamps. How many of each does he have?

$$\begin{aligned} x &= 49\text{¢ stamps} \\ y &= 20\text{¢ stamps} \\ z &= 3\text{¢ stamps} \end{aligned} \quad \begin{aligned} .49x + .20y + .03z &= 23.55 \\ x + y &= 56 \rightarrow x = 56 - y \\ z &= y + 9 \end{aligned}$$

40 - 49¢ stamps
16 - 20¢ stamps
25 - 3¢ stamps

$$\begin{aligned} .49(56 - y) + .20y + .03(y + 9) &= 23.55 \\ 27.44 - .49y + .2y + .03y + .27 &= 23.55 \\ -.26y + 27.71 &= 23.55 \\ -.26y &= -4.16 \\ y &= 16 \end{aligned} \quad \begin{aligned} z &= 16 + 9 \\ z &= 25 \\ \underline{x = 56 - 16} & \\ x &= 40 \end{aligned}$$

9. The three medals earned in the Olympics are the gold, silver, and bronze medals. In the 2014 Winter Olympics, the United States had 28 total medal winners. There were two more gold medals than silver medals and the number of bronze medals was four less than the sum of the number of gold and silver medals. How many winners were there of each type of medal?

$$\begin{aligned} x &= \text{gold} \\ y &= \text{silver} \\ z &= \text{bronze} \end{aligned} \quad \begin{aligned} x + y + z &= 28 \\ x &= y + 2 \\ z &= x + y - 4 \rightarrow z = (y + 2) + y - 4 \\ z &= 2y - 2 \end{aligned}$$

9 golds, 7 silvers,
and 12 bronze

$$\begin{aligned} (y + 2) + y + (2y - 2) &= 28 \\ 4y &= 28 \\ y &= 7 \end{aligned} \quad \begin{aligned} x &= 7 + 2 \\ x &= 9 \\ z &= 2(7) - 2 \\ z &= 12 \end{aligned}$$

10. Enrique has a pay-per-use plan on his cell phone. It costs \$0.10 per minute talk, \$0.20 per text message, and \$0.25 per picture message. His total bill last month was \$30.95. The total number of text and picture messages sent was 87. The cost for the use of minutes was \$1.60 less than the cost of the text messages. Find the number of talk minutes, text messages, and picture messages he used.

$$\begin{aligned} x &= \text{talk} \\ y &= \text{text} \\ z &= \text{picture} \end{aligned} \quad \begin{aligned} .10x + .20y + .25z &= 30.95 \\ y + z &= 87 \rightarrow z = 87 - y \\ .10x &= .20y - 1.60 \rightarrow x = 2y - 16 \end{aligned}$$

128 talk min,
72 texts, and
15 pic. messages

$$\begin{aligned} .10(2y - 16) + .20y + .25(87 - y) &= 30.95 \\ .2y - 1.6 + .2y + 21.75 - .25y &= 30.95 \\ .15y + 20.15 &= 30.95 \\ .15y &= 10.8 \\ y &= 72 \end{aligned} \quad \begin{aligned} z &= 87 - 72 = 15 \\ x &= 2(72) - 16 \\ &= 128 \end{aligned}$$

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Unit 2: Linear Functions & Systems

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Homework 7: Systems with Three Variables

** This is a 2-page document! **

Solve each system of equations. Clearly identify your solution.

$$1. \begin{cases} 9x+3y-4z=37 \\ 4x+3y+7z=16 \\ x-5y+8z=-31 \end{cases} \begin{matrix} A \\ B \end{matrix}$$

$$A: \begin{array}{r} 9x+3y-4z=37 \\ -(4x+3y+7z=16) \\ \hline 5x-11z=21 \end{array}$$

$$B: \begin{array}{r} (4x+3y+7z=16) \cdot 5 \\ (x-5y+8z=-31) \cdot 3 \\ \hline 20x+15y+35z=80 \\ + 3x-15y+24z=-93 \\ \hline 23x+59z=-13 \end{array}$$

$$5x-11(-1)=21$$

$$5x=10$$

$$x=2$$

$$\begin{array}{r} 2-5y+8(-1)=-31 \\ -5y-6=-31 \\ -5y=-25 \\ y=5 \end{array}$$

$$\begin{array}{r} 23 \cdot (5x-11z=21) \\ -5 \cdot (23x+59z=-13) \\ \hline 115x-253z=483 \\ + -115x-295z=65 \\ \hline -548z=548 \\ z=-1 \end{array}$$

$$(2, 5, -1)$$

$$2. \begin{cases} 3x+4y-6z=-60 \\ 7x-5y+z=-37 \\ 2x+3y-z=-29 \end{cases} \begin{matrix} A \\ B \end{matrix}$$

$$A: \begin{array}{r} 3x+4y-6z=-60 \\ (7x-5y+z=-37) \cdot 6 \\ \hline 3x+4y-6z=-60 \\ + 42x-30y+6z=-222 \\ \hline 45x-26y=-282 \end{array}$$

$$B: \begin{array}{r} 7x-5y+z=-37 \\ + 2x+3y-z=-29 \\ \hline 9x-2y=-66 \end{array}$$

$$9x-2(-3)=-66$$

$$9x+6=-66$$

$$9x=-72$$

$$x=-8$$

$$\begin{array}{r} 2(-8)+3(-3)-z=-29 \\ -25-z=-29 \\ z=4 \end{array}$$

$$(-8, -3, 4)$$

$$\begin{array}{r} 45x-26y=-282 \\ -5 \cdot (9x-2y=-66) \\ \hline 45x-26y=-282 \\ -45x+10y=330 \\ \hline -16y=48 \\ y=-3 \end{array}$$

$$3. \begin{cases} 10x+7y-2z=46 \\ 3x-2y+9z=22 \\ 5x+y-3z=28 \end{cases} \begin{matrix} B \\ A \end{matrix}$$

$$A: \begin{array}{r} 3x-2y+9z=22 \\ (5x+y-3z=28) \cdot 2 \\ \hline 3x-2y+9z=22 \\ + 10x+2y-6z=56 \\ \hline 13x+3z=78 \end{array}$$

$$B: \begin{array}{r} 10x+7y-2z=46 \\ (5x+y-3z=28) \cdot -7 \\ \hline 10x+7y-2z=46 \\ + -35x-7y+21z=-196 \\ \hline -25x+19z=-150 \end{array}$$

$$\begin{array}{r} 13(6)+3z=78 \\ 78+3z=78 \\ 3z=0 \\ z=0 \end{array}$$

$$\begin{array}{r} 5(6)+y-3(0)=28 \\ 30+y=28 \\ y=-2 \end{array}$$

$$(6, -2, 0)$$

$$\begin{array}{r} (13x+3z=78) \cdot 19 \\ (-25x+19z=-150) \cdot -3 \\ \hline -247x-57z=-1482 \\ + -75x+57z=-450 \\ \hline -322x=-1932 \\ x=6 \end{array}$$

4. One week, the music store sold 2 trumpets, 3 clarinets, and 5 violins for \$1,240. The next week, they sold 3 trumpets, 1 clarinet, and 4 violins for \$1,027. The following week, they sold 5 trumpets, 7 clarinets, and 2 violins for \$2,091. Find the cost of a trumpet, clarinet, and violin.

x = trumpets
 y = clarinets
 z = violins

$$\begin{cases} 2x + 3y + 5z = 1240 & A \\ 3x + y + 4z = 1027 & B \\ 5x + 7y + 2z = 2091 & C \end{cases}$$

\$174 per trumpet,
 \$149 per clarinet,
 \$89 per violin

$$\begin{array}{r} A: 2x + 3y + 5z = 1240 \\ (3x + y + 4z = 1027) \cdot -3 \\ \hline 2x + 3y + 5z = 1240 \\ + -9x - 3y - 12z = -3081 \\ \hline -7x - 7z = -1841 \end{array}$$

$$\begin{array}{r} 16 \cdot (-7x - 7z = -1841) \\ -7 \cdot (16x - 26z = -5098) \\ \hline -112x - 112z = -29456 \\ + 112x + 182z = -35686 \\ \hline 70z = -6230 \\ z = 89 \end{array}$$

$$\begin{array}{r} B: (3x + y + 4z = 1027) \cdot -7 \\ 5x + 7y + 2z = 2091 \\ \hline -21x - 7y - 28z = -7189 \\ + 5x + 7y + 2z = 2091 \\ \hline -16x - 26z = -5098 \end{array}$$

$$\begin{array}{r} -7x - 7(89) = -1841 \\ -7x = -1218 \\ x = 174 \end{array}$$

$$\begin{array}{r} 2(174) + 3y + 5(89) = 1240 \\ 3y + 793 = 1240 \\ 3y = 447 \\ y = 149 \end{array}$$

5. Caleb earns points on his credit card that he can use towards future purchases. He earns four points per dollar spent on flights, two points per dollar spent at hotels, and one point per dollar spent on all other purchases. Last year, he charged a total of \$9,480 and earned 14,660 points. The amount of money spent on flights was \$140 more than twice the amount of money spent on hotels. Find the amount of money spent on each type of purchase.

x = flights
 y = hotels
 z = other

$$\begin{array}{r} 4x + 2y + z = 14660 \\ x + y + z = 9480 \\ x = 2y + 140 \end{array}$$

$$\begin{array}{r} x = 2(680) + 140 \\ x = 1500 \end{array}$$

$$4(2y + 140) + 2y + z = 14660 \rightarrow 10y + z = 14100$$

$$(2y + 140) + y + z = 9480 \rightarrow -(3y + z = 9340)$$

$$\begin{array}{r} 10y + z = 14100 \\ -(3y + z = 9340) \\ \hline 7y = 4760 \\ y = 680 \end{array}$$

$$\begin{array}{r} 1500 + 680 + z = 9480 \\ 2180 + z = 9480 \\ z = 7300 \end{array}$$

\$1500 on flights,
 \$680 on hotels, and
 \$7300 on other expenses

Name: _____

Algebra 2

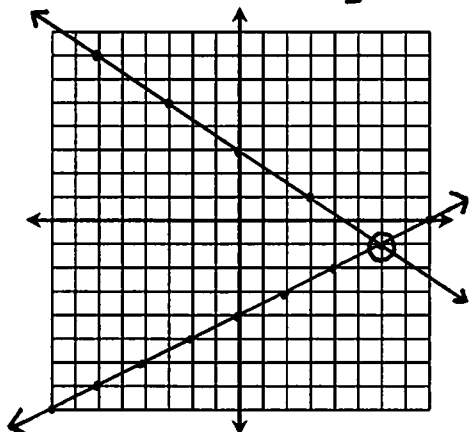
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Unit 2: Linear Functions & Systems

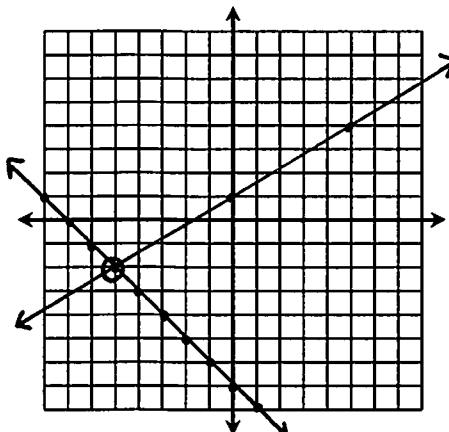
Quiz 2-3: Systems of Equations

For questions 1-2, solve the system by graphing.

$$1. \begin{cases} 2x + 3y = 9 \rightarrow y = -\frac{2}{3}x + 3 \\ x - 2y = 8 \rightarrow y = \frac{1}{2}x - 4 \end{cases}$$



$$2. \begin{cases} 6x = 10y - 10 \rightarrow y = \frac{3}{5}x + 1 \\ x + y + 7 = 0 \rightarrow y = -x - 7 \end{cases}$$



1. (6, -1)

2. (-5, -2)

3. (1, -8)

4. (10, 3)

5. (-4, 7)

6. (-6, 0)

7. No Solution

8. (9, -5)

For questions 3-4, solve the system using the substitution method.

$$3. \begin{cases} 9x - y = 17 \rightarrow y = 9x - 17 \\ 5x + 4y = -27 \end{cases}$$

$$5x + 4(9x - 17) = -27$$

$$5x + 36x - 68 = -27$$

$$41x = 41$$

$$x = 1$$

$$y = 9(1) - 17$$

$$y = -8$$

$$(1, -8)$$

$$4. \begin{cases} 2x - 4y = 8 \rightarrow x = 2y + 4 \\ 7x - 3y = 61 \end{cases}$$

$$7(2y + 4) - 3y = 61$$

$$14y + 28 - 3y = 61$$

$$11y = 33$$

$$y = 3$$

$$x = 2(3) + 4$$

$$x = 10$$

$$(10, 3)$$

For questions 5-8, solve the system using the elimination method.

$$5. \begin{cases} (8x + y = -39) \cdot 5 \\ 2x - 5y = 27 \end{cases}$$

$$40x + 5y = -195$$

$$+ 2x - 5y = 27$$

$$42x = -168$$

$$x = -4$$

$$(-4, -7)$$

$$6. \begin{cases} (3x - 7y = -18) \cdot 4 \\ (4x - 2y = -24) \cdot 3 \end{cases}$$

$$12x - 28y = -72$$

$$+ 12x - 6y = 72$$

$$-34y = 0$$

$$y = 0$$

$$12x - 28y = -72$$

$$+ 12x - 6y = 72$$

$$-22y = 0$$

$$y = 0$$

$$4x - 2(0) = -24$$

$$4x = -24$$

$$x = -6$$

$$(-6, 0)$$

$$7. \begin{cases} 12y = 17 - 9x \rightarrow 9x + 12y = 17 \\ -4y - 3x = 31 \rightarrow (-3x - 4y = 31) \cdot 3 \end{cases}$$

$$9x + 12y = 17$$

$$+ -9x - 12y = 93$$

$$0 \neq 110$$

No Solution

$$8. \begin{cases} 11y + 1 = -6x \rightarrow (6x + 11y = -1) \cdot 4 \\ 0 = 43 - 7x - 4y \rightarrow (7x + 4y = 43) \cdot -11 \end{cases}$$

$$24x + 44y = -4$$

$$+ -77x - 44y = -473$$

$$-53x = -477$$

$$x = 9$$

$$11y + 1 = -6(9)$$

$$11y = -55$$

$$y = -5$$

$$(9, -5)$$

For questions 9-11, solve using a system of equations with your method of choice.

9. Casey bought a 15.4-pound turkey and an 11.6-pound ham for Thanksgiving and paid \$38.51. Her friend Jane bought a 10.2-pound turkey and a 7.3-pound ham from the same store and paid \$24.84. Find the cost per pound of turkey.

$$\begin{array}{rcl} x = \text{turkey} & 7.3(15.4x + 11.6y = 38.51) \\ y = \text{ham} & -11.6(10.2x + 7.3y = 24.84) \\ \hline & 112.42x + 84.68y = 281.123 \\ & + -118.32x - 84.68y = -288.144 \\ \hline & -5.9x = -7.021 \\ & x = 1.19 \end{array}$$

9. \$1.19/lb.

10. 67 nickels,

19 quarters

11. \$3500

10. Owen has a collection of nickels and quarters worth \$8.10. If the nickels were quarters and the quarters were nickels, the value would be \$17.70. How many of each type of coin does Owen have?

$$\begin{array}{rcl} x = \text{nickels} & (.05x + .25y = 8.10) \cdot -5 \\ y = \text{quarters} & .25x + .05y = 17.70 \\ \hline & -.25x - 1.25y = -40.50 \\ & + .25x + .05y = 17.70 \\ \hline & -1.2y = -22.8 \\ & y = 19 \end{array}$$

$$\begin{array}{rcl} .05x + .25(19) = 8.10 \\ .05x + 4.75 = 8.10 \\ .05x = 3.35 \\ x = 67 \end{array}$$

11. Michelle invested \$12,000 in stocks and bonds, paying 4.8% and 7% annual interest, respectively. If the total interest earned for the year was \$763, how much was invested in stocks?

$$\begin{array}{rcl} x = \text{stocks} & x + y = 12000 \rightarrow x = 12000 - y \\ y = \text{bonds} & .048x + .07y = 763 \\ & .048(12000 - y) + .07y = 763 \\ & 576 - .048y + .07y = 763 \\ & .022y = 187 \\ & y = 8500 \end{array}$$

$$\begin{array}{rcl} x = 12000 - 8500 \\ x = 3500 \end{array}$$

Solve the system of three variables below.

$$12. \begin{cases} x - 7y + 3z = 17 \\ 5x + 2y - 2z = -57 \\ 3x - 10y - z = -11 \end{cases} \begin{matrix} A \\ B \end{matrix}$$

$$\begin{array}{rcl} A: & 5x + 2y - 2z = -57 \\ & (3x - 10y - z = -11) \cdot -2 \\ \hline & 5x + 2y - 2z = -57 \\ & + -6x + 20y + 2z = 22 \\ \hline & -x + 22y = -35 \end{array}$$

12. $(-9, -2, 4)$

$$\begin{array}{rcl} -x + 22(-2) = -35 \\ -x - 44 = -35 \\ -x = 9 \\ x = -9 \end{array}$$

$$\begin{array}{rcl} -9 - 7(-2) + 3z = 17 \\ -9 + 14 + 3z = 17 \\ 3z = 12 \\ z = 4 \end{array}$$

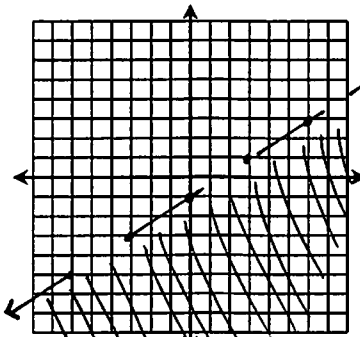
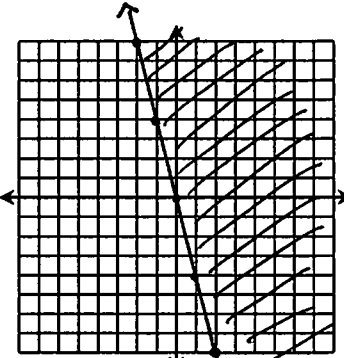
$$\begin{array}{rcl} B: & x - 7y + 3z = 17 \\ & (3x - 10y - z = -11) \cdot 3 \\ \hline & x - 7y + 3z = 17 \\ & + 9x - 30y - 3z = -33 \\ \hline & 10x - 37y = -16 \end{array}$$

$$\begin{array}{rcl} 10 \cdot (-x + 22y = -35) \\ 10x - 37y = -16 \\ \hline -10x + 220y = -350 \\ + 10x - 37y = -16 \\ \hline 183y = -366 \\ y = -2 \end{array}$$

$(-9, -2, 4)$

Name:	Date:
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Topic:	Class:
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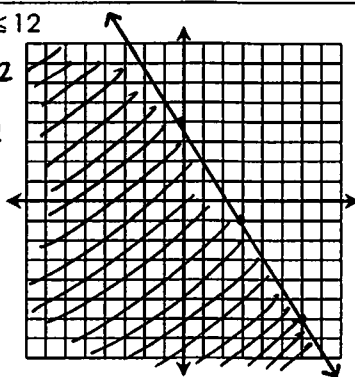
Main Ideas/Questions	Notes/Examples			
LINEAR Inequalities	A linear inequality is similar to a linear equation, however with an inequality symbol instead. Example: Given $y \geq -2x + 9$, determine which points satisfy the inequality: <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 33%; vertical-align: top;"> a) (7, 4) $4 \geq -2(7) + 9$ $4 \geq -14 + 9$ $4 \geq -5$ Yes </td><td style="width: 33%; vertical-align: top;"> b) (-1, 3) $3 \geq -2(-1) + 9$ $3 \geq 2 + 9$ $3 \geq 11$ No </td><td style="width: 33%; vertical-align: top;"> c) (10, -5) $-5 \geq -2(10) + 9$ $-5 \geq -20 + 9$ $-5 \geq -11$ Yes </td></tr> </table>	a) (7, 4) $4 \geq -2(7) + 9$ $4 \geq -14 + 9$ $4 \geq -5$ Yes	b) (-1, 3) $3 \geq -2(-1) + 9$ $3 \geq 2 + 9$ $3 \geq 11$ No	c) (10, -5) $-5 \geq -2(10) + 9$ $-5 \geq -20 + 9$ $-5 \geq -11$ Yes
a) (7, 4) $4 \geq -2(7) + 9$ $4 \geq -14 + 9$ $4 \geq -5$ Yes	b) (-1, 3) $3 \geq -2(-1) + 9$ $3 \geq 2 + 9$ $3 \geq 11$ No	c) (10, -5) $-5 \geq -2(10) + 9$ $-5 \geq -20 + 9$ $-5 \geq -11$ Yes		
Real World EXAMPLE	John sells t-shirts. Plain shirts cost \$5 and graphic shirts cost \$7. In order to match his product costs, he must make at least \$1,200. a) Write an inequality to represent this situation. $5p + 7g \geq 1200$ b) If he sells 150 plain shirts and 60 graphic shirts, will he be able to cover the product costs? $5(150) + 7(60) \geq 1200$ $750 + 420 \geq 1200$ $1170 \geq 1200$ No c) If he sells 90 graphic shirts, what's the minimum number of plain shirts he must sell to break even? $5p + 7(90) = 1200$ $5p + 630 = 1200$ $5p = 570$ $p = 114$ 114 plain shirts			
GRAPHING Linear Inequalities Graphing linear inequalities is a visual way of showing the ordered pairs that satisfy the inequality.	① Put the inequality in SLOPE-INTERCEPT FORM. Be sure to flip the inequality if you multiply or divide by a negative. ② GRAPH the line. Use a dashed line for $<$ and $>$ symbols and a solid line for $\leq$ and $\geq$ symbols. ③ SHADE above the line for $>$ or $\geq$ symbols and below the line for $<$ or $\leq$ symbols. Graph each linear inequality. 1. $y < \frac{2}{3}x - 1$ ↑ dotted, below  2. $y \geq -4x$ ↑ solid, above 			

$$3. \ 5x + 3y \leq 12$$

$$3y \leq -5x + 12$$

$$y \leq -\frac{5}{3}x + 4$$

↑
solid,
below

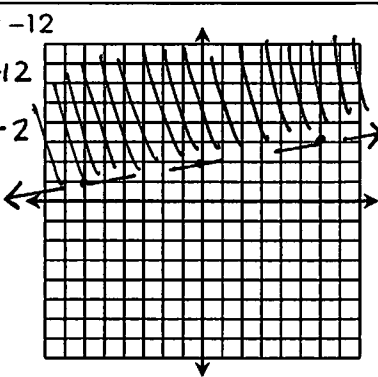


$$4. \ x - 6y < -12$$

$$-6y < -x - 12$$

$$y > \frac{1}{6}x + 2$$

↑
dotted,
above

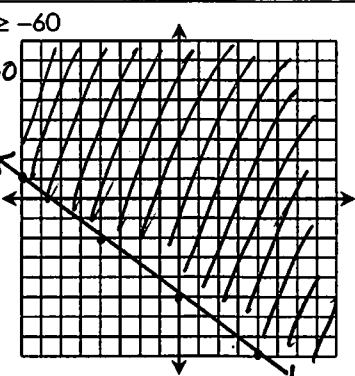


$$5. \ 9x + 12y \geq -60$$

$$12y \geq -9x - 60$$

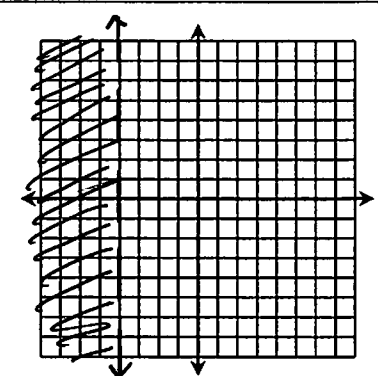
$$y \geq -\frac{3}{4}x - 5$$

↑
solid,
above



$$6. \ x < -4$$

↑
dotted,
left



7. Stephanie spends \$0.10 per text message sent and \$0.25 per picture message sent on her cell phone. Her monthly budget for messaging is \$30.

a) Write a linear inequality to represent this situation, then graph.

$$.10x + .25y \leq 30$$

$x = \text{text}$

$y = \text{picture}$

$$.25y \leq -.10x + 30$$

$$y \leq -.4x + 120$$

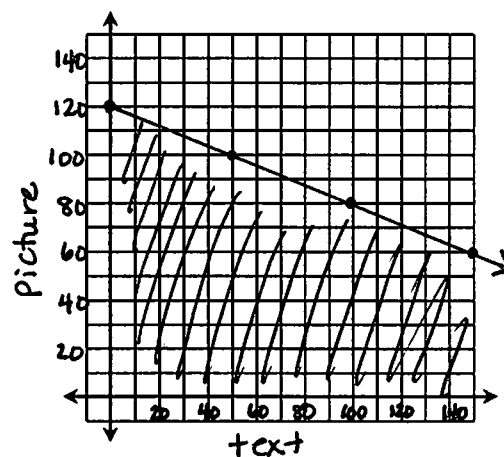
$$y \leq -\frac{2}{5}x + 120$$

b) Using your graph, give two possible combinations of text and picture messages she can send and stay within her budget.

80 text and 40 picture

140 text and 10 picture

* answers
may vary



8. Scott makes \$12 per hour babysitting and \$8 per hour working at the grocery store. He will need to make at least \$600 to buy a new computer.

a) Write a linear inequality to represent this situation, then graph.

$$12x + 8y \geq 600$$

$x = \text{babysitting}$

$y = \text{grocery store}$

$$8y \geq -12x + 600$$

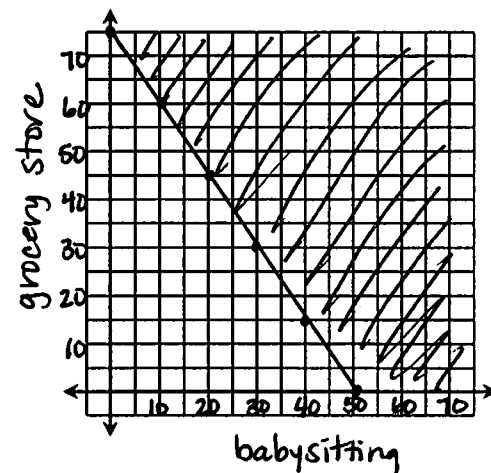
$$y \geq -\frac{3}{2}x + 75$$

b) Using your graph, give two possible combinations of the number of hours Scott can babysit and work in the grocery store to ensure he can afford his new computer.

70 hr. babysitting and
0 hr. at grocery store

30 hr. babysitting and
35 hr. at grocery store

* answers
may vary

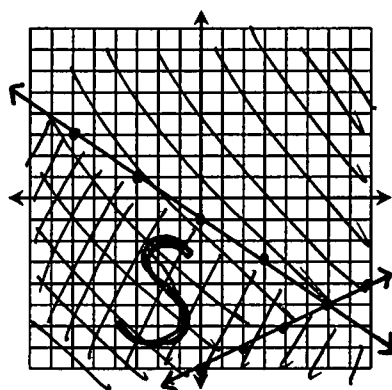


Name:	Date:
Topic:	Class:

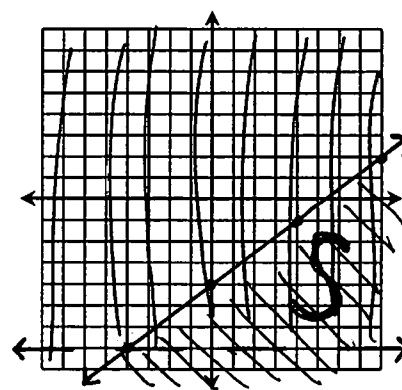
Main Ideas/Questions	Notes/Examples
SYSTEMS of Linear Inequalities	The solution to a system of linear inequalities is the set of ordered pairs that satisfy ALL the linear inequalities. The solution can be shown by graphing.
	1 GRAPH and SHADE each linear inequality.
	2 The SOLUTION to the system is where the shading overlaps.

Graph each system of linear inequalities. Clearly indicate the solution region.

1.
$$\begin{cases} y \leq -\frac{2}{3}x - 1 \\ y \geq \frac{1}{2}x - 8 \end{cases}$$



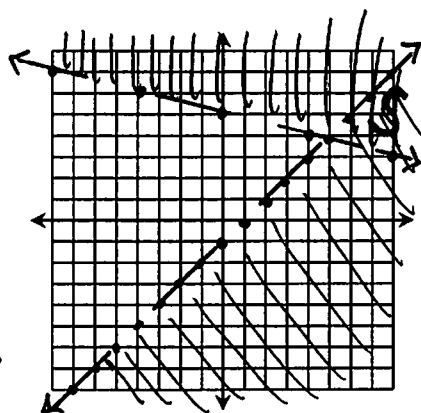
2.
$$\begin{cases} y \leq \frac{3}{4}x - 4 \\ y > -7 \end{cases}$$



3.
$$\begin{cases} x - y > 1 \\ x + 4y > 20 \end{cases}$$

$$\begin{aligned} x - y &> 1 \\ -y &> -x + 1 \\ y &< x - 1 \end{aligned}$$

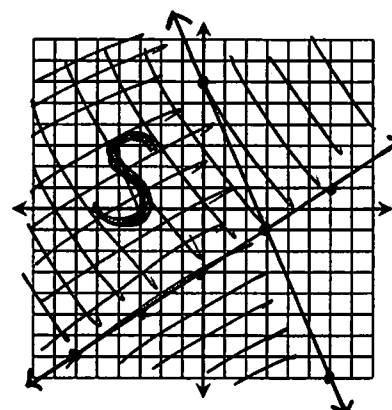
$$\begin{aligned} x + 4y &> 20 \\ 4y &> -x + 20 \\ y &> -\frac{1}{4}x + 5 \end{aligned}$$



4.
$$\begin{cases} 7x + 3y \leq 18 \\ 2x - 3y \leq 9 \end{cases}$$

$$\begin{aligned} 7x + 3y &\leq 18 \\ 3y &\leq -7x + 18 \\ y &\leq -\frac{7}{3}x + 6 \end{aligned}$$

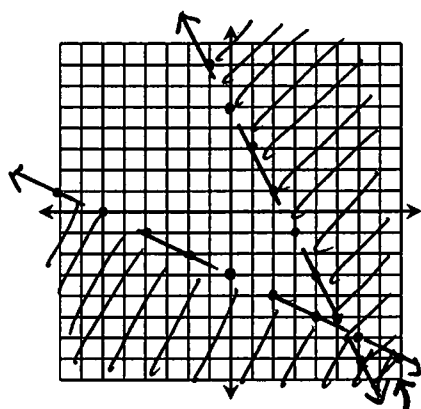
$$\begin{aligned} 2x - 3y &\leq 9 \\ -3y &\leq -2x + 9 \\ y &\geq \frac{2}{3}x - 3 \end{aligned}$$



5.
$$\begin{cases} x + 2y < -6 \\ -4x - 2y < -10 \end{cases}$$

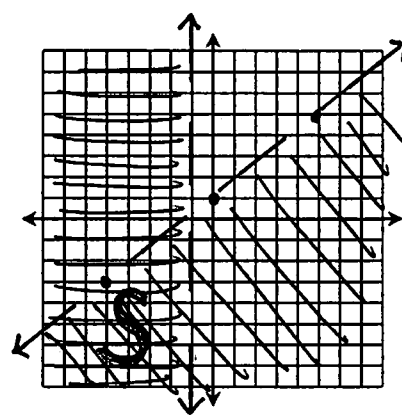
$$\begin{aligned} x + 2y &< -6 \\ 2y &< -x - 6 \\ y &< -\frac{1}{2}x - 3 \end{aligned}$$

$$\begin{aligned} -4x - 2y &< -10 \\ -2y &< 4x - 10 \\ y &> -2x + 5 \end{aligned}$$



6.
$$\begin{cases} 4x - 5y > -5 \\ x < -1 \end{cases}$$

$$\begin{aligned} 4x - 5y &> -5 \\ -5y &> -4x - 5 \\ y &< \frac{4}{5}x + 1 \end{aligned}$$



7. Which points are solutions to the system of inequalities below? Check all that apply.

$$\begin{cases} x - 6y > -48 \\ 2x + 3y < 9 \end{cases}$$

(1, 1):

$$\begin{aligned} 1 - 6 &> -48 && \text{yes} \\ 2 + 3 &< 9 && \text{yes} \end{aligned}$$

(3, 4):

$$\begin{aligned} 3 - 24 &> -48 && \text{yes} \\ 6 + 12 &< 9 && \text{No} \end{aligned}$$

(-8, -3):

$$\begin{aligned} -8 + 18 &> -48 && \text{yes} \\ -16 - 9 &< 9 && \text{yes} \end{aligned}$$

(9, -2):

$$\begin{aligned} 9 + 12 &> -48 && \text{yes} \\ 18 - 6 &< 9 && \text{No} \end{aligned}$$

(-2, 6):

$$\begin{aligned} -2 - 36 &> -48 && \text{yes} \\ -4 + 18 &< 9 && \text{No} \end{aligned}$$

(4, -7):

$$\begin{aligned} 4 + 42 &> -48 && \text{yes} \\ 8 - 21 &< 9 && \text{yes} \end{aligned}$$

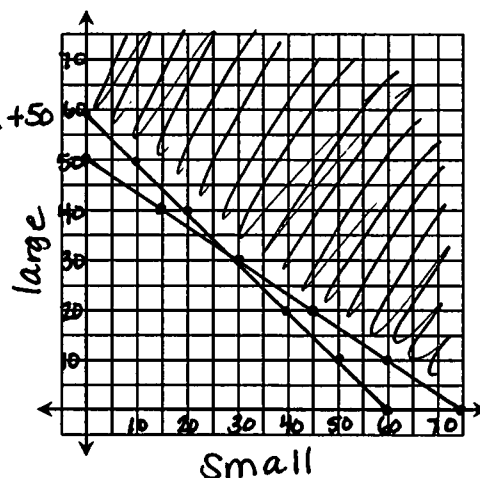
☒ (1, 1)	☐ (-2, 6)
☐ (9, -2)	☒ (-8, -3)
☐ (3, 4)	☒ (4, -7)

8. To raise money to attend Boy Scout camp, Alec is selling popcorn. Small bags of popcorn sell for \$4 each and large bags sell for \$6 each. He needs to earn at least \$300 to attend camp. He will also earn a badge if he sells at least 60 bags total of any combination.

a) Write a system of linear inequalities to represent this situation, then graph.

x = small
 y = large

$$\begin{aligned} 4x + 6y &\geq 300 \rightarrow y \geq -\frac{2}{3}x + 50 \\ x + y &\geq 60 \rightarrow y \geq -x + 60 \end{aligned}$$



b) Using your graph, give two possible combinations of bags of popcorn that Alec can sell to attend camp and earn his badge.

(10, 60) - 10 small and 60 large

(50, 40) - 50 small and 40 large

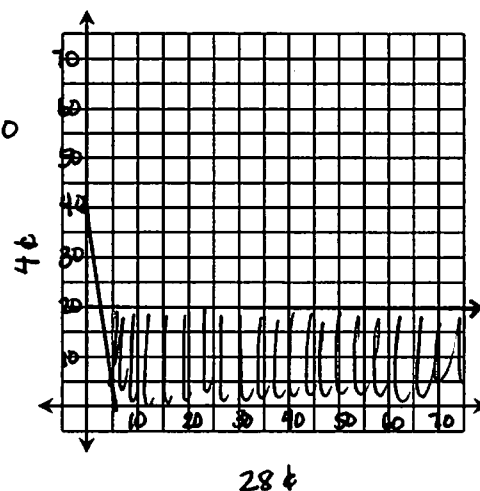
* ans. may vary

9. Greg needs at least \$1.60 in stamps to mail a package. He has 28¢ and 4¢ stamps. He can use no more than twenty 4¢ stamps as he only has one book left.

a) Write a system of linear inequalities to represent this situation, then graph.

x = 28¢
 y = 4¢

$$\begin{aligned} .28x + .04y &\geq 1.60 \rightarrow y \geq -7x + 40 \\ y &\leq 20 \end{aligned}$$



b) Using your graph, give two possible combinations of stamps that Greg can use to ensure he mails his package with the correct postage.

(5, 15) - 5 - 28¢ stamps and 15 - 4¢ stamps

(20, 0) - 20 - 28¢ stamps and 0 - 4¢ stamps

* ans. may vary

Name: _____

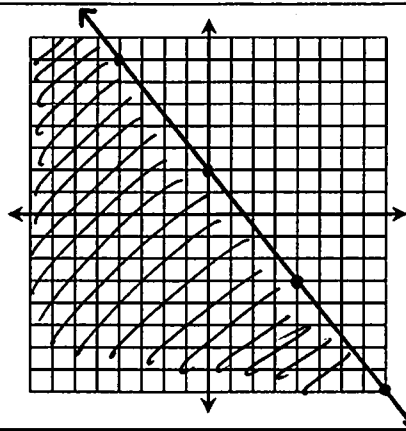
Unit 2: Linear Functions & Systems

Date: _____ Per: _____

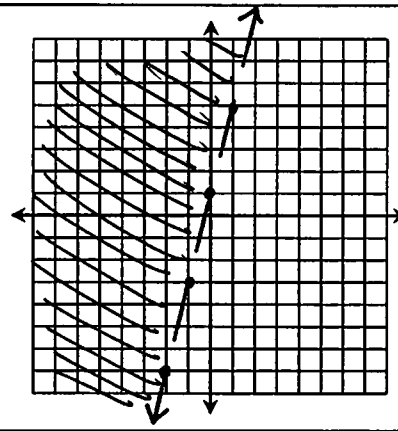
Homework 8: Linear Inequalities

**** This is a 2-page document! ******Graph each linear inequality.**

1. $y \leq -\frac{5}{4}x + 2$



2. $4x - y < -1$
 $-y < -4x - 1$
 $y > 4x + 1$



3. In football, a kicker can score one point for the extra-point on a touchdown or three points for a field goal. Logan is a kicker for his high school football team. The record number of points scored by a kicker in a single game is 24. Logan is hoping to beat this record in his next game.

- a) Write a linear inequality to represent this situation, then graph.

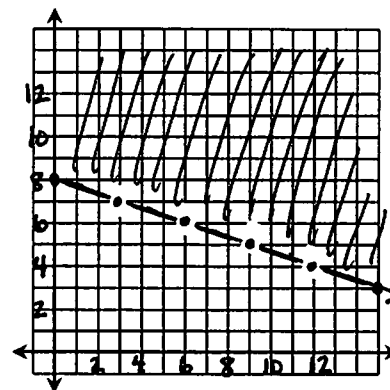
x = extra point
 y = field goal

$$x + 3y > 24$$

$$3y > -x + 24$$

$$y > -\frac{1}{3}x + 8$$

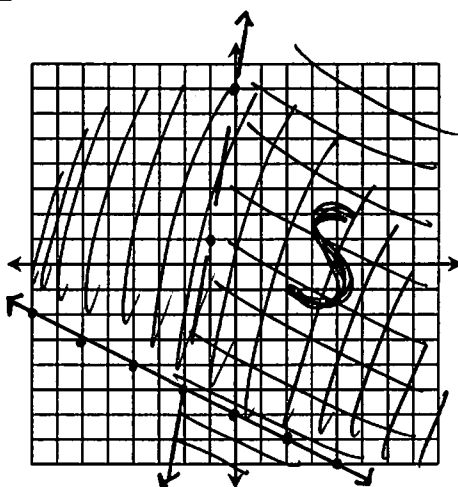
- b) Using your graph, give two possible combinations extra-point kicks and field goals Logan could score to beat the current record.

**Graph each system of linear inequalities. Clearly indicate the solution region.**

4. $\begin{cases} x + 2y \geq -12 \\ y < 6x + 7 \end{cases}$

$$2y \geq -x - 12$$

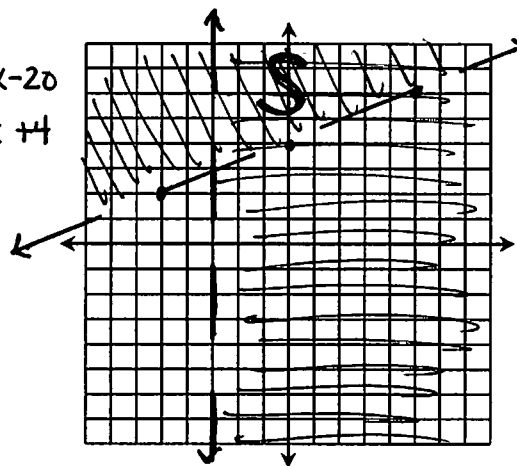
$$y \geq -\frac{1}{2}x - 6$$



5. $\begin{cases} 2x - 5y < -20 \\ x > -3 \end{cases}$

$$-5y < -2x - 20$$

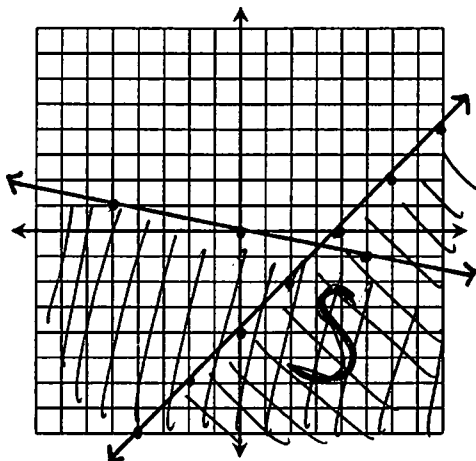
$$y > \frac{2}{5}x + 4$$



$$6. \begin{cases} x - y \geq 4 \\ x + 5y \leq 0 \end{cases}$$

$$-y \geq -x + 4 \\ y \leq x - 4$$

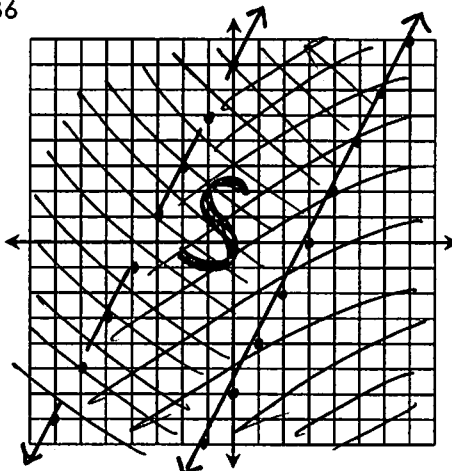
$$5y \leq -x \\ y \leq -\frac{1}{5}x$$



$$7. \begin{cases} 8x - 4y > -28 \\ 6y \geq 12x - 36 \end{cases}$$

$$-4y > -8x - 28 \\ y < 2x + 7$$

$$y \geq 2x - 6$$



8. Brandon is shopping at Old Navy during a sale. All shorts are \$16 each and all t-shirts are \$10 each. He has \$100 to spend and would like to purchase at least 2 pairs of shorts.

a) Write a system of linear inequalities to represent this situation, then graph.

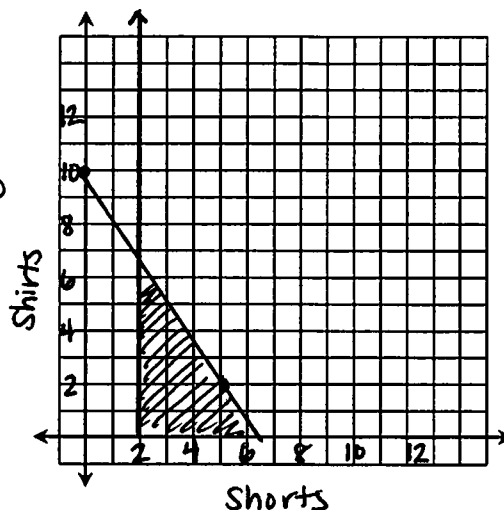
$x = \text{shorts}$

$y = \text{shirts}$

$$16x + 10y \leq 100$$

$$x \geq 2$$

$$y \leq -\frac{8}{5}x + 10$$



* ans. may vary b) Using your graph, give two possible combinations of shorts and t-shirts that Brandon can buy.

$(3, 4)$ - 3 shorts and 4 shirts

$(5, 1)$ - 5 shorts and 1 shirt

9. Darcy is buying apples and oranges for a large fruit basket to give away as a door prize at a charity event. Apples cost \$0.24 each and oranges cost \$0.80 each. She has \$12 to spend and would like to purchase at least 20 pieces of fruit in total.

a) Write a system of linear inequalities to represent this situation, then graph.

$x = \text{apples}$

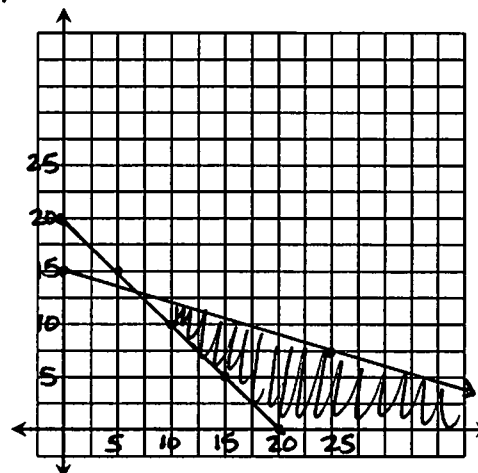
$y = \text{oranges}$

$$.24x + .80y \leq 12$$

$$x + y \geq 20$$

$$y \geq -x + 20$$

$$y \leq -\frac{3}{10}x + 15$$



* ans. may vary b) Using your graph, give two possible combinations of apples and oranges Darcy can buy.

$(15, 10)$ - 15 apples and 10 oranges

$(20, 5)$ - 20 apples and 5 oranges

Unit 2 Test Study Guide

(Linear Functions & Systems)

Name: _____

Date: _____ Per: _____

Topic 1: Relations & Functions

Find the domain and range of each relation. Then determine if the relation is a function.

1.	2.	3.	4.										
<table border="1"><thead><tr><th>x</th><th>y</th></tr></thead><tbody><tr><td>5</td><td>5</td></tr><tr><td>6</td><td>6</td></tr><tr><td>7</td><td>7</td></tr><tr><td>8</td><td>8</td></tr></tbody></table>	x	y	5	5	6	6	7	7	8	8			
x	y												
5	5												
6	6												
7	7												
8	8												
$D: \{5, 6, 7, 8\}$	$D: \{-2, -1, 3, 5\}$	$D: -4 \leq x < 1$	$D: x \geq -3$										
$R: \{5, 6, 7, 8\}$	$R: \{-7, 0, 4\}$	$R: -3 < y \leq 5$	$R: \mathbb{R}$										
Function? Yes	Function? Yes	Function? Yes	Function? No										

Use $f(x) = -4x + 1$, $g(x) = x^2 - 8x + 21$, and $h(x) = |9 - 3x|$ for questions 5 – 10.

5. Find $g(-5)$ $g(-5) = (-5)^2 - 8(-5) + 21$ $= 25 + 40 + 21$ $= \boxed{86}$	6. Find $f\left(\frac{11}{12}\right)$ $f\left(\frac{11}{12}\right) = -4\left(\frac{11}{12}\right) + 1$ $= -\frac{11}{3} + 1$ $= \boxed{-\frac{8}{3}}$	7. Find $h(8) - f(-7)$ $h(8) = 9 - 3(8) = 9 - 24 = 15$ $f(-7) = -4(-7) + 1 = 29$ $15 - 29 = \boxed{-14}$
8. Find $g(x + 4)$ $g(x + 4) = (x + 4)^2 - 8(x + 4) + 21$ $= x^2 + 8x + 16 - 8x - 32 + 21$ $= \boxed{x^2 + 5}$	9. If $f(x) = 53$, find x . $-4x + 1 = 53$ $-4x = 52$ $x = \boxed{-13}$	10. If $h(x) = 39$, find x . $ 9 - 3x = 39$ $9 - 3x = 39$ $9 - 3x = -39$ $-3x = 30$ $-3x = -48$ $x = -10$ $x = 16$ $x = \boxed{\{-10, 16\}}$

Topic 2: Linear Equations: Slope-Intercept & Standard Form

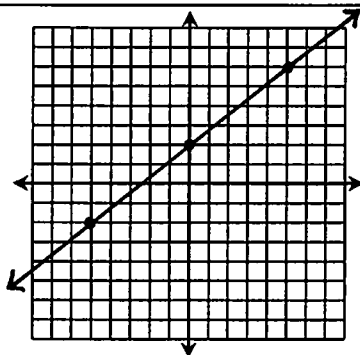
Write each equation in slope-intercept form, when necessary, then graph the line.

11. $y = -\frac{2}{3}x + 3$ 	12. $\frac{5}{2}x - 4y = -8$ $5x - 8y = -16$ $-8y = -5x - 16$ $y = \frac{5}{8}x + 2$
---------------------------------	---

13. $4x - 5y = -10$

$$-5y = -4x - 10$$

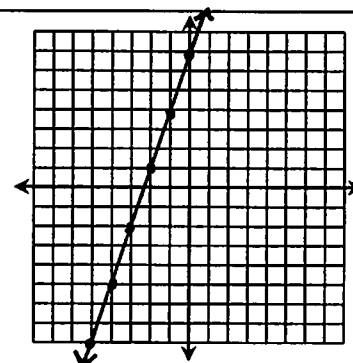
$$y = \frac{4}{5}x + 2$$



14. $12x = 4y - 28$

$$12x + 28 = 4y$$

$$3x + 7 = y$$



Give an example of a line that is parallel and a line that is perpendicular to each given line.

15. $9x + 6y = -6$

$$6y = -9x - 6$$

$$y = -\frac{3}{2}x - 1$$

Parallel: $y = -\frac{3}{2}x + 4$

Perpendicular: $y = \frac{2}{3}x - 7$

16. $y = -5$

Parallel: $y = 1$

Perpendicular: $x = 3$

Find the x- and y-intercepts of each line, then graph the line.

17. $y = -5x - 3$

X-int:

$$0 = -5x - 3$$

$$3 = -5x$$

$$-\frac{3}{5} = x$$

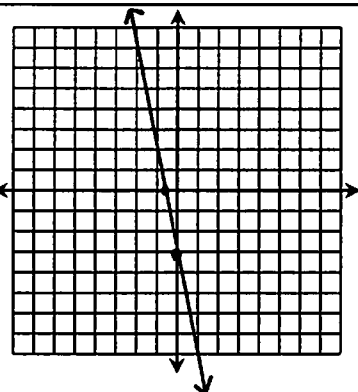
$$(-\frac{3}{5}, 0)$$

Y-int:

$$y = -5(0) - 3$$

$$y = 0 - 3$$

$$y = -3$$



18. $4y = 10x - 24$

X-int:

$$4(0) = 10x - 24$$

$$24 = 10x$$

$$\frac{12}{5} = x$$

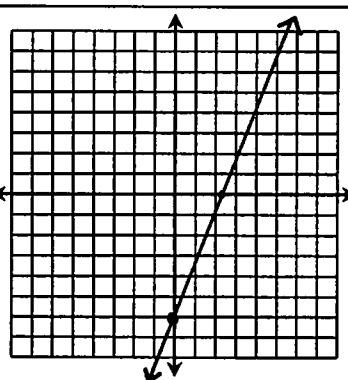
$$(\frac{12}{5}, 0)$$

Y-int:

$$4y = 10(0) - 24$$

$$4y = -24$$

$$y = -6 \quad (0, -6)$$



Topic 3: Writing Linear Equations & Applications

Write the equation in SLOPE-INTERCEPT FORM with the given information.

19. Passes through $(-8, 3)$ with a slope of -2

$$y - 3 = -2(x + 8)$$

$$y - 3 = -2x - 16$$

$$y = -2x - 13$$

20. Passes through $(-4, -2)$ and $(8, 7)$

$$m = \frac{7 - (-2)}{8 - (-4)} = \frac{9}{12} = \frac{3}{4}$$

$$y + 2 = \frac{3}{4}(x + 4)$$

$$y + 2 = \frac{3}{4}x + 3$$

$$y = \frac{3}{4}x + 1$$

21. Passes through $(-6, 4)$ and parallel to $x + y = 8$.

$$\begin{aligned} x + y &= 8 \\ \downarrow \\ y &= -x + 8 \\ m &= -1 \end{aligned}$$

$$\begin{aligned} y - 4 &= -1(x + 6) \\ y - 4 &= -x - 6 \\ \boxed{y} &= \boxed{-x - 2} \end{aligned}$$

22. Passes through $(-4, 1)$ and perpendicular to the line passing through $(-5, 2)$ and $(10, 8)$.

$$m = \frac{8-2}{10+5} = \frac{6}{15} = \frac{2}{5} \Rightarrow m = -\frac{5}{2}$$

$$\begin{aligned} y - 1 &= -\frac{5}{2}(x + 4) \\ y - 1 &= -\frac{5}{2}x - 10 \end{aligned}$$

$$\boxed{y = -\frac{5}{2}x - 9}$$

DEFINE VARIABLES and WRITE EQUATIONS to represent the following situations, then solve.

23. For commission as a realtor, Michelle earns \$349 plus 3% of the purchase price for each home she helps buy or sell. If she earned \$8,965 in commission on a certain home, find its purchase price.

p = purchase price

C = commission

$$C = 0.03p + 349$$

$$8965 = 0.03p + 349$$

$$8616 = 0.03p$$

$$p = 287,200$$

$\boxed{\$287,200}$

24. On Ryan's last social studies test, there were two types of questions: true/false worth 2 points each and multiple choice questions worth 4 points each. If Ryan earned 86 points on the test and answered 18 multiple choice questions correctly, how many true/false questions did he answer correctly?

x = true/false

y = mult. choice

S = score

$$S = 2x + 4y$$

$$86 = 2x + 4(18)$$

$$86 = 2x + 72$$

$$14 = 2x$$

$$x = 7$$

$\boxed{7 \text{ true/false questions}}$

25. The table below shows the altitude of a plane once it begins its descent to a runway.

Time (minutes)	Altitude (feet)
0	28,500
1	26,378
2	24,105
3	21,774
4	19,452
5	17,991

- a) Write an equation to model the data using linear regression.

$$y = -2161.54x + 28437.19$$

- b) Find the height of the plane after 8 minutes. $\rightarrow x = 8$

$$y = -2161.54(8) + 28437.19$$

$$\boxed{y = 11,144.87 \text{ ft}}$$

- c) How long will it take the plane to land? $\rightarrow y = 0$

$$0 = -2161.54x + 28437.19$$

$$2161.54x = 28437.19$$

$$x = 13.15$$

$\boxed{\sim 13 \text{ min}}$

Topic 4: Systems of Equations

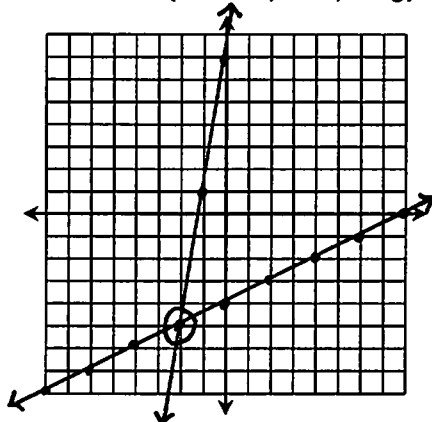
26. $\begin{cases} x - 2y = 8 \\ 6x - y = -7 \end{cases}$

(Solve by Graphing)

$$\begin{aligned} x - 2y &= 8 \\ -2y &= -x + 8 \\ y &= \frac{1}{2}x - 4 \end{aligned}$$

$$\begin{aligned} 6x - y &= -7 \\ 6x + 7 &= y \end{aligned}$$

$(-2, -5)$



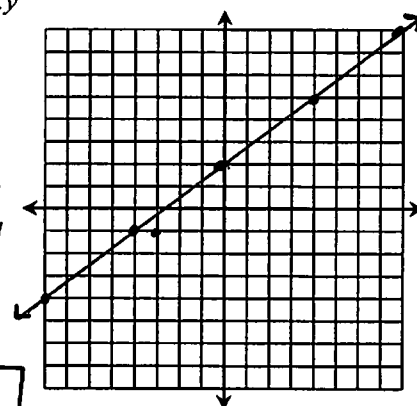
27. $\begin{cases} 3y - 6 = 4x \\ 16x + 24 = 12y \end{cases}$

(Solve by Graphing)

$$\begin{aligned} 3y - 6 &= 4x \\ 3y &= 4x + 6 \\ y &= \frac{4}{3}x + 2 \end{aligned}$$

$$\begin{aligned} 16x + 24 &= 12y \\ \frac{4}{3}x + 2 &= y \end{aligned}$$

Infinite Sol.



For questions 28 and 29, solve using the substitution method.

28. $\begin{cases} 5x - 4y = 9 \\ x + 7y = -6 \end{cases} \rightarrow x = -7y - 6$

$$\begin{aligned} 5(-7y - 6) - 4y &= 9 \\ -35y - 30 - 4y &= 9 \\ -39y &= 39 \\ y &= -1 \end{aligned}$$

$$\begin{aligned} x + 7(-1) &= -6 \\ x - 7 &= -6 \\ x &= 1 \end{aligned}$$

$(1, -1)$

29. $\begin{cases} 2x + 3y = -35 \\ 8x - y = -23 \end{cases} \rightarrow 8x + 23 = y$

$$\begin{aligned} 2x + 3(8x + 23) &= -35 \\ 2x + 24x + 69 &= -35 \\ 26x &= -104 \\ x &= -4 \end{aligned}$$

$$\begin{aligned} 8(-4) - y &= -23 \\ -32 - y &= -23 \\ -y &= 9 \\ y &= -9 \end{aligned}$$

$(-4, -9)$

For questions 30 and 31, solve using the elimination method.

30. $\begin{cases} 3x + 10 = 14y \\ 8x - 7y = 34 \end{cases}$

$$\begin{aligned} 3x - 14y &= -10 \\ (8x - 7y = 34) \cdot -2 \\ \hline 3x - 14y &= -10 \\ + -16x + 14y &= -68 \\ \hline -13x &= -78 \\ x &= 6 \end{aligned}$$

$$\begin{aligned} 3(6) + 10 &= 14y \\ 28 &= 14y \\ 2 &= y \end{aligned}$$

$(6, 2)$

31. $\begin{cases} 18x = 12y + 7 \\ -8y + 21 = -12x \end{cases}$

$$\begin{aligned} (18x - 12y = 7) \cdot 2 \\ (12x - 8y = -21) \cdot -3 \\ \hline 36x - 24y &= 14 \\ + -36x + 24y &= 63 \\ \hline 0 &\neq 77 \end{aligned}$$

No Solution

32. Ben has a collection of quarters and nickels worth \$5.35. If the number of nickels is five less than twice the number of quarters, find the number of each coin.

x = quarters
 y = nickels

$$.25x + .05y = 5.35$$

$$y = 2x - 5$$

$$.25x + .05(2x - 5) = 5.35$$

$$.25x + .10x - .25 = 5.35$$

$$.35x = 5.60$$

$$x = 16$$

$$y = 2(16) - 5$$

$$y = 32 - 5$$

$$y = 27$$

16 quarters and
 27 nickels

33. Aliyah bought four composition notebooks and three packs of pencils from the school bookstore and paid \$10.93. Laura bought seven composition notebooks and two packs of pencils and paid \$13.31. If each pencil pack contains ten pencils, what is the unit price per pencil?

x = Comp. notebooks
 y = pencils

$$4x + 3y = 10.93 \Rightarrow -8x - 6y = -21.86$$

$$7x + 2y = 13.31 \quad + 21x + 6y = 39.93$$

$$13x = 18.07$$

$$x = 1.39$$

$$4(1.39) + 3y = 10.93$$

$$3y = 5.37$$

$$y = 1.79$$

Each pencil cost
 \$ 0.179

34. Solve the system below using your method of choice:

$$\begin{cases} 4x + 2y - 5z = 47 \\ x - 2y + 6z = -10 \\ 9x - 7y - z = 75 \end{cases} \begin{matrix} A \\ B \end{matrix}$$

$$\begin{array}{r} 4x + 2y - 5z = 47 \\ + x - 2y + 6z = -10 \\ \hline 5x + z = 37 \end{array}$$

$$\begin{array}{r} 7 \cdot (4x + 2y - 5z = 47) \\ 2 \cdot (9x - 7y - z = 75) \\ \hline 28x + 14y - 35z = 329 \\ + 18x - 14y - 2z = 150 \\ \hline 46x - 37z = 479 \end{array}$$

$$5(8) + z = 37$$

$$40 + z = 37$$

$$z = -3$$

$$8 - 2y + 6(-3) = -10$$

$$-2y - 10 = -10$$

$$-2y = 0$$

$$y = 0$$

$$37 \cdot (5x + z = 37)$$

$$46x - 37z = 479$$

$$185x + 37z = 1369$$

$$+ 46x - 37z = 479$$

$$231x = 1848$$

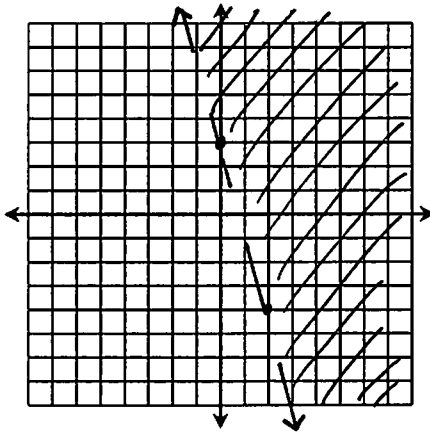
$$x = 8$$

(8, 0, -3)

Topic 5: Linear Inequalities & Systems of Linear Inequalities

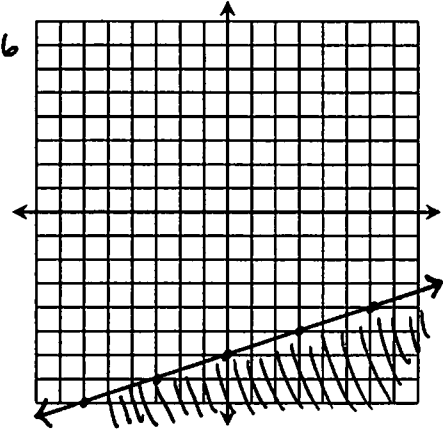
Show the solution to each linear inequality and system of linear inequalities by graphing.

35. $y > -\frac{7}{2}x + 3$



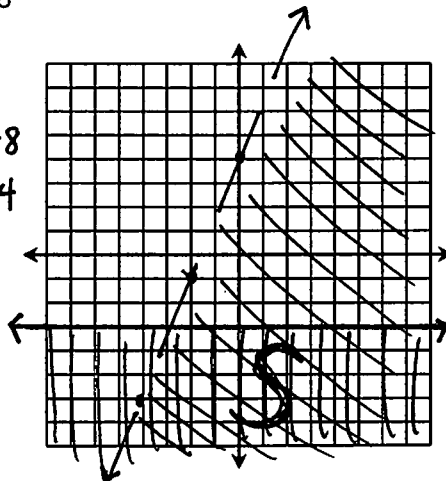
36. $2x - 6y \geq 36$

$$\begin{aligned} -6y &\geq -2x + 36 \\ y &\leq \frac{1}{3}x - 6 \end{aligned}$$



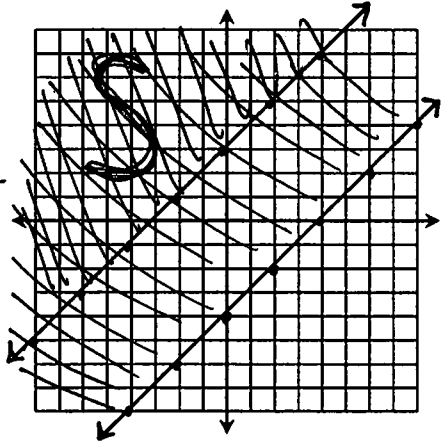
37. $\begin{cases} 5x - 2y > -8 \\ y \leq -3 \end{cases}$

$$\begin{aligned} 5x - 2y &> -8 \\ -2y &> -5x - 8 \\ y &< \frac{5}{2}x + 4 \end{aligned}$$



38. $\begin{cases} x - y \leq 4 \\ 3y \geq 3x + 9 \end{cases}$

$$\begin{aligned} x - y &\leq 4 \\ -y &\leq -x + 4 \\ y &\geq x - 4 \\ 3y &\geq 3x + 9 \\ y &\geq x + 3 \end{aligned}$$



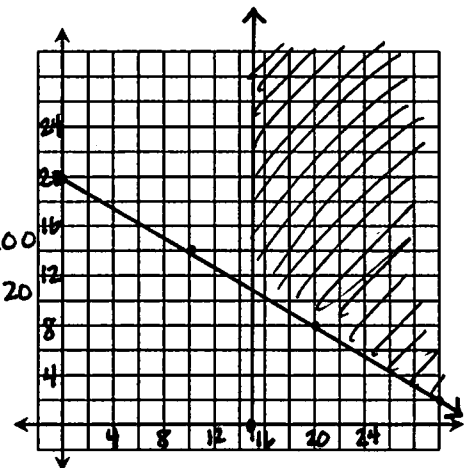
39. As part of Kayla's exercise program, she either runs 6 miles/day or rides her bike 10 miles/day. Her new goal is to cover a minimum distance of 200 miles, with at least 15 of the days running. She would like to determine the number of days it would take to accomplish this.

a) Write a system of linear inequality to represent this situation, then graph.

x = running
 y = biking

$$\begin{aligned} 6x + 10y &\geq 200 \\ x &\geq 15 \end{aligned}$$

$$\begin{aligned} 10y &\geq -6x + 200 \\ y &\geq -\frac{3}{5}x + 20 \end{aligned}$$



b) Using your graph, give two possible combinations of days Kayla can run and ride her bike to meet her goal.

(20, 10) - 20 mi. running and 10 mi. biking

(24, 6) - 24 mi. running and 6 mi. biking

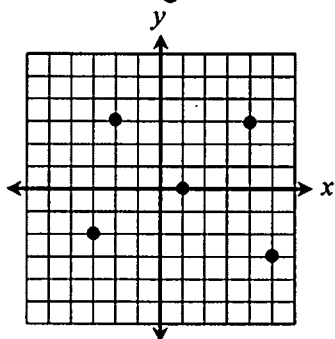
Name: _____

Unit 2 Test

Date: _____ Per: _____

Linear Functions & Systems

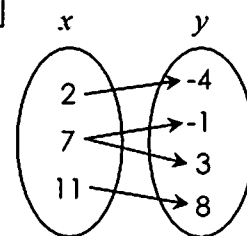
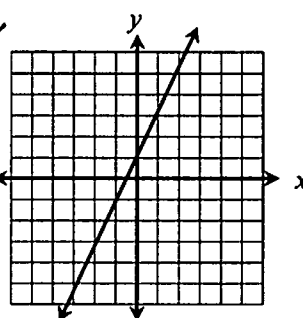
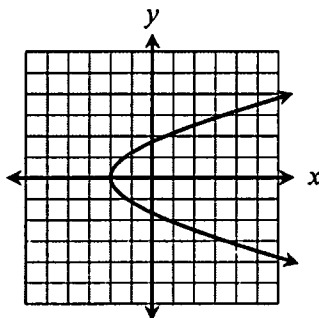
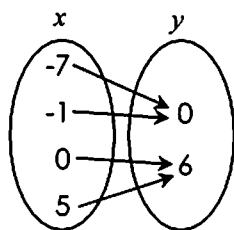
1. Find the range of the relation below, then determine whether the relation is a function.



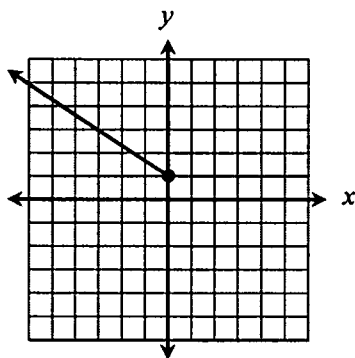
- A. $\{-3, -2, 1, 4, 5\}$; Function
- B. $\{-3, -2, 1, 4, 5\}$; Not a function
- C. $\{-3, -2, 0, 3\}$; Function
- D. $\{-3, -2, 0, 3\}$; Not a function

C

2. Which relations represent a function? Check all that apply.

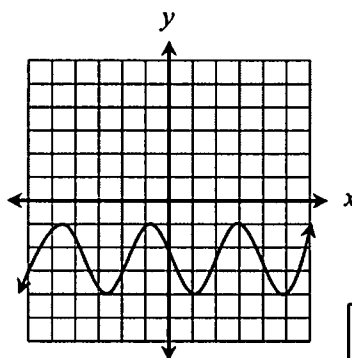


3. What is the domain of the graph below?



$$x \leq 0$$

4. What is the range of the graph below?



$$-4 \leq y \leq -1$$

5. Given the function below, find $f(7)$.

$$f(x) = -x^2 + 9x$$

$$\begin{aligned} f(7) &= -(7)^2 + 9(7) \\ &= -49 + 63 \\ &= 14 \end{aligned}$$

14

6. Given the functions below, find $b(-2) - a(12)$.

$$a(x) = |13 - 2x|; b(x) = 8x - 14$$

$$b(-2) = 8(-2) - 14 = -30$$

$$a(12) = |13 - 2(12)| = |13 - 24| = 11$$

$$-30 - 11 = -41$$

-41

7. Given the function below, find $p(x-1)$.

$$\begin{aligned} p(x) &= x^2 - 7x \\ p(x-1) &= (x-1)^2 - 7(x-1) \\ &= x^2 - 2x + 1 - 7x + 7 \\ &= x^2 - 9x + 8 \end{aligned}$$

$$x^2 - 9x + 8$$

8. Given the function below, if $h(x) = -1$, find x .

$$\begin{aligned} h(x) &= -6x + 20 \\ -1 &= -6x + 20 \\ -21 &= -6x \\ \frac{-21}{-6} &= \frac{-6x}{-6} \end{aligned}$$

$$\frac{7}{2} = x$$

$$x = \frac{7}{2}$$

9. Write the equation below in **standard form**.

$$\begin{aligned} 4 \cdot \left[\frac{3}{4}x - 3y = -21 \right] \\ 3x - 12y = -84 \end{aligned}$$

$$3x - 12y = -84$$

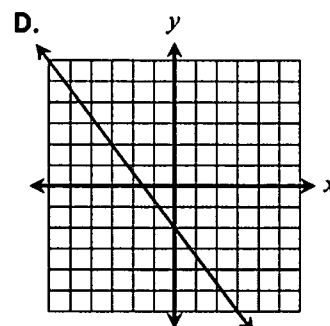
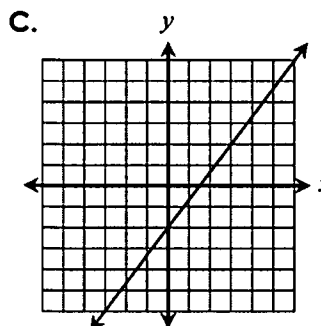
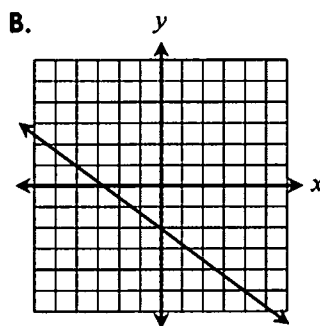
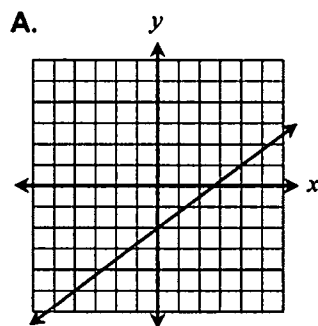
10. Write the equation below in **slope-intercept form**.

$$\begin{aligned} 14x &= 6y - 12 \\ 14x + 12 &= 6y \\ \frac{7}{3}x + 2 &= y \end{aligned}$$

$$y = \frac{7}{3}x + 2$$

11. Which graph best represents the equation $12x - 9y = 18$?

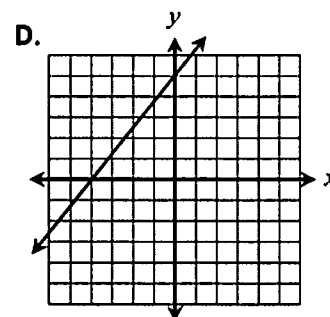
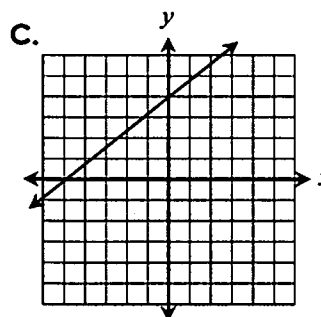
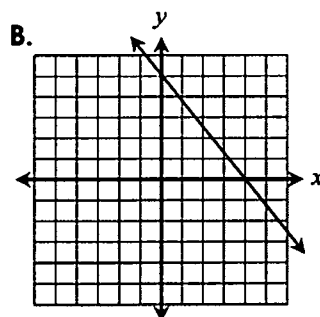
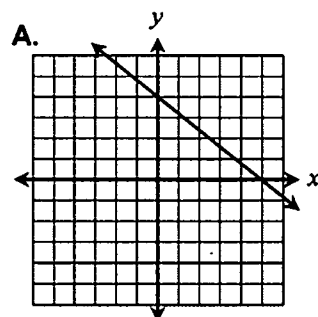
$$\begin{aligned} -9y &= -12x + 18 \\ y &= \frac{4}{3}x - 2 \end{aligned}$$



C

12. Which graph best represents the equation $-15y - 12x = -60$?

$$\begin{aligned} -15y &= 12x - 60 \\ y &= -\frac{4}{5}x + 4 \end{aligned}$$



A

13. Find the x- and y-intercepts of the equation below. Write your answers as ordered pairs.

$$8y = -9x - 12$$

x-int:
 $8(0) = -9x - 12$
 $12 = -9x$
 $-\frac{4}{3} = x$

y-int:
 $8y = -9(0) - 12$
 $8y = -12$
 $y = -\frac{3}{2}$

x-int.: $\left(-\frac{4}{3}, 0\right)$ y-int.: $\left(0, -\frac{3}{2}\right)$

14. Which of the following lines is **perpendicular** to the equation given below?

$$\begin{aligned} -9x &= 15 - 3y \\ -9x - 15 &= -3y \\ 3x + 5 &= y \end{aligned}$$

A. $6x + 2y - 8 = 0$

B. $x - 3y = -3$

C. $12x - 24 = 4y$

D. $6y = 42 - 2x$

$y = 1 - \frac{1}{3}x$

D

15. Give an example of a linear equation that is **parallel** to the y-axis.

* Ans. may vary *

$x = -1$

16. Write a linear equation in **slope-intercept form** that passes through the point $(-1, -8)$ and is **parallel** to the line $15x + 3y = 21$.

$$\begin{aligned} y + 8 &= -5(x + 1) \\ y + 8 &= -5x - 5 \\ y &= -5x - 13 \end{aligned}$$

$$\begin{aligned} 3y &= -15x + 21 \\ y &= -5x + 7 \end{aligned}$$

$y = -5x - 13$

17. Write a linear equation in **slope-intercept form** that passes through the points $(-11, -5)$ and $(1, -2)$.

$$m = \frac{-2 + 5}{1 + 11} = \frac{3}{12} = \frac{1}{4}$$

$$y + 2 = \frac{1}{4}(x - 1)$$

$$y + 2 = \frac{1}{4}x - \frac{1}{4}$$

$$y = \frac{1}{4}x - \frac{9}{4}$$

$y = \frac{1}{4}x - \frac{9}{4}$

18. For each boat he sells, Mick earns \$149 in addition to 2% of the purchase price of the boat as commission. If p represents the purchase price of the boat, which equation represents Mick's commission, c ?

A. $c = 149 + .2p$

B. $c = 149 + .02p$

C. $p = 149 + .2c$

D. $p = 149 + .02c$

B

19. Angela fully charged her laptop, then unplugged it and tracked the battery level at certain times under constant use. The table below shows her findings. Find an equation for the line of best fit, then estimate the battery level after two hours of constant use unplugged.

$x = 120$

Time	Battery Level (%)
2:00	100%
2:10	94%
2:25	87%
2:40	83%
3:00	79%
3:10	76%

$$y = -0.32x + 97.57$$

$$\begin{aligned} y &= -0.32(120) + 97.57 \\ &= 59.17 \end{aligned}$$

Equation:

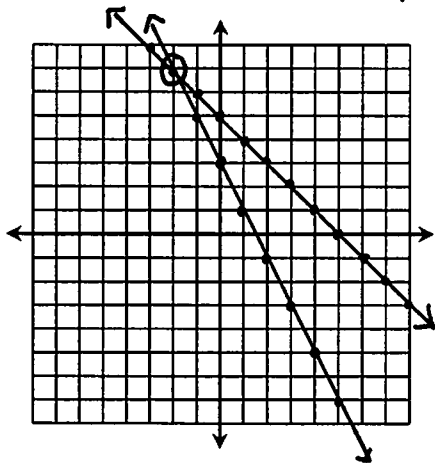
$y = -0.32x + 97.57$

Solution:

$\sim 59\%$

20. Solve the system below by graphing.

$$\begin{cases} y = 3 - 2x \rightarrow y = -2x + 3 \\ -4x - 4y = -20 \rightarrow -4y = 4x - 20 \\ y = -x + 5 \end{cases}$$



$(-2, 7)$

21. Solve the system below by substitution.

$$\begin{cases} 6x - y = -23 \rightarrow 6x + 23 = y \\ 2x + 5y = -13 \end{cases}$$

$$\begin{aligned} 2x + 5(6x + 23) &= -13 \\ 2x + 30x + 115 &= -13 \\ 32x &= -128 \\ x &= -4 \end{aligned}$$

$$\begin{aligned} 6(-4) - y &= -23 \\ -24 - y &= -23 \\ -y &= 1 \\ y &= -1 \end{aligned}$$

$(-4, -1)$

22. Solve the system below by elimination.

$$\begin{aligned} \begin{cases} x = y - 4 \\ 7x - 2y = 17 \end{cases} &\rightarrow \begin{aligned} -2 \cdot (x - y = -4) \\ 7x - 2y = 17 \end{aligned} \\ &\quad \underline{-2x + 2y = 8} \\ &\quad + 7x - 2y = 17 \\ &\quad \hline &\quad 5x = 25 \\ &\quad x = 5 \end{aligned}$$

$$\begin{aligned} 5 &= y - 4 \\ 9 &= y \end{aligned}$$

$(5, 9)$

23. Solve the system below by elimination.

$$\begin{aligned} \begin{cases} 2y = -24 - 8x \\ 6x + 18 = 5y \end{cases} &\rightarrow \begin{aligned} (8x + 2y = -24) \cdot 3 \\ (6x - 5y = -18) \cdot -4 \end{aligned} \\ &\quad \underline{24x + 6y = -72} \\ &\quad + \underline{-24x + 20y = 72} \\ &\quad \hline &\quad 26y = 0 \\ &\quad y = 0 \end{aligned}$$

$$\begin{aligned} 6x + 18 &= 5(0) \\ 6x &= -18 \\ x &= -3 \end{aligned}$$

$(-3, 0)$

24. Which system of equations has an infinite solution?

A. $\begin{cases} y = 2x - 7 \\ 2x + y = 5 \end{cases}$

$$y = 2x - 7$$

$$y = -2x + 5$$

B. $\begin{cases} x = 3y + 15 \\ 12y = 4x - 60 \end{cases}$

$$y = \frac{1}{3}x - 5$$

$$y = \frac{1}{3}x - 5$$

C. $\begin{cases} 10x + 8y = -16 \\ -5x = 4y + 28 \end{cases}$

$$y = -\frac{5}{4}x - 2$$

$$y = -\frac{5}{4}x - 7$$

D. $\begin{cases} y = 8 \\ x = 8 \end{cases}$

B

25. Jasmine bought 1.3 pounds of ham and 0.8 pounds of cheese from the deli and paid \$11.28. She went back the following week and bought 1.5 pounds of ham and 1.2 pounds of cheese and paid \$14.76. If the prices remained the same, find the price per pound of ham and cheese.

$x = \text{ham}$
 $y = \text{cheese}$

$$\begin{array}{r} 3 \cdot (1.3x + 0.8y = 11.28) \\ -2 \cdot (1.5x + 1.2y = 14.76) \\ \hline 3.9x + 2.4y = 33.84 \\ + -3.0x - 2.4y = -29.52 \\ \hline 0.9x = 4.32 \\ x = 4.8 \end{array}$$

$$\begin{array}{r} 1.3(4.8) + 0.8y = 11.28 \\ 0.8y = 5.04 \\ y = 6.3 \end{array}$$

Ham:

\$4.80/lb

Cheese:

\$6.30/lb

26. Solve the system below.

$$\begin{cases} 3x - 5y + 2z = -53 \\ x - 7y - 4z = -37 \\ 5x + 9y + 2z = 57 \end{cases}$$

$$\begin{array}{r} (3x - 5y + 2z = -53) \cdot 2 \\ x - 7y - 4z = -37 \\ \hline 6x - 10y + 4z = -106 \\ + x - 7y - 4z = -37 \\ \hline 7x - 17y = -143 \end{array}$$

$$\begin{array}{r} x - 7y - 4z = -37 \\ (5x + 9y + 2z = 57) \cdot 2 \\ \hline x - 7y - 4z = -37 \\ + 10x + 18y + 4z = 114 \\ \hline 11x + 11y = 77 \end{array}$$

$$\begin{array}{r} 11x + 11(8) = 77 \\ 11x + 88 = 77 \\ 11x = -11 \\ x = -1 \end{array}$$

$$\begin{array}{r} -1 - 7(8) - 4z = -37 \\ -57 - 4z = -37 \\ -4z = 20 \\ z = -5 \end{array}$$

$$\begin{array}{r} 11 \cdot (7x - 17y = -143) \\ -7 \cdot (11x + 11y = 77) \\ \hline 77x - 187y = -1573 \\ + -77x - 77y = -539 \\ \hline -264y = -2122 \\ y = 8 \end{array}$$

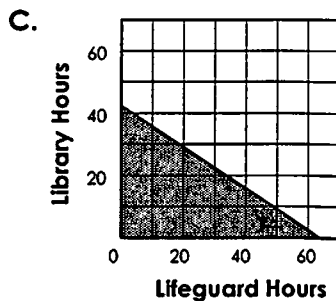
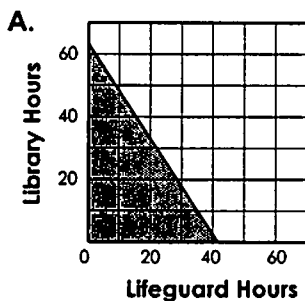
$(-1, 8, -5)$

27. Which ordered pairs represent solutions to the linear inequality below? Check all that apply.

$$x - 4y < -12$$

$1 + 32 < -12$	$-8 + 12 < -12$	$-12 < -12$
☐ (1, -8)	☐ (-8, -3)	☐ (-12, 0)
☒ (3, 5)	☒ (-7, 9)	☐ (-4, 2)
$3 - 20 < -12$	$-7 - 36 < -12$	$-4 - 8 < -12$

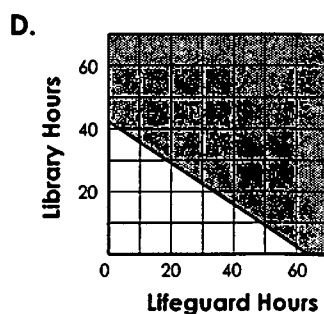
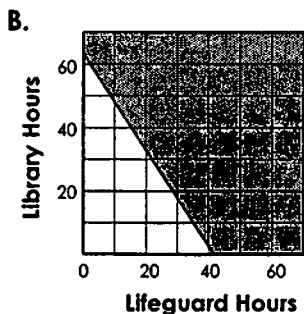
28. Jon makes \$12 per hour as a lifeguard and \$8 per hour working at the library. This summer, he would like to make at least \$500 working his two jobs. Which graph represents the number of hours he must work at each job to reach his goal?



$$12x + 8y \geq 500$$

$$8y \geq -12x + 500$$

$$y \geq -\frac{3}{2}x + 62.5$$



B

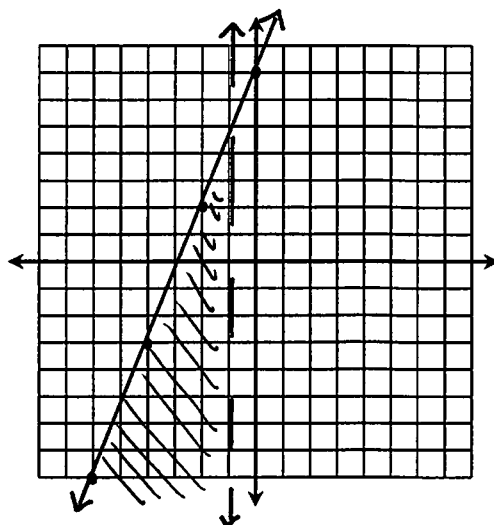
29. Graph the system of linear inequalities. Then, choose the point that represents a solution to the system.

$$\begin{cases} 5x - 2y \geq -14 \\ x < -1 \end{cases}$$

$$5x - 2y \geq -14$$

$$-2y \geq -5x - 14$$

$$y \leq \frac{5}{2}x + 7$$



- A. (-6, 1)
B. (4, 0)
C. (1, 8)
D. (-3, -7)

D

30. To save money for prom this weekend, Tom is going to walk his neighbor's dog for \$6 per hour and wash cars for \$7.50 per hour. His mother said he can work no more than 15 hours to ensure he keeps up with his homework. Tom needs to make at least \$75 to cover prom expenses. If d represents the number of hours walking his neighbor's dog and c represents the number of hours washing cars, which system of inequalities represents this situation?

A. $\begin{cases} 6d + 7.5c \leq 15 \\ d + c \geq 75 \end{cases}$

B. $\begin{cases} 6d + 7.5c \geq 15 \\ d + c \leq 75 \end{cases}$

C. $\begin{cases} 6d + 7.5c \leq 75 \\ d + c \geq 15 \end{cases}$

D. $\begin{cases} 6d + 7.5c \geq 75 \\ d + c \leq 15 \end{cases}$

D

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
LINEAR PROGRAMMING	A method which uses linear equations and inequalities to determine the optimal solution of a certain situation
<i>Finding the OPTIMAL SOLUTION</i>	1 Use the CONSTRAINTS to write a SYSTEM OF INEQUALITIES. Constraints are the limitations on a problem. 2 GRAPH the system, clearly showing the FEASIBLE REGION. 3 Write an OBJECTIVE FUNCTION. This is the quantity you would like to minimize or maximize. 4 TEST THE VERTICES of the feasible region to find the OPTIMAL SOLUTION.

Example 1: The smallest plane in the Coast2Coast Airlines fleet offers first class and tourist class seats, and can hold a maximum of 50 passengers. In order to charter a flight, the airline must sell at least 8 first class tickets and at least 20 tourist class tickets. The company makes \$40 profit for each first class ticket and \$30 profit for each tourist class ticket sold. In order for the company to maximize its profits on a single flight, how many first class and tourist class tickets should they sell?

CONSTRAINTS:

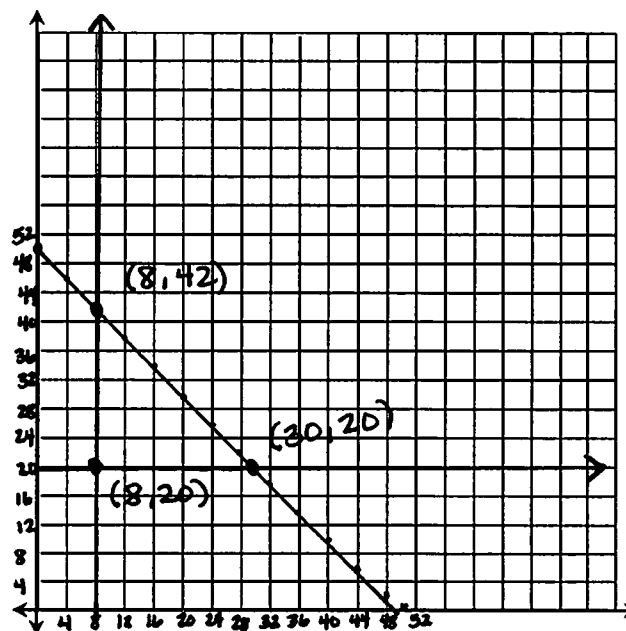
x = first class
 y = tourist class

$$x + y \leq 50 \rightarrow y \leq -x + 50$$

$$x \geq 8$$

$$y \geq 20$$

GRAPH:



OBJECTIVE FUNCTION: $P = 40x + 30y$

TEST VERTICES:

$(8, 20)$

$$P = 40(8) + 30(20)$$

$$= 920$$

$(8, 42)$

$$P = 40(8) + 30(42)$$

$$= 1580$$

$(30, 20)$

$$P = 40(30) + 30(20)$$

$$= 1800$$

SOLUTION: $(30, 20)$

Profits are maximized by selling 30 first class tickets and 20 tourist class tickets.

Example 2: Natalie is a hair stylist. When making appointments for her clients, she allots 30 minutes for each cut and two hours for each cut and color. She has ten hours available for appointments each day and prefers to see no more than 8 total clients. Each routine cut costs \$40 and each cut and color costs \$100. How many of each type of appointment should she schedule each day to maximize her income?

CONSTRAINTS:

$$x = \text{cut}$$

$$y = \text{cut} + \text{color}$$

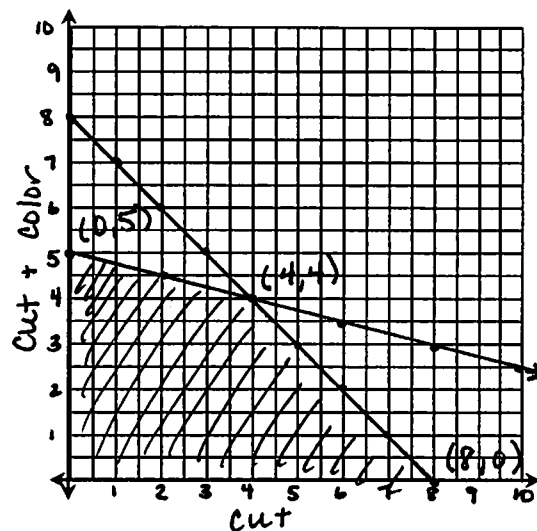
$$\frac{1}{2}x + 2y \leq 10 \rightarrow y \leq -\frac{1}{4}x + 5$$

$$x + y \leq 8 \rightarrow y \leq -x + 8$$

$$x \geq 0$$

$$y \geq 0$$

GRAPH:



OBJECTIVE FUNCTION: $P = 40x + 100y$

TEST VERTICES:

$$(0, 5)$$

$$P = 40(0) + 100(5) = 500$$

$$(4, 4)$$

$$P = 40(4) + 100(4) = 560$$

$$(8, 0)$$

$$P = 40(8) + 100(0) = 320$$

SOLUTION:

Natalie will maximize her profits by doing 4 cuts and 4 cuts w/ color.

Example 3: Baking a tray of chocolate chip cookies takes three cups of milk and two cups of flour. Baking a tray of oatmeal raisin cookies takes one cup of milk and four cups of flour. The baker has 9 cups of milk and 16 cups of flour available. He makes \$2.50 profit per tray of chocolate chip cookies and \$2 profit per tray of oatmeal raisin. How many trays of each type of cookies should he make to maximize his profit?

$$x = \text{Choc. chip}$$

$$y = \text{oat. raisin}$$

$$\text{Milk} = 3x + y \leq 9 \rightarrow y \leq -3x + 9$$

$$\text{Flour} = 2x + 4y \leq 16 \rightarrow y \leq -\frac{1}{2}x + 4$$

$$x \geq 0$$

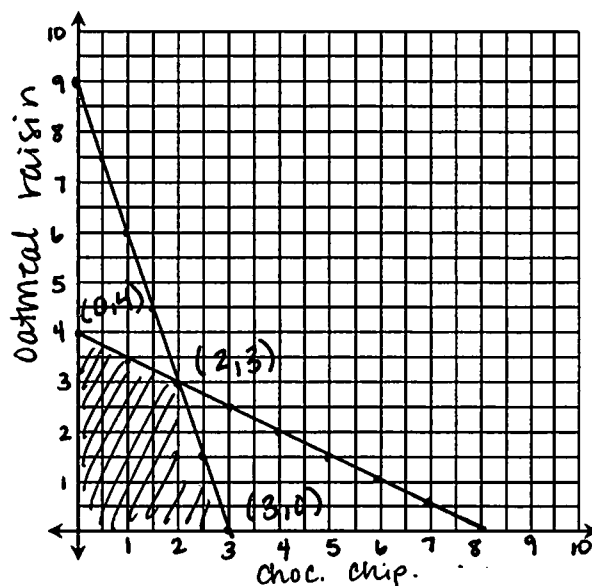
$$y \geq 0$$

$$P = 2.5x + 2y$$

$$(0, 4) : P = 2.5(0) + 2(4) = 8$$

$$(2, 3) : P = 2.5(2) + 2(3) = 11$$

$$(3, 0) : P = 2.5(3) + 2(0) = 7.5$$



He should make 2 trays of chocolate chip and 3 trays of oatmeal raisin.

LINEAR PROGRAMMING

Choice Board Assignment

Name: _____

Date: _____ Per: _____

Directions: Choose any **two** problems from the linear programming choice board. Organize and show all work to support your final answer. Give a written summary at the end with your recommendation.

Choice # 1

x = mountain bikes

y = road bikes

$$4x + 3y \leq 480 \rightarrow y \leq -\frac{4}{3}x + 280$$

$$x + y \geq 250 \rightarrow y \geq -x + 250$$

$$x \geq 0$$

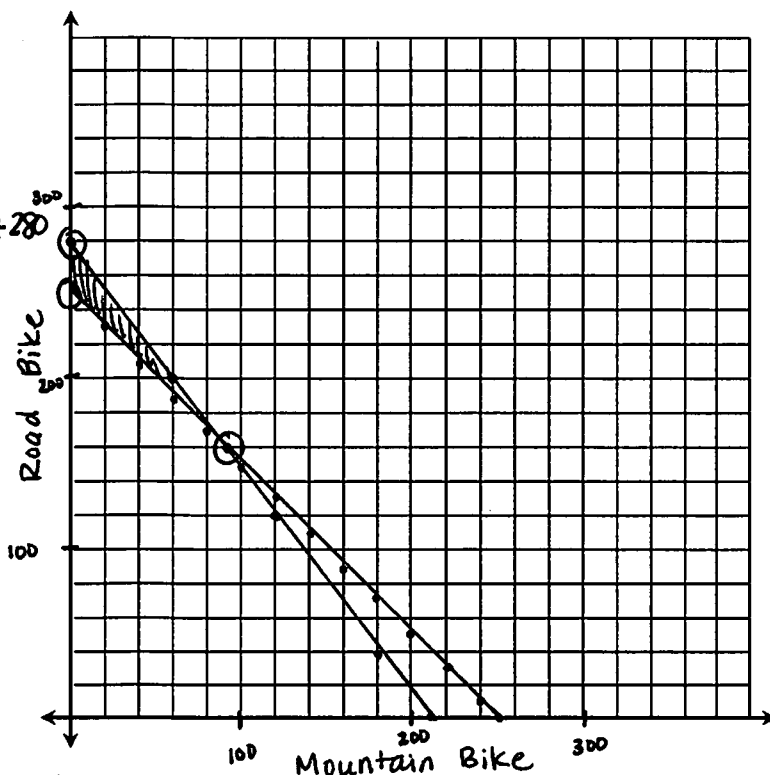
$$y \geq 0$$

$$P = 75x + 60y \text{ (Max)}$$

$$(0, 250): 75(0) + 60(250) = 15,000$$

$$(0, 280): 75(0) + 60(280) = 16,800$$

$$(90, 160): 75(90) + 60(160) = 16,350$$



Summary: Streamline Bicycles will maximize profits by producing 280 road bikes and 0 mountain bikes each week.

Choice # 2

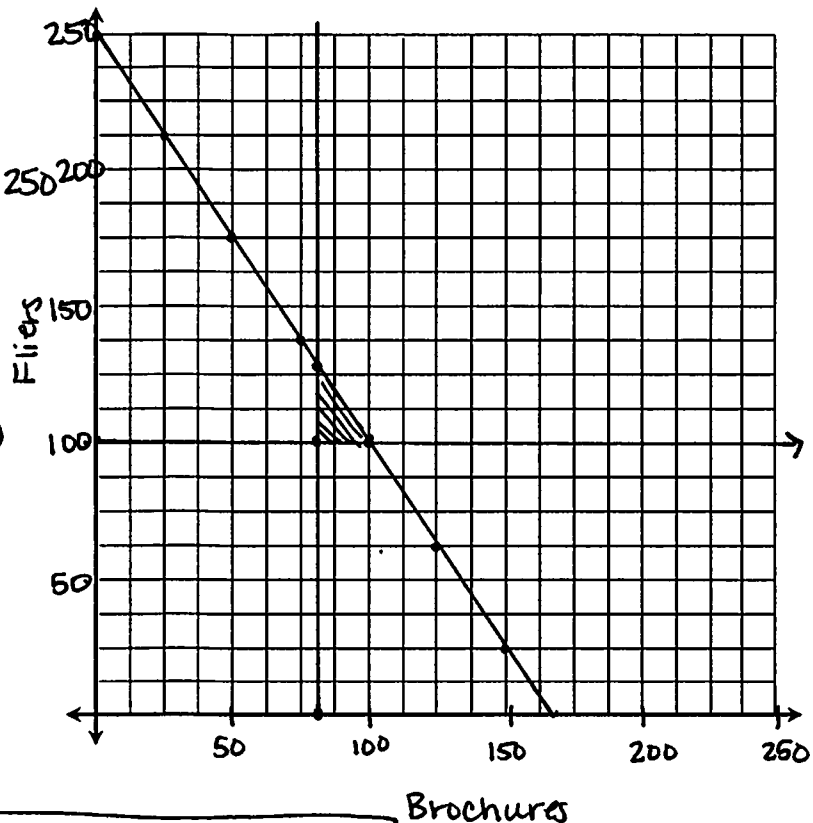
$x =$ brochures
 $y =$ fliers

$$3x + 2y \leq 500 \rightarrow y \leq -\frac{3}{2}x + 250$$

$$x \geq 80$$

$$y \geq 100$$

$$C = 0.10x + 0.06y \text{ (Min.)}$$



$$(80, 100): C = 0.10(80) + 0.06(100) = 14$$

$$(80, 130): C = 0.10(80) + 0.06(130) = 15.8$$

$$(100, 100): C = 0.10(100) + 0.06(100) = 16$$

Summary: Beach Cruises can minimize costs by
printing 80 brochures and 100 fliers.

Choice # 3

$x = \text{reserved}$

$y = \text{lawn}$

$$x + y \leq 20000 \rightarrow y \leq -x + 20000$$

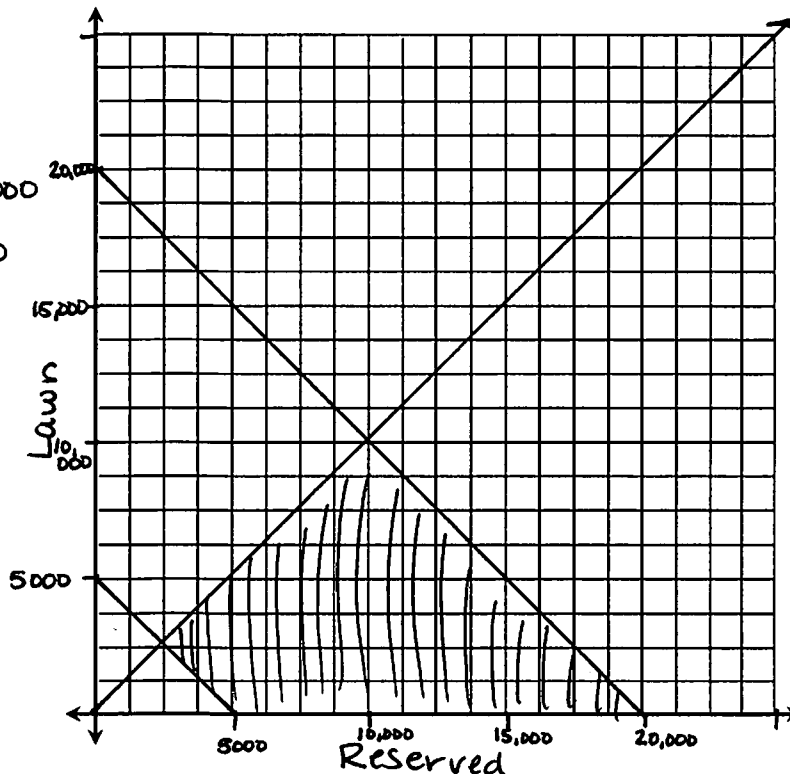
$$x + y \geq 5000 \rightarrow y \geq -x + 5000$$

$$y \leq x$$

$$x \geq 0$$

$$y \geq 0$$

$$P = 65x + 40y \text{ (Max.)}$$



$$(5000, 0) : 65(5000) + 40(0) = 325,000$$

$$(2500, 2500) : 65(2500) + 40(2500) = 262,500$$

$$(10,000, 10,000) : 65(10000) + 40(10000) = 1,050,000$$

$$(20,000, 0) : 65(20000) + 40(0) = 1,300,000$$

Summary: The amphitheater will maximize profits by selling 20,000 reserved seats and 0 lawn seats.

Choice # 4

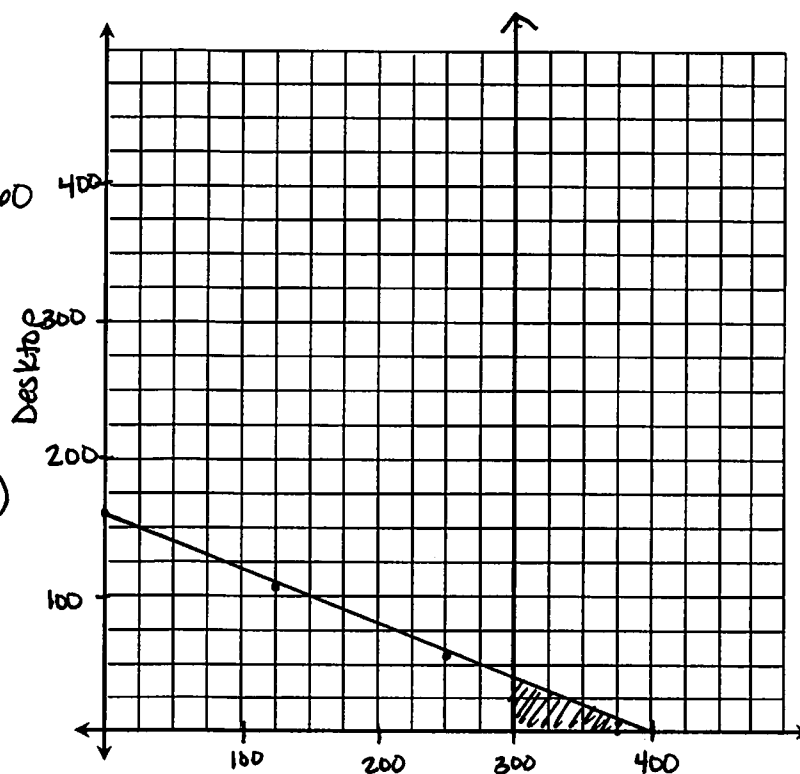
$x = \text{laptop}$
 $y = \text{desktop}$

$$2x + 5y \leq 800 \rightarrow y \leq -\frac{2}{5}x + 160$$

$$x \geq 300$$

$$y \geq 0$$

$$P = 225x + 280y \text{ (Max.)}$$



$$(300, 0): 225(300) + 280(0) = 67,500 \text{ Laptop}$$

$$(300, 40): 225(300) + 280(40) = 78,700$$

$$(400, 0): 225(400) + 280(0) = 90,000$$

Summary: JP Computers will maximize their profits
if they load 400 laptops and 0 desktop
computers.

Choice # 5

$x = \text{Snowboards}$

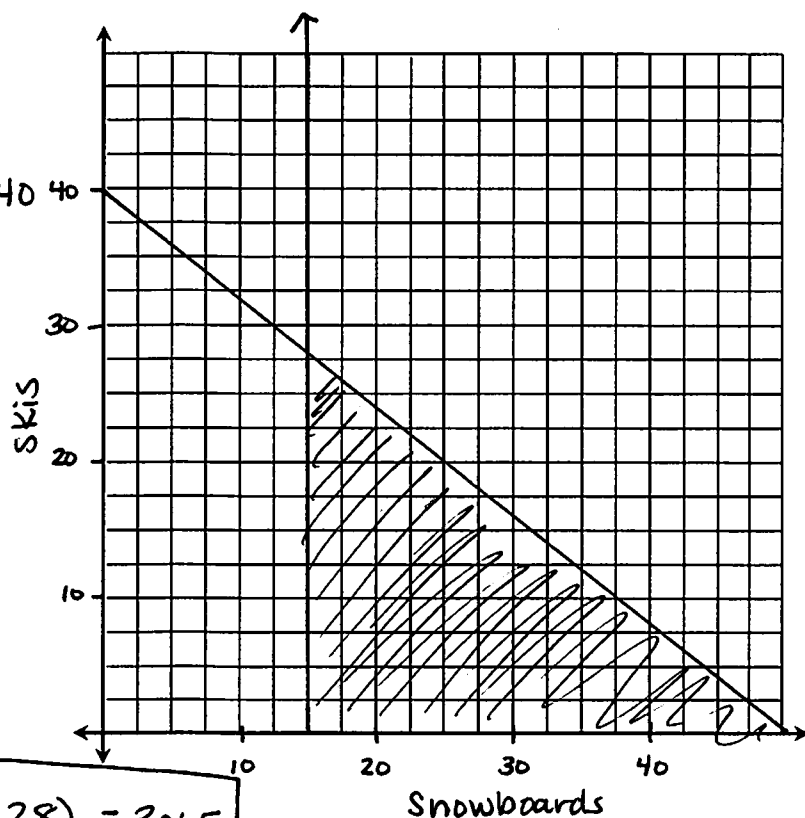
$y = \text{Skis}$

$$8x + 10y \leq 400 \rightarrow y \leq -\frac{4}{5}x + 40$$

$$x \geq 15$$

$$y \geq 0$$

$$P = 55x + 80y \text{ (Max)}$$



$$(15, 28): 55(15) + 80(28) = 3065$$

$$(15, 0): 55(15) + 80(0) = 825$$

$$(50, 0): 55(50) + 80(0) = 2750$$

Summary: The Polar Sports Company will maximize profits by producing 15 snowboards and 28 sets of skis each week.

Choice # 6

$x = \text{Buffalo}$

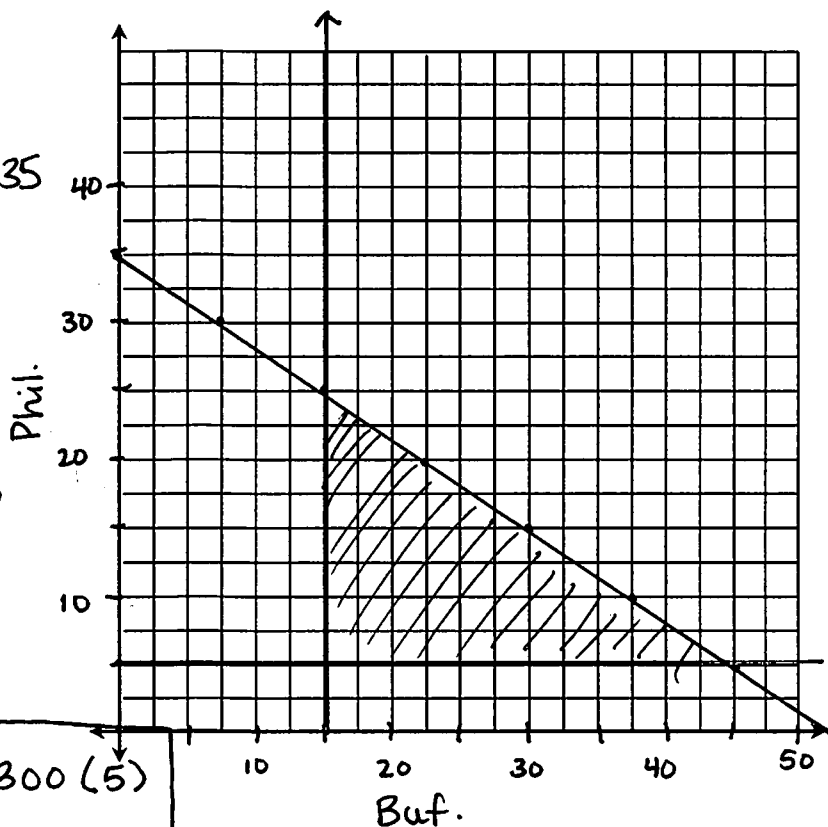
$y = \text{Philadelphia}$

$$2x + 3y \leq 105 \rightarrow y \leq -\frac{2}{3}x + 35$$

$$x \geq 15$$

$$y \geq 5$$

$$C = 180x + 300y \text{ (Min)}$$



$$(15, 5) : C = 180(15) + 300(5) \\ = 4200$$

$$(15, 25) : C = 180(15) + 300(25) \\ = 10,200$$

$$(45, 5) : C = 180(45) + 300(5) \\ = 9,600$$

Summary: In order to minimize costs, Ben should take 15 trips to Buffalo and 5 trips to Philadelphia.

CREDITS

I use clipart and
fonts in my products by:



Art with Jenny K



Many thanks to these
talented artists!